Shaping Persistent Representations from Independent Interactions

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ABSTRACT

World models learn environment dynamics from interaction experience. These dynamics depend on the current state and actions, as well as on properties that persist across interactions. Yet standard predictive training can reduce error using local evidence alone, without organizing persistent information into reusable context. We introduce **SPRII**, a training principle that uses relations between interactions as weak supervision for persistent context while retaining the learner's native objective. For example, different trajectories of the same system share persistent properties even when their states and actions differ. SPRII uses such relations to guide context learning without numerical property labels. Two composable components act on related interactions: Align encourages context agreement while preserving variation; Cross predicts one interaction's future from the other's context. Our analysis distinguishes three linked questions: what persistent information is accessible in the learned context (**Formation**), how that context influences a fixed predictor (**Use**), and whether it reduces task error (**Value**). Success at one stage does not guarantee success at the next. Controlled experiments show that more reliable relations improve representation organization, but changing the pairing rule so paired interactions share another property can reduce access to one that remains shared. Context substitutions change predictions at fixed model weights, while the benefit from history depends on prediction horizon and readout. Evaluations span thirteen settings, including controlled physical systems, public dynamics tasks, robotic and tactile data, and partner interaction, across multiple learner families. Across five primary comparisons spanning JEPA, CPC and RSSM learners, SPRII reduces prediction error by 14.6% on average relative to the corresponding baselines.

The project page and code are publicly available.

1 INTRODUCTION

World models infer how environments evolve to support planning, decision-making, and counterfactual reasoning (Hafner et al., 2019; 2020; Zhou et al., 2025). In physical domains, an interaction sequence interweaves two distinct kinds of information: transient interaction states that vary from moment to moment, and persistent underlying properties, such as masses, friction coefficients, or spring constants, that remain invariant across encounters with the same system. An object at rest reveals little about how quickly it will slow after a push; an earlier interaction can provide that information. Standard autoregressive or recurrent dynamics predictors optimize per-step or short-horizon prediction objectives over individual trajectories. When recent motion already predicts the target well, this objective creates an inductive bias toward dominant transient cues and local kinematic shortcuts. Consequently, accurate prediction need not yield reusable representations of persistent system invariants, leaving models vulnerable when novel physical regimes or longer prediction horizons demand information beyond local motion (Figure 1).

Existing approaches follow two broad strategies. Explicit system identification estimates physical parameters, while meta-learning learns to adapt across systems (Rakelly et al., 2019; Zintgraf et al., 2020); these approaches can depend on privileged supervision, specialized model assumptions, or

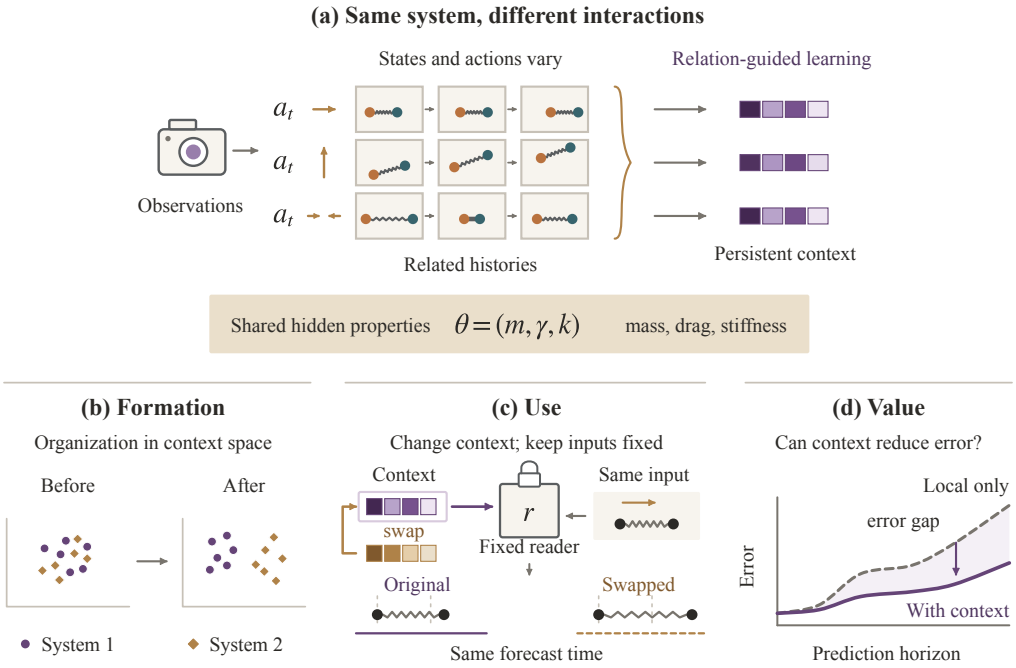


Figure 1: **From related interactions to persistent context.** All panels are schematic. (a) Interactions vary in state and action while sharing hidden system properties $\theta = (m, \gamma, k)$. Relations between observed histories guide context learning without numerical property labels. Three separate questions examine the learned context: (b) Formation: what persistent information is organized? (c) Use: does swapping context change the predicted motion with the reader, local input and forecast time fixed? (d) Value: can context reduce task error? Success at one does not establish the others.

costly inner-loop adaptation. Context-conditioned models infer context from histories for prediction (Lee et al., 2020; Shaj et al., 2022; Kirchmeyer et al., 2022). Yet history-conditioned prediction alone provides no explicit cross-interaction pressure to organize persistent invariants for reuse. Learned context representations can collapse into transient summaries. Even when physical information is accessible, a predictor need not use it effectively (Tan et al., 2026). This gap points to a broader question: *how should a world model learn and use what remains stable across changing interactions?* We study one concrete instance: learning structured, reusable persistent representations across distinct interactions without privileged physical labels.

To bridge this gap, we introduce **SPRII** (Shaping Persistent Representations from Independent Interactions), a relation-supervised training principle. Rather than relying on numerical property labels, SPRII exploits weak relational pairings indicating that separately realized interactions (hereafter, independent interactions) share persistent properties—for example, that two trajectories originate from the same underlying physical system. Building on the value of interaction relations (Gondal et al., 2021; Guo et al., 2022), SPRII operationalizes this supervision through two complementary, composable objectives: **Align** encourages representational consistency across related trajectories while maintaining variation, organizing shared persistent properties into coherent latent manifolds; **Cross** routes context extracted from one interaction to predict the forward dynamics of another related interaction, pushing the dynamics model to exploit persistent context rather than rely solely on local kinematic shortcuts. These components build on agreement regularization and cross-trajectory prediction (Bardes et al., 2022; Chen et al., 2026) and preserve each learner's native objective. This formulation makes the relation itself—its reliability and shared semantics—an explicit training design variable, allowing us to trace how relation design reshapes the formation, use and predictive value of persistent information. SPRII integrates into predictive, recurrent, contrastive, and supervised learners through an existing history state or a dedicated encoder (Figure 2).

Persistent information can be learned without being used, and used without improving prediction. We distinguish three linked stages: whether shared system properties become organized and accessible in context (**Formation**), whether a fixed predictor relies on that context (**Use**), and when it reduces error

for a given task, horizon, and readout (**Value**). Probes, fixed-model context substitutions, and task comparisons test each stage (Figure 1); Appendix I gives analytical counterparts.

We apply SPRII across controlled physical systems, public benchmarks, real robot recordings, and non-physical tasks, with predictive, recurrent, contrastive, and supervised learners. It reduces cold-start prediction error by 22.3% relative to Native JEPA on 256 sealed SpringWorld systems. On held-out CoPhy episodes (Baradel et al., 2020), prediction error falls by 11.4–15.4% across three Collision learners and by 9.9% on Balls4 RSSM, both relative to the history-conditioned Structure control. Averaged equally across these five primary comparisons, the reduction is 14.6% (Table 1). Controlled interventions reveal how relations shape each stage. In Formation, reliable pairings strengthen latent organization, while changing the pairing rule to share additional properties can suppress access to existing invariants—a phenomenon explained by capacity-limited spectral selection in our linear reference model (Proposition 3). In Use, substituting donor context into a fixed downstream reader trained on frozen representations steers its forecasts toward the donor's physical dynamics, demonstrating functional reliance on persistent context. In Value, history gains grow at longer horizons in SpringWorld, and lightweight readout adaptation extracts further predictive benefit from frozen context without end-to-end retraining (Figures 3–5).

Our key contributions are:

- **A relation-supervised training principle.** SPRII uses weak relations between independent interactions, through complementary Align and Cross objectives, to organize reusable persistent information without ground-truth physical labels.
- **A Formation–Use–Value decomposition.** We distinguish whether persistent information is formed in the representation, used by a fixed predictor, and valuable for downstream prediction, and probe each stage with operational measurements and targeted interventions, showing that the three can dissociate.
- **Mechanistic findings.** Controlled interventions show how relational pairings govern persistent structure, how donor context steers fixed predictions, and how horizon and readout shape task benefit. A linear reference model explains how limited capacity can create competition among shared properties.
- **Broad validation.** Across multiple learner families, we evaluate SPRII in thirteen settings spanning controlled physical systems, public benchmarks, real robot and tactile recordings, and non-physical tasks. Detailed Formation–Use–Value analyses focus on selected settings.

2 A Relation-Supervised Training Principle

2.1 Interactions, relations and persistent properties

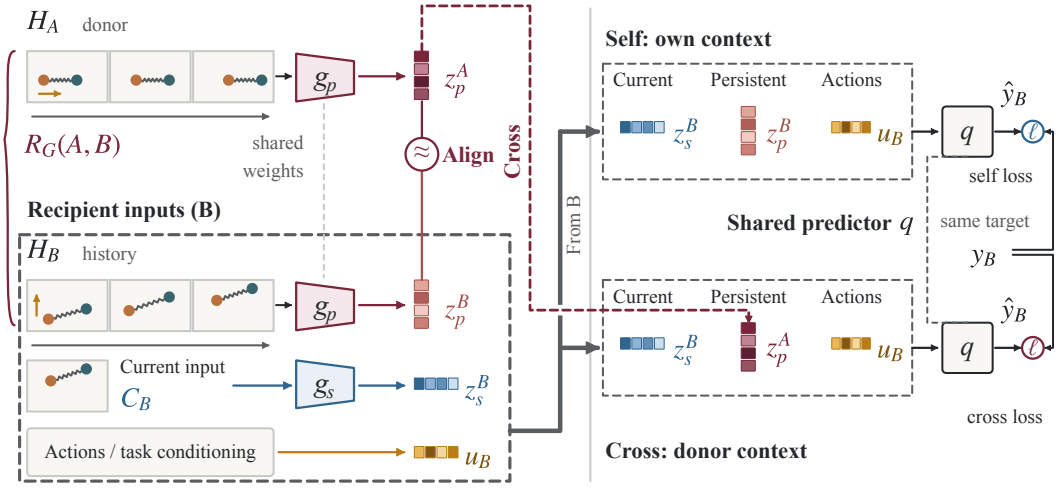
To predict a new spring trajectory (the recipient), a model can use an earlier trajectory of the same system (the donor). Denote these interactions by B and A. Recipient B has history H_B, current input C_B, permitted task conditioning u_B and target y_B; donor A supplies history H_A. The relation $\mathcal{R}_G(A, B)$ identifies shared persistent information, for example through system, task or partner identity. In controlled systems, G specifies the shared subset of θ, such as mass while drag varies.

Training samples a recipient, then a related donor, defining pair distribution Π_G. Relation membership supervises persistent context by identifying which interactions are expected to share information, without supplying numerical property targets. Here, the term *independent interactions* denotes separately realized episodes or trajectories. The relation defines what is shared, while the sampling rule selects the legal donor. Sharing an additional factor can therefore change both the shared information and the selected pairs. Appendix B.1 specifies each relation and sampling law.

2.2 A persistent–current interface

SPRII separates history-derived context from recipient-specific current information through a shared history encoder g_p and a current-input encoder g_s:

$$z_p^A = g_p(H_A), \qquad z_p^B = g_p(H_B), \qquad z_s^B = g_s(C_B). \tag{1}$$



(a) Histories and current inputs **(b) Prediction with shared q**

Figure 2: **Relation-supervised learning of persistent context.** Thick arrows carry recipient inputs (a) into both prediction branches (b). (a) Align regularizes contexts paired by $\mathcal{R}_G$. (b) Both calls share predictor q, current input C_B and conditioning u_B; Cross replaces only z_p^B with related donor context z_p^A. y_B is the loss target. Both components augment the native objective. Trajectories and tokens are schematic; Figure 8 shows the JEPA implementation.

Here z_s denotes current-input features, such as image embeddings, rather than the physical state itself. A dedicated encoder or a projection of an existing history state implements g_p in one forward pass. Context can encode several shared conditions in distributed form; prediction uses it directly, while evaluation probes assess its information content. For action-conditioned prediction, u_B contains the permitted future actions, masks and horizon. The predictor q combines these inputs with a persistent context, whose source can be the recipient or a related donor.

2.3 Align and Cross

Consistency across related histories. Align brings related contexts together while preserving variation. We implement $\mathcal{L}_{\text{align}}$ with VICReg (Bardes et al., 2022): paired-context agreement, per-dimension variance and a cross-dimension covariance penalty. The relation determines which pairs receive this pressure (Appendix B.1).

Prediction with context from another interaction. Cross trains prediction with related donor context. Self and cross-interaction paths share the recipient's current inputs and task conditioning:

$$\begin{aligned} \hat{y}_B^{\text{self}} &= q(z_s^B, z_p^B, u_B), \\ \hat{y}_B^{A\to B} &= q(z_s^B, z_p^A, u_B). \end{aligned} \tag{2}$$

Switching the context source from H_B to H_A gives the Cross objective:

$$\mathcal{L}_{\text{cross}} = \mathbb{E}_{(A,B)\sim\Pi_G}\left[\ell_{\text{task}}\left(\hat{y}_B^{A\to B}, y_B\right)\right]. \tag{3}$$

The recipient target supplies supervision. Because current inputs may already suffice, we test reliance on context directly through the interventions in Section 4.2.

Joint training. The two operations augment the native objective,

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_{\text{p}}\mathcal{L}_{\text{align}} + \lambda_{\text{x}}\mathcal{L}_{\text{cross}}, \tag{4}$$

where $\mathcal{L}_{\text{base}}$ retains the learner's prediction loss and regularizers. The composition is host-dependent: recipes use Align, Cross or both. In our NOD and Overcooked instantiations, the native objective already uses other interactions, so we add Align. Appendix B.3 summarizes each learner's recipe, with operations and weights fixed before evaluation; Algorithm 1 gives one training update. Here, P denotes the recipient distribution and $Q_G(A \mid B)$ the related-donor sampling kernel.

Algorithm 1 One SPRII training update.

Inputs: recipient sampler P, donor kernel Q_G, encoders and task readout q.

1 **Sample** recipients and related donor histories:
$B \sim P, \qquad H_A \sim Q_G(A \mid B).$

2 **Encode** the histories and current recipient input:
$z_{\mathrm{p}}^A = g_{\mathrm{p}}(H_A), \qquad z_{\mathrm{p}}^B = g_{\mathrm{p}}(H_B), \qquad z_{\mathrm{s}}^B = g_{\mathrm{s}}(C_B).$

3 **Compute** the native objective $\mathcal{L}_{\mathrm{base}}$ on ordinary recipient inputs.

4 **Initialize** $\mathcal{L}_{\mathrm{align}} \leftarrow 0$, $\mathcal{L}_{\mathrm{cross}} \leftarrow 0$.

5 **if Align is enabled:**
$\mathcal{L}_{\mathrm{align}} \leftarrow \mathrm{VICReg}(Z_{\mathrm{p}}^A, Z_{\mathrm{p}}^B).$

6 **if Cross is enabled:**
$\hat{y}_B^{A\to B} \leftarrow q(z_{\mathrm{s}}^B, z_{\mathrm{p}}^A, u_B).$
$\mathcal{L}_{\mathrm{cross}} \leftarrow \mathrm{mean}\ \ell_{\mathrm{task}}(\hat{y}_B^{A\to B}, y_B).$

7 **Combine** the native and enabled relation objectives:
$\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{base}} + \lambda_{\mathrm{p}}\mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{x}}\mathcal{L}_{\mathrm{cross}}.$

8 **Backpropagate** $\mathcal{L}$ and apply one optimizer update.

2.4 Applying the principle to existing learners

In JEPA, an observation encoder and causal Transformer combine observation and action histories into z_p; an MLP of the last observation embedding produces z_s. The predictor uses z_p, z_s, future actions and a horizon embedding to predict the recipient's future embedding. CPC and RSSM retain their native contrastive and variational objectives; supervised dynamics and action learners retain their task losses (Appendices B.1–B.3; Figure 8).

At inference, SpringWorld encodes an independent same-system history without gradient updates; other settings identify donors by system, task or partner. SpringWorld's physical prediction and Use/Value tests use a task reader r fitted to frozen encoder features. This reader is separate from q, which predicts embeddings during source training.

3 Predictive performance and generality

We evaluate prediction on controlled and public dynamics tasks, then relation supervision across robotic, tactile and recurring-partner settings (Appendices A, B.4). SpringWorld shows elastic-coupling dynamics in 128×128 images; PokeWorld shows an actuated finger pushing an object. Both vary mass, drag and stiffness across systems while holding them fixed within each system. SpringWorld tests held-out prediction, Use and Value; PokeWorld varies which factors pairs share.

A *source* is a trained representation model; a reader predicts from its frozen features. Controls are the history-conditioned host (Native), a split history interface without relation objectives (Structure), and random pairing with the same relation objective (Random); Structure isolates the interface and Random isolates the relation. Readers share targets and permitted inputs within each comparison. Appendix B.6 specifies aggregation and selection; native embedding losses remain separate.

3.1 Predictive performance

A 96-frame same-system history conditions a SpringWorld query with no preceding motion history; frozen-source readers predict state increments over the next 0.8 seconds. On 256 sealed systems (held out until the final evaluation), all nine source/reader pairs favor SPRII over Native and the trajectory-decoupled prediction (TDS) baseline adapted to same-system donors (Table 1A). CoPhy gains span three Collision learners and Balls RSSM, with budgets matched within each learner (Table 1B). The 14.6% average combines SpringWorld's sealed Native-host comparison with CoPhy's Structure controls, which isolate the increment from relation supervision (Appendix C).

Table 1: **Primary predictive performance on SpringWorld and CoPhy.** Lower is better; bold marks the lowest comparable mean.[1]

A. SpringWorld: single-state queries, 256 sealed systems

Source recipe	MSE ↓	SPRII reduction	95% CI
Native JEPA	0.138 ± 0.002	22.3%	[13.2, 30.7]%
TDS	0.130 ± 0.013	17.5%	[8.4, 25.9]%
SPRII (Align + Cross)	**0.107** ± 0.004	Reference	

B. CoPhy: physical trajectory MSE

	Collision			Balls4
Source recipe	**CPC**	**RSSM**	**JEPA**	**RSSM**
Native	0.248 ± 0.008	0.268 ± 0.007	0.250 ± 0.004	1.292 ± 0.037
Structure	0.223 ± 0.006	0.228 ± 0.005	0.215 ± 0.007	1.280 ± 0.031
Random	0.221 ± 0.006	0.222 ± 0.007	0.265 ± 0.003	1.327 ± 0.011
SPRII (Cross)	**0.189** ± 0.007	**0.202** ± 0.007	**0.184** ± 0.005	**1.154** ± 0.025

SPRII lowers mean error relative to released NOD in all three Burgers distributions (Table 2A). With matched additional updates, SPRII retains a 4.1% ID gain while remaining near-tied with continued NOD on OOD (Appendix C.4). Applied to an amortized variant of CoDA, SPRII lowers mean zero-step error (Table 2A; Appendix C.5). Using a common reader design across the D-Clean methods, SPRII gives the lowest mean error at all four prediction horizons, including 33.7% lower error than CaDM at 32 steps (Table 2B). Together, these results show predictive gains across controlled visual dynamics, public physical benchmarks, field dynamics and multiple learner families.

Table 2: **Predictive comparisons on Burgers and D-Clean.** Lower is better; bold marks the lowest mean within each comparable host or reader group.[2]

A. Burgers: native-trajectory / zero-step MSE ($\times 10^{-3}$)

Host learner	Source recipe	ID	Viscous OOD	Inviscid OOD
NOD (arXiv'26)	Released configuration	9.096 ± 0.544	8.012 ± 0.815	11.857 ± 0.628
	+ SPRII	**8.999** ± 0.798	**7.487** ± 0.183	**11.510** ± 0.360
Amortized CoDA (ICML'22)	Plain	3.483 ± 0.567	6.390 ± 1.436	5.883 ± 1.700
	+ SPRII (Cross)	**3.250** ± 0.729	**6.060** ± 1.185	**5.501** ± 1.496

B. D-Clean: common-reader raw-state MSE ($\times 10^{-4}$)

Method	h1	h4	h16	h32
State-space TDS	0.1034	0.8009	4.557	13.43
FCRL (ICML'21)	0.1228 ± 0.0112	0.7620 ± 0.0809	4.833 ± 0.492	13.34 ± 1.014
CaDM (ICML'20)	0.0332 ± 0.0018	0.1474 ± 0.0081	0.7711 ± 0.0081	2.273 ± 0.0628
DALI context (NeurIPS'25)	0.3201 ± 0.0357	2.732 ± 0.3115	17.26 ± 1.991	45.92 ± 5.362
SPRII	**0.0314** ± 0.0029	**0.1166** ± 0.0051	**0.5232** ± 0.0117	**1.507** ± 0.1355

3.2 Generality beyond controlled physical systems

Relations extend to recorded robotic and tactile interaction. Relations can also express task, object and partner identity. In RH20T robot manipulation, where relations group episodes of the same task, Cross lowers error on all five force/torque and TCP forecasting endpoints relative to Structure under matched donors (Appendix F.3). In real Baxter grasps, the pairing rule selects which property

[1]Table 1: SpringWorld uses standardized state-increment MSE, three sources × three readers, source mean ± SD and paired-system CIs. CoPhy reports physical trajectory MSE (mean ± SD over three joint seeds), on 1,994 Collision episodes and 1,000 Balls4 recipients (Appendices C, B.6).

[2]Table 2: Burgers reports mean ± SD over three sources, with 45/500/100 ID/viscous/inviscid queries. D-Clean uses 100 systems and three sources × three readers; SD is across source means. DALI is a context-component adaptation with 160,000 source updates; other D-Clean sources use 20,000. This is not an equal-total-compute comparison. Training and evaluation protocols are in Appendices C.3–C.5.

becomes accessible: hardness accuracy is 1.000 under same-hardness relations versus 0.333 under same-shape relations, and the ordering reverses for shape balanced accuracy: 1.000 under same-shape relations versus 0.500 under same-hardness relations (Appendix E.3).

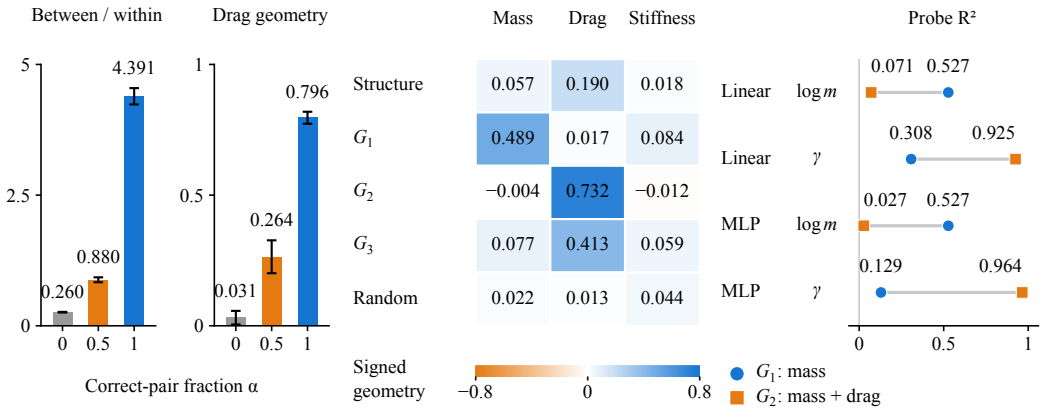
Persistent information can also describe a recurring partner. In cooperative Overcooked play, same-partner alignment lowers mean partner-behavior probe error at every history budget and, with full history, raises familiar-partner return in all three training seeds (Appendix H.2). These settings change both the observations and what persists: task context in robot manipulation, object properties in touch, and behavior across encounters with a partner. Each retains its own task, with relations as a common form of supervision. Gains in property readout and task performance need not grow together across evaluated settings (Appendix A.1); Section 4 examines formation, use and value separately.

4 Formation, Use, and Value

We examine Formation through context geometry and probes, Use through fixed-predictor context substitutions, and Value through task-reader and horizon comparisons.

4.1 Formation: relations shape strength and accessible content

PokeWorld permits controlled changes to pairing accuracy and to the properties pairs share, with source architecture and observation budgets fixed.



(a) Reliability **(b) Factor geometry** **(c) Factor accessibility**

Figure 3: **Relations shape the strength and content of persistent representations.** (a) Pair reliability strengthens separation and drag geometry; whiskers: source SD. (b) Partial geometry. (c) Linear/MLP probe R^2. $G_1 = m$, $G_2 = (m, \gamma)$, $G_3 = (m, \gamma, k)$. Means cover three sources. Geometry uses 400 confirmation systems; probes use 400 validation systems (Appendix E.1).

Reliable relations strengthen organization. On 400 confirmation systems, raising the correct-pair fraction from 0 to 1 increases the between/within-system context-distance ratio from 0.260 to 4.39. Partial drag geometry, measured by distance correlation controlling the other factors, rises from 0.031 to 0.796 (Figure 3(a)). Observation budgets, donor/query marginals and model settings are fixed.

Changing the pairing rule redirects accessibility. Changing $G_1 = m$ to $G_2 = (m, \gamma)$ keeps mass shared but lowers its linear probe R^2 from 0.527 to 0.071; drag rises from 0.308 to 0.925. These probes use per-system context centroids (Figure 3(c)). Nonlinear probes and all three source replicas reproduce both directions, showing that changing the pairing rule redirects accessibility.

Under limited context capacity, the learner must select among history directions that remain predictive beyond current inputs and conditioning. Our linear reference model captures this trade-off:

$$M_G = K^\top K - \beta D_G, \qquad \beta \geq 0,$$

where $K^\top K$ scores predictive information remaining after current inputs and conditioning are accounted for, D_G measures disagreement between paired histories, and β controls their trade-off. With context dimension d smaller than history dimension p, the selected subspace is a top-d eigenspace of M_G. Changing the relation can therefore redirect this subspace even when an earlier property remains shared (Appendix I.2, Proposition 3; Figure 26).

Align organizes context; Cross adds benefit. SpringWorld's component study uses 178 systems for geometry and 100 development systems (600 cases; one reader per source) for task error, separate from the sealed evaluation in Table 1. From Structure's separation ratio 0.240 and reader error 0.172, Align reaches 2.04 and 0.123; Cross reaches 0.252 and 0.168; both reach 2.36 and 0.110 (Appendix E.2). Align supplies most of the organization and task gain; adding Cross lowers error in all three sources. In Collision, CPC, RSSM and JEPA use Cross to train the donor-context route.

4.2 Use: persistent context enters fixed predictive computation

We next substitute donor context to test its effect on a fixed predictor. In SpringWorld, we intervene on a task reader fitted to frozen features, separately from the source predictor q.

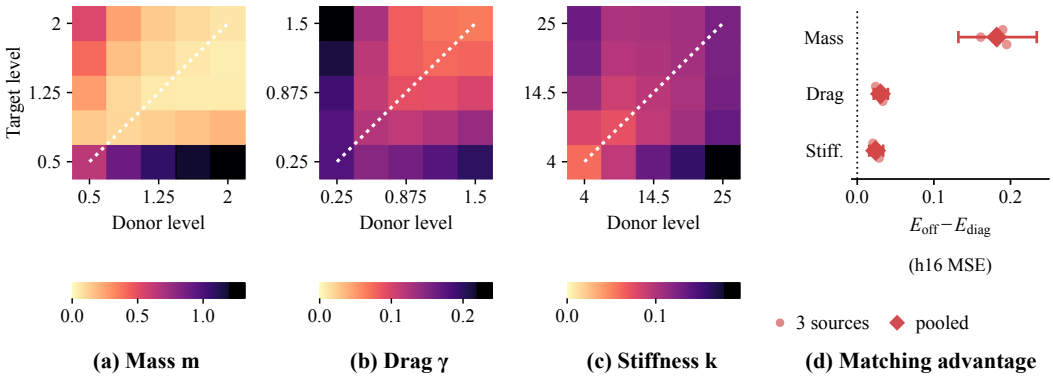


Figure 4: **Donor physics structures fixed predictive responses.** (a)–(c) Donor–target landscapes of h16 standardized-increment MSE, averaged over 64 development systems and three Align + Cross sources; color scales differ. Only donor context changes the fixed reader's prediction. (d) Mean off-diagonal minus diagonal error; positive favors matching. Points: sources; diamonds: means; intervals: pointwise 95% paired-system CIs, conditional on fitted models (Appendix F.1).

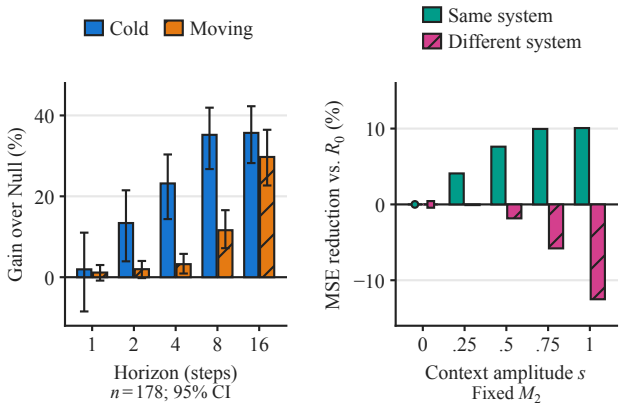
Donor physics changes which dynamics a fixed predictor fits. In SpringWorld, we hold the source, fitted reader, recipient initial state and future actions fixed, then vary one donor factor. Independently varying the corresponding simulator target factor yields a 5×5 donor–target landscape for each of 64 development systems. Simulator targets serve only to compute prediction error. Matching donor and target physics lowers mean 16-step (h16) error for mass, drag and stiffness in every source (Figure 4(a)–(d)). Donor context therefore steers the fixed reader's predictions toward the corresponding physical dynamics.

Align strengthens the advantage of matching context. Using the same matching-advantage measure as Figure 4(d), which shows the joint recipe, we extend the comparison to Structure, Align and Cross. The Align effect, averaged over the presence and absence of Cross, is positive for all three factors. Each recipe uses its own source and fitted reader. Appendix F.1 gives complete effects and a decomposition of the same readers' physical output responses.

Relation design has a predictive counterpart. On 144 PokeWorld validation systems, a fixed reader's predictions respond more strongly along mass under G_1 and drag under G_2 (Appendix F.2).

4.3 VALUE: PREDICTION CONDITIONS AND THE READER SHAPE BENEFIT

With source features fixed, we test context value across horizons and readers (Appendix I.3).



(a) Horizon value **(b) Context value**

(c1) Residual readout (MSE)

Source	M_1	M_2	Gain (%)
0	0.608	0.550	9.6
1	0.627	0.585	6.8
2	0.620	0.574	7.5
Overall	**0.619**	**0.570**	**7.9**

(c2) Persistent vs. Decode

Task	MSE change (%) ↓
D-Clean	**−22.8**
SpringWorld	**+2.1**

Figure 5: **Persistent context has prediction- and readout-dependent value.** (a) SpringWorld gain over Null readers as horizon increases; whiskers: pointwise 95% system-bootstrap intervals over 178 systems. (b) Scaling context in fixed M_2 changes MSE relative to recipient-only R_0; $s = 0$ gives R_0. (c1) Query-only M_1 versus context-conditioned M_2 on SpringWorld development systems: source means over three readers with development-selected checkpoints. (c2) MSE change from Decode to Persistent (%); negative favors direct context. In (c2), endpoints are D-Clean h32 and SpringWorld cold-start h16 (Appendix G).

History value changes with the prediction horizon. Null readers receive no history; Persistent readers receive learned context, Decode readers receive decoded physical parameters, and Oracle readers receive true parameters. SpringWorld's Cold and Moving queries both provide zero observed transitions and differ in initial motion. Persistent-over-Null gain rises from 1.9% at h1 to 35.7% at h16 for Cold, and from 1.2% to 29.7% for Moving (Figure 5(a)). All comparisons use the same 178 systems and fitted grid; horizon changes the target while query evidence stays fixed.

Decodable parameters do not determine the best readout. Persistent and Decode readers receive the same donor context, either directly or through decoded physical parameters, with a shared frozen query encoder. In D-Clean, direct context lowers h32 error by 22.8% relative to Decode despite independently validated decoder $R^2 > 0.996$. In SpringWorld, direct context instead has 2.1% higher error than Decode (Figure 5(c2)). The preferred context interface therefore depends on the task, even when physical parameters are accurately decodable (paired effects and readers: Appendix G.2).

Readout adaptation recovers value from frozen context. We freeze source features and a fitted recipient-only base R_0, then fit query-only (M_1) or context-conditioned (M_2) residual readers with common targets and schedules. On SpringWorld development systems, M_2 improves on M_1 by 7.9% (mean MSE: 0.570 vs. 0.619; Figure 5(c1)); all nine source/reader pairs improve (Appendices B.5, G.4).

With fixed M_2, scaling its persistent input by s lowers error for same-system context and raises it for different-system context (Figure 5(b)). Zero amplitude equals the frozen recipient-only base, distinct from the separately fitted M_1. These substitutions show sensitivity to system-specific content. Training and evaluating the same reader architecture with mismatched context raises error in all nine source/reader pairs (Appendix G.4), confirming the importance of correctly associated history.

In PokeWorld with fully reliable relations, organization and drag accessibility are strong, yet the context-conditioned reader has higher mean h16 error than its query-only counterpart at the 5,000-update development endpoint (Appendix G.4). Appendix I.4 formalizes available task value and fitted-reader realization.

5 Related Work

Context-conditioned dynamics models infer environment factors from transitions (Lee et al., 2020; Röder et al., 2025; Shaj et al., 2022). CoDA adapts context codes by test-time optimization (Kirchmeyer et al., 2022). NOD conditions prediction on a separate same-system trajectory, a construction closely related to Cross (Chen et al., 2026). GG-ODE regularizes environment codes contrastively (Huang et al., 2023); RIA infers shared-environment relations through interventional prediction to organize dynamics context (Guo et al., 2022). FCRL learns representations from same-function observation sets (Gondal et al., 2021); view-design studies examine how positive relations shape the information preserved across views (Tian et al., 2020; von Kügelgen et al., 2021). VICReg supplies Align's agreement objective (Bardes et al., 2022).

SPRII combines relation supervision with existing predictive objectives and treats relation reliability and shared semantics as explicit training design choices. Controlled comparisons then distinguish the persistent information made accessible by a relation, its functional use by a fixed predictor, and its task benefit under different readouts (Appendix J).

6 Conclusion

SPRII uses relations between interactions to help world models organize persistent information into reusable context. Its composable Align and Cross components retain the learner's native objective and require no numerical property labels. Across controlled physical systems, public benchmarks, robotic and tactile recordings and recurring-partner tasks, relation supervision improves prediction or makes the corresponding persistent information more accessible. The Formation, Use and Value framework separates three questions: what information is accessible in the context, how a fixed predictor uses it, and whether that use improves task performance.

The controlled studies show that more reliable relations improve representation organization, but changing the pairing rule to share additional properties can reduce access to information that remains shared. Context substitutions reveal sensitivity to persistent information at fixed model weights, while horizon and readout comparisons show that its predictive benefit depends on the task and how the context is used. Relation design and readout design are therefore complementary: one shapes the information available in context, and the other determines how effectively it serves a task.

In practice, system, object or partner identities provide a starting point for defining related histories; probes, context interventions and task comparisons can then assess what those relations achieve. More broadly, SPRII offers a concrete route toward world models that learn and use what remains stable across changing interactions. Future work could improve the reliability of relations learned from interaction logs, extend persistent context to properties that change over longer timescales, and investigate its role in planning.

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A Findings and Evaluation Guide

A.1 Cross-setting evidence overview

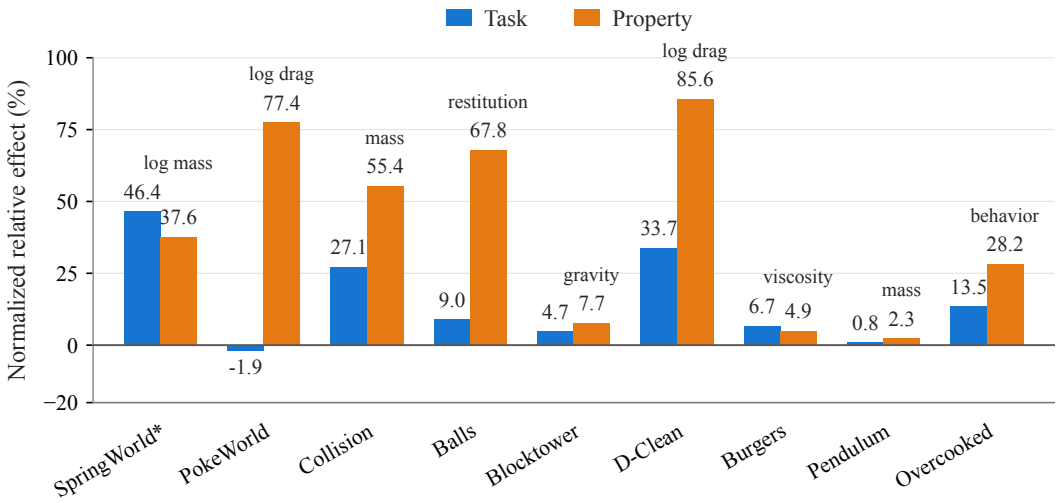


Figure 6: **Normalized task and property effects for selected comparisons.** Each setting pairs a downstream task metric with a persistent-property readout under the same stated model comparison and source set. Percentages normalize heterogeneous metrics for visualization; they are not a common scale for cross-setting ranking or averaging. Positive values favor SPRII under the metric-specific definition. These comparisons include development results; * marks a single-source example. Figure 7 reports the corresponding original metrics.

Figure 6 summarizes selected paired task/property comparisons, with both measurements available under the same declared comparison. Across several settings, task benefit and property accessibility change by different amounts, illustrating why Formation and realized Value are measured separately.

Normalization. Error/cost effects are $100(1 - E_{\mathrm{SPRII}}/E_{\mathrm{ref}})$; probe R^2 effects are $100(R^2_{\mathrm{SPRII}} - R^2_{\mathrm{ref}})/(1 - R^2_{\mathrm{ref}})$; return effects are $100(R_{\mathrm{SPRII}} - R_{\mathrm{ref}})/R_{\mathrm{ref}}$ for a positive reference. Where multiple readers are fitted, raw measurements are first averaged within each source, then equally across sources, before computing relative effects (Appendix B.6).

A.2 MAIN FINDINGS AND EVIDENCE

Table 3 connects the main findings to what each establishes and to its supporting experiments.

Table 3: **Main findings and supporting evidence.** The evidence column links each observation to its supporting experiments.

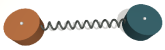
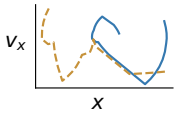
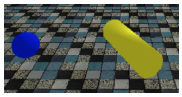
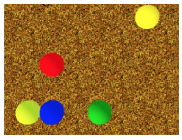
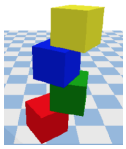
Finding	What this establishes	Evidence
• **More reliable relations** strengthen the organization of shared information.	Relation quality changes how persistent information is organized.	App. E.1
• Changing the instantiated $G_1 \to G_2$ **pairing rule** keeps mass shared but shifts accessibility from mass to drag.	Relation design can redirect accessible content rather than monotonically adding shared factors.	App. E.1; I.2
• In SpringWorld, **Align** drives most of the observed organization; **Cross** adds further task benefit.	The components make different contributions to organization and prediction.	App. E.2
• **Changing donor context** redirects a fixed predictor along the corresponding physical properties.	Fixed-weight substitutions show that the consuming predictor functionally relies on persistent context.	App. F.1; F.2
• In SpringWorld, history brings larger gains at **longer prediction horizons** under the same current-evidence budget.	The value of the same history depends on the prediction condition as well as the information represented.	App. G.1
• **Direct context** performs better in D-Clean; **decoded physical parameters** perform slightly better in SpringWorld.	Accessible physical information does not determine the best downstream interface.	App. G.2
• **Changing the readout** recovers value from frozen SpringWorld context, but not from every evaluated source family.	Realized value depends on the consuming reader as well as the frozen representation.	App. G.4

Practical use. SPRII applies when repeated encounters share information that remains stable enough to be useful across interactions. A practical application requires three choices: identify what is expected to persist (an object, physical regime, task or partner), provide a history/context route through which earlier experience reaches the predictor, and ensure that relevant prior experience is available when a new prediction is made. Observations, targets and learner architectures may differ; the settings below instantiate these choices without assuming a single deployment pattern.

A.3 EVALUATION SETTINGS

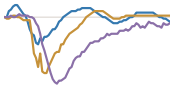
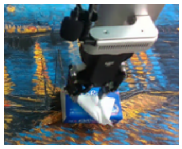
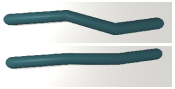
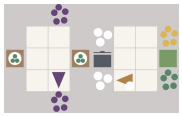
Evaluated instance identifies the learner, enabled operations and principal controls. A component study compares the interface alone (Structure), Align, Cross and their combination; application-specific choices are detailed in Appendix B.3.

Table 4: **Evaluation settings and their principal tests.** Images illustrate the settings, not model predictions.

Setting	Relation	Experience → prediction	Evaluated instance	Primary evidence
Controlled environments				
SpringWorld 	**Shared:** Mass m, drag γ, stiffness k. **Varied:** Interaction history.	**Input:** Independent history; current query/actions. **Target:** Future state increments.	**Learner:** JEPA. **Instance:** Align + Cross. **Controls:** Native; TDS. **Components:** Structure; Align; Cross.	**Primary:** Sealed new-system prediction. **Further:** Geometry; Use; Value.
PokeWorld 	**Shared:** Selected factors: m; (m, γ); (m, γ, k). **Varied:** Correct-pair fraction α, separately.	**Input:** Independent image/action history; current query. **Target:** Future observation embedding.	**Learner:** JEPA. **Instances:** Structure; Align; Cross; Align + Cross. **Controls:** Relation and objective composition.	**Primary:** Formation: relation reliability and shared factors. **Further:** Probes; Use; task readouts.
D-Clean 	**Shared:** Drag γ. **Varied:** Initial state; force sequence.	**Input:** Independent state/action history; recipient state. **Target:** Future prediction.	**Learner:** JEPA. **Source:** Align + Cross. **Readouts:** Null; Persistent; Decode; Oracle. **Comparisons:** TDS; NOD; FCRL; CaDM; DALI.	**Primary:** Formation; Value. **Further:** Closest methods.
Benchmarks, recorded data, and control settings				
CoPhy *Collision* 	**Shared:** Matched objects; scene-specific rule. **Varied:** Support interaction.	**Input:** Three support histories; three query frames. **Target:** Future trajectory.	**Learners:** CPC; RSSM; JEPA. **Instance:** Cross. **Controls:** Native; Structure; Random. **Extensions:** Other learners in H.1.	**Primary:** Held-out prediction. **Further:** Learner matrix.
CoPhy *Balls* 	**Shared:** Matched objects; scene-specific rule. **Varied:** Support interaction.	**Input:** Three support histories; three query frames. **Target:** Ball trajectories.	**Instances:** JEPA, CPC, RSSM: Cross; CoPhyNet: Align + Cross. **Controls:** Native; Random; Structure where tested.	**Primary:** Held-out RSSM prediction. **Further:** Memory probes and Use; learner matrix.
CoPhy *Blocktower* 	**Shared:** Matched objects; scene-specific rule. **Varied:** Support interaction.	**Input:** Three support histories; three query frames. **Target:** Multibody motion.	**Instances:** JEPA, CPC, RSSM: Cross; CoPhyNet: Align + Cross. **Controls:** Native; Random; Structure where tested.	**Primary:** Multi-body prediction. **Further:** Reader design.

Continued on the next page.

Table 4: **Evaluation settings and their principal tests (continued).**

Setting	Relation	Experience → prediction	Evaluated instance	Primary evidence
NOD *Burgers* 	**Shared:** Viscosity ν. **Varied:** Initial field.	**Input:** Independent trajectory; recipient field. **Target:** 1D field evolution.	**Learner:** Released NOD. **Instance:** Relation-trained stage; prediction-only continuation. **Controls:** NOD; Random; continued NOD; GEPS/CoDA (50 adaptation steps).	**Primary:** Public operator prediction. **Further:** Viscosity probes; fixed-donor use.
NOD *FHN* 	**Shared:** Reaction parameters (k, β). **Varied:** Initial field.	**Input:** Independent field histories; recipient field. **Target:** 2D field evolution.	**Learner:** NOD-Hier. **Instance:** Align; prediction-only adaptation. **Comparison:** Equal-history NOD.	**Primary:** 2D prediction; history aggregation. **Further:** Factor probes.
Baxter 	**Shared:** Hardness or shape. **Varied:** The other factor.	**Input:** Separate tactile grasp. **Target:** Future tactile representation.	**Learner:** Predictive encoder. **Instance:** Align + Cross. **Control:** Random relation.	**Primary:** Tactile factor selectivity. **Further:** Hardness and shape readouts.
RH20T 	**Shared:** Task identity. **Varied:** Recorded episode.	**Input:** Multimodal history; current inputs/actions. **Target:** Force/torque; TCP motion.	**Learner:** Supervised predictor. **Instances:** Cross; Align + Cross. **Controls:** Structure; pairing; inputs.	**Primary:** Robot forecasting and task-specific donor use.
Swimmer 	**Recipient:** Fixed system. **Donor:** Own-system or wrong-system update.	**Input:** Persistent update; current query/action. **Target:** Prediction through fixed weights.	**Learners:** Frozen JEPA/GRU. **Intervention:** Update substitution. **Training:** No new Align/Cross source fit.	**Primary:** Fixed-update use.
Overcooked 	**Shared:** Fixed partner identity. **Varied:** Disjoint episode.	**Input:** Prior episodes; current query. **Target:** Ego-action prediction.	**Learner:** AD action learner. **Instance:** Align only. **Control:** Matched history slot; variance/covariance terms.	**Primary:** Partner-behavior readout. **Further:** Cooperation across history budgets.
Pendulum 	**Shared:** System dynamics. **Varied:** Episodes; mass and length across systems.	**Input:** Transition history; current state. **Target:** Dynamics for online control.	**Learner:** CaDM interface. **Instance:** ID-selected SPRII recipe. **Controls:** CaDM; Random relation.	**Primary:** Return and success. **Further:** Same-checkpoint factor probes and Use.

A.4 ORIGINAL-METRIC COMPARISONS

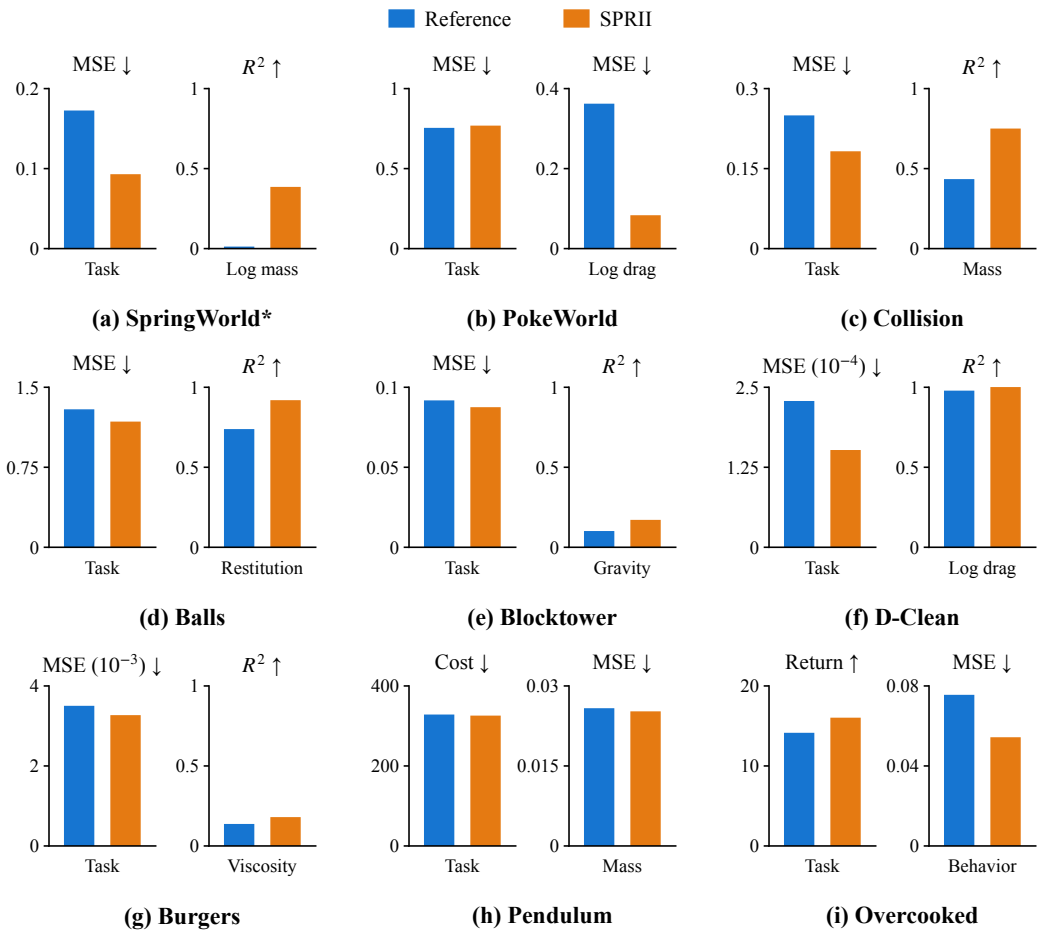


Figure 7: **Original-metric task and property comparisons.** Left: task; right: property. Blue: reference; orange: SPRII; arrows indicate preferred directions. Comparisons are (a) Native JEPA → Align + Cross; (b) random → correct pairing; (c) Native CPC → Cross; (d) Native RSSM → Cross; (e) Native JEPA → Cross; (f) CaDM → SPRII; (g) Plain CoDA → Cross; (h) CaDM → SPRII; (i) H+VC → Align. Task endpoints are (a) cold-start MSE, (b) h16 state MSE, (c–e) trajectory MSE, (f) h32 MSE, (g) ID MSE, (h) ID control cost and (i) familiar-partner return. Panels label metrics and targets. Collision and Blocktower use independent support S; Balls uses reader memory M.

Scope and aggregation. The panels are setting-specific comparisons, not a pooled suite statistic. SpringWorld* uses development source 0/reader 0 on 100 systems, separate from sealed evaluation (Table 35). PokeWorld averages all three sources and three readers per source on 400 validation systems at the 5,000-update endpoint, pairing the same sources' log-drag probes (Appendix G.4). Collision and Blocktower average two development sources; Balls averages three, with its memory probe including the fitted support projection (Table 60; Appendix F.4). D-Clean uses three sources with three readers each on its validation bank; Burgers uses three-source, zero-step held-out ID results (Appendix C.3; Tables 18, 21). Pendulum reports three-seed ID control and same-checkpoint probes (Appendix G.5). Overcooked reports three-seed full-history familiar-partner results: net cooperation return includes action costs, and the property is partner behavior (Table 65).

B METHOD AND EXPERIMENTAL IMPLEMENTATION

This section specifies how the instances in Table 4 are built and measured. We first give the common relation objectives and training update, then the controlled JEPA implementation and learner-specific interfaces. Appendix B.4 defines the environments and data splits; Appendices B.5–B.6 define the probes, readers, interventions and aggregation used in the results.

B.1 RELATIONS, OBJECTIVES AND TRAINING UPDATE

The relation $\mathcal{R}_G$ specifies what histories share; the kernel $Q_G(A \mid B)$ specifies how a legal donor is selected. Drawing a recipient from P induces the pair law

$$\Pi_G(A, B) = P(B)Q_G(A \mid B). \tag{5}$$

Relation semantics and sampling are separate choices: $\mathcal{R}_G$ defines the shared conditions, whereas Q_G selects actual legal pairs. Our main setting pairs separately realized interactions. The formulation also admits broader pairing constructions, such as overlapping clips, whose shared episode structure offers different opportunities to rely on local cues. Partial-factor sharing is distinct from this sampling choice: two separately realized interactions may share selected factors while others differ. Their pairing kernels need not have nested supports, even when the shared-factor definitions are nested.

The factorized PokeWorld experiment uses

$$G_1 = m, \qquad G_2 = (m, \gamma), \qquad G_3 = (m, \gamma, k). \tag{6}$$

The first kernel changes drag and stiffness, the second changes stiffness, and the third changes the rollout while preserving the complete system. Baxter's hardness and shape relations provide a real-data counterpart. The network receives observations and task inputs; grouping metadata is used to construct the pairs. Parameter regression probes are fitted outside representation training.

Objectives and gradients. For paired code matrices $A, B \in \mathbb{R}^{n \times d}$, the alignment loss is

$$I(A, B) = \frac{1}{nd}\|A - B\|_F^2, \tag{7}$$

$$V(A, B) = \frac{1}{2d}\sum_{j=1}^{d}\left[\max\{0, 1 - \sqrt{\mathrm{Var}(A_{:j}) + \epsilon}\} + \max\{0, 1 - \sqrt{\mathrm{Var}(B_{:j}) + \epsilon}\}\right], \tag{8}$$

$$C(A, B) = \frac{1}{d}\sum_{i \neq j}\left[\mathrm{Cov}(A)_{ij}^2 + \mathrm{Cov}(B)_{ij}^2\right], \tag{9}$$

$$\mathcal{L}_{\text{align}} = 25I + 25V + C, \qquad \epsilon = 10^{-4}. \tag{10}$$

Variances and covariances use the sample denominator $n - 1$. Related branches are evaluated jointly for this loss; no stop-gradient is inserted in Align. The base learner determines target-encoder gradients for its prediction loss. In the controlled D-Clean/PokeWorld models, both prediction and target embeddings are produced by the online observation encoder.

The loss weights and pairing kernels are part of each instantiated training recipe. For the controlled factorial, the predictor and representation dimensions are shared across Structure, Align, Cross, and Align + Cross. For transferred learners, the native comparator and the structure-matched comparator answer complementary questions: the total effect of the adaptation and the increment from its relation term. Configuration choices and common evaluation settings are reported with each environment. Each application uses a fixed configuration or validation-selected recipe. Appendix B.6 specifies the checkpoint and selection rules; the profiles below state the enabled operations. History codes are substituted rather than added; fixed-predictor evaluations make no gradient updates.

Algorithm 1 retains the application's native loss and regularizers. Its task loss, readout, normalization and optimizer are application-specific. Disabled relation terms are zero; a legal donor obeys the temporal and input constraints and leaves the recipient target unchanged.

B.2 SHARED CONTROLLED-JEPA IMPLEMENTATION

Figure 8 details the controlled JEPA instantiation of the general interface. The controlled D-Clean and PokeWorld models use histories of 24 observations and predict encoded targets at horizons

$\{1, 4, 16\}$. In split variants, $z_s \in \mathbb{R}^{64}$ is an MLP of the last observation embedding. The persistent encoder concatenates encoded observations with 32-dimensional action features. A leading zero action aligns each later observation with its preceding action; learned position embeddings are added before processing the sequence with a four-layer causal Transformer of width 192, eight attention heads, and four-times-width feed-forward blocks, producing $z_p \in \mathbb{R}^{64}$. The predictor receives a 128-dimensional context, padded future actions and masks, and a 32-dimensional horizon embedding; its two hidden layers have width 256. The monolithic Native replaces the split encoders with a 128-dimensional history context while retaining the predictor capacity.

D-Clean observations are 4D states encoded by a three-layer MLP. PokeWorld renders a 64×64 current frame and temporal-difference channel. Its visual encoder uses four stride-2 convolutional blocks with channels $32, 64, 128, 256$, followed by a linear projection to 128 dimensions and batch normalization. Models train for 20,000 steps with AdamW, learning rate 3×10^{-4}, weight decay 0.05, 500 warmup steps followed by cosine decay, bfloat16 autocasting, and gradient clipping at 1.0. A paired batch contains 48 systems and 96 windows ordered as donor/query halves.

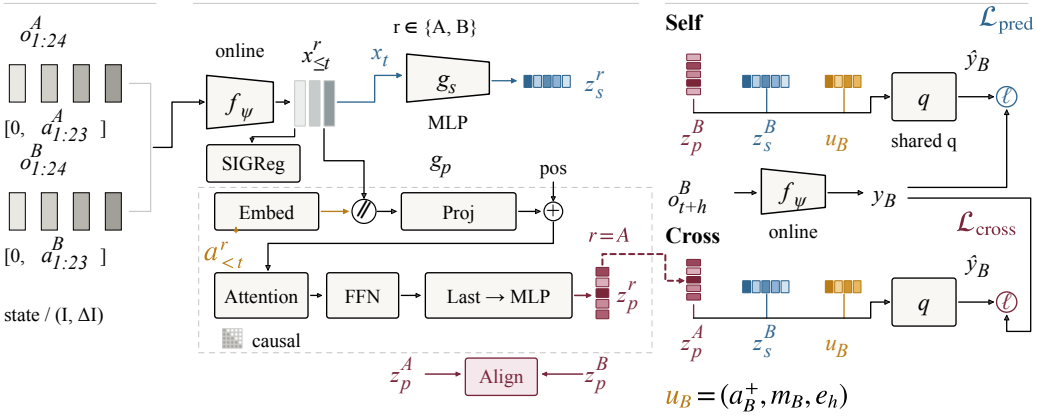


Figure 8: **Controlled-JEPA implementation of SPRII.** (a) Related observation/action histories. (b) A shared observation encoder feeds separate current and persistent branches; Align compares the persistent codes. (c) Self and Cross share the predictor and recipient target. Cross replaces only the persistent input with the donor's code. Self-prediction and SIGReg together form the native objective in Figure 2. Tokens and the causal-mask crop are schematic.

SIGReg uses 17 quadrature knots and 1,024 random directions, with $\lambda_{\text{reg}} = 0.02$ (Balestriero & LeCun, 2025). Persistence uses VICReg weights $25/25/1$ for invariance, variance, and covariance, unbiased per-branch variance, and $\epsilon = 10^{-4}$ (Bardes et al., 2022). The controlled objective weights $\lambda_{\text{p}} = 1.0$ and $\lambda_{\text{x}} = 0.1$ were frozen before the three-source factorial. Rel-InfoNCE uses the InfoNCE objective lineage of contrastive predictive coding and the two-view group-positive construction of supervised contrastive learning (van den Oord et al., 2018; Khosla et al., 2020), with the separate three-seed validation rule in Appendix E.1.

B.3 Cross-learner instantiation

The persistent–current interface is a structural inductive bias rather than a shared network architecture. A history encoder or an existing context state supplies the persistent route; recipient-specific inputs remain available to the native predictor. Table 5 summarizes each learner's native predictive route, measured representation and enabled relation operations.

The controlled family tests both operations separately and jointly. The CoPhy source-0 cross-scene development matrix uses one recipe per learner family. The separately validation-selected Collision JEPA recipe is specified in Appendix B.4. Overcooked's native action objective already consumes previous episodes, so its added operation is Align; NOD likewise retains its trajectory-decoupled prediction objective. Its alignment phase and prediction-only continuation are specified in Appendix C.4. Structure isolates the interface; Random changes the relation with the operation fixed.

Table 5: **Predictive routes and relation operations across learners.** U denotes complete source state, S independent support and M trained reader memory; each is probed separately.

Instance	Native predictive route and measured representation	Relation operations
• **Controlled JEPA**	24-step history P_{64}, current S_{64}; future online embedding MSE + SIGReg. D-Clean state MLP; Poke visual CNN.	Structure, Align, Cross, Align + Cross.
• **Spring JEPA**	96-frame visual/action history; embedding MSE + SIGReg; frozen source and separately fitted physical readers.	Align + Cross, with component controls.
• **CoPhy JEPA/CPC**	Frozen 784D perception features; latent prediction or future contrast; native dynamics state retained.	Cross in the main matrix.
• **CoPhy RSSM**	Reconstruction, prior prediction and KL; $U = [P_{64}, h_{128}, z_{32}]$, versus 160D Native U.	Cross; U, S, M measured separately.
• **CoPhyNet**	Coordinate MSE and stability classification; original 32D code split as P_{16}, T_{16}; reader receives full U.	Align + Cross on P.
• **Baxter**	40-step tactile history, P_{64}, S_{64}; future embedding MSE + SIGReg.	Align + Cross; hardness/shape relations.
• **RH20T**	Fused visual/sensor history; joint latent, FT and TCP MSE + SIGReg.	Cross / Align + Cross.
• **Overcooked AD**	Ego-action cross-entropy; disjoint prior episodes pooled in a 32D slot.	Align only; $\lambda_x = 0$.
• **NOD Burgers/FHN**	Released trajectory-decoupled/hierarchical field operators; native conditional prediction retained.	Align; prediction-only continuation recorded separately.
• **Amortized CoDA**	History-inferred context through a frozen CoDA decoder; native field prediction.	Cross; four-arm protocol in Appendix C.5.
• **Pendulum / CaDM**	Dynamics context supplied through the existing control interface; return, success and same-checkpoint probes.	Align; ID-selected configuration in Appendix G.5.
• **Swimmer**	Recipient-specific persistent update through frozen JEPA/GRU predictors.	Fixed-input substitution; no new Align/Cross source training.

B.4 Data, splits and admissible inputs

Each profile specifies the observations, target, relation and data split. Table 4 links to these profiles; Table 7 separates the evaluation populations and consuming routes. CoPhy's three scenes and NOD's two field systems count separately among the thirteen settings.

SpringWorld

Environment and information. SpringWorld is a controlled visual elastic-coupling environment in which mass m, drag γ, and stiffness k persist across independent interactions. The main ranges are $m \in [0.5, 2]$ kg, $\gamma \in [0.25, 1.5]\,\mathrm{s}^{-1}$, and $k \in [4, 25]$ N/m. Observations comprise 128×128 grayscale images and causal frame differences. A 96-frame donor spans 4.75 seconds; the first frame difference is zero. Targets are eight-dimensional position/velocity increments standardized using training statistics. The cold-start query has no preceding motion history, and prediction covers the next 0.8 seconds. Physical labels are used for evaluation and downstream supervised heads.

The development readout protocol uses 144 training systems and 178 validation systems: 64 continuous new systems, 36 unseen discrete combinations, 30 new interpolated levels, and 48 single-factor extrapolation systems. The original development cold-start panel uses the first two groups, 100 systems and 600 cases. The full mixture uses 16,020 cases across the four groups and its fixed query conditions. They are distinct evaluation profiles rather than different training-set sizes.

Base learner and adaptation. The JEPA reference is a monolithic predictor; its SPRII instance introduces separate history and current-context branches and adds Align with weight 1 and Cross with weight 0.1. The two use the same data bank, 10,000 source updates, 48 pairs per batch, AdamW

learning rate 3×10^{-4}, and common regularization and schedule. The final-step checkpoint is used. Structure, Align, and Cross isolate the interface and individual operations.

All source representations are frozen before the common readout is trained for 10,000 updates with batch 256 and the same head seed. The readout has a 128-dimensional context interface, 64-dimensional slot, and two 256-unit hidden layers. Supervised GRU, Transformer, TCN, and DeepSets encoders provide strong reference representations under their own training recipes; they are reported alongside the matched JEPA comparisons.

Observation normalization uses a fixed pre-update statistics snapshot at each source-training update; training-history features update that snapshot only afterward. The representation is frozen for downstream readers and probes. The 96-frame donor protocol is specific to SpringWorld, not the 24-frame controlled D-Clean/PokeWorld input.

D-Clean

D-Clean is an analytically integrated two-dimensional inertial system, $\dot{v} = F/m - \gamma v$, with $m = 1$ and $\Delta t = 0.05$. A system fixes drag and varies initial state and force segments across eight independent 64-step interactions. The train/validation/test split contains 1,000/200/200 disjoint systems. The 4D state histories provide a clean setting for measuring factor accessibility and prediction separately. The history, rather than a single current state alone, supplies evidence about drag. The original source comparison is retained in Table 29; the state-space baseline and same-donor physical bottleneck use their own downstream populations.

Independent-method comparison. NOD uses its released MassSpring network components with D-Clean action inputs; CaDM uses a PyTorch implementation of its released model components, and FCRL is implemented from the paper. Each representation is frozen before fitting an identical downstream reader. Each method uses three source seeds and three reader seeds per source, with 20,000 source updates and 20,000 reader updates. Evaluation uses 100 systems from the existing validation report population, raw-state MSE and horizons $1, 4, 16, 32$. Readers are averaged within source before computing the three-source mean and sample SD. All source fits are included, including NOD source 1, whose code is constant. This comparison measures predictive value under the common reader; the Burgers comparison uses NOD's native field predictor. The corresponding source probe and fixed-reader donor interventions use the same fitted models.

PokeWorld

PokeWorld renders an actuated finger interacting with an object, with mass, drag and contact stiffness fixed within each system. Observations are current images and frame differences, both 64×64; actions are available to the history encoder and predictor. Four independent 64-step rollouts vary initial states and action modes. The bank has 2,000/400/400 disjoint train/validation/test systems.

The factorized bank contains $10 \times 10 \times 28 = 2,800$ physical tuples. A frozen allocation assigns 20/4/4 stiffness tuples in each mass–drag block to train/validation/confirmation, producing 2,000/400/400 systems. Exact tuples are disjoint across splits, while factor levels occur in each split.

Relations $G_1 = m$, $G_2 = (m, \gamma)$ and $G_3 = (m, \gamma, k)$ define which factors are shared. Training samples one relation-eligible donor per query and update. For G_3, the eligible histories are the other three rollouts of its system. The relation-eligible pool and the finite candidates retained by a sampler are different objects; Appendix E.1 reports both and describes the candidate-selection rule. Random uses balanced system derangements. The fidelity intervention fixes a pool of eight independent rollouts per training system (R8) and mixes correct and random relations at $\alpha \in \{0, 0.5, 1\}$ while preserving donor/query marginals. All model settings remain common within each intervention. The acquisition and relation-control banks have distinct protocols and measurements (Appendix E).

CoPhy

Environment and task. CoPhy provides visually observed systems whose physical properties govern different initial conditions and interactions (Baradel et al., 2020). We treat its three scene families as distinct environments in the suite:

Balls. Balls emphasizes rebound dynamics and physical-factor readout from ball trajectories.

Collision. Collision emphasizes contact-dependent motion and is the held-out task used for the main prediction comparison.

Blocktower. Blocktower tests multibody stability and falling in a different interaction geometry.

Shared protocol. The three scenes share a visual front end and episode construction. We construct cross-experience prediction episodes with three independent support histories (S3) and the first three query frames (query3). Prediction starts at query frame four, for 27 frames in Balls/Blocktower and 12 in Collision. Complete validation contains 2,000 Balls, 4,000 Collision, and 8,088 Blocktower episodes. The original CoPhy task and this multi-support prediction task have different input and target intervals; the reported matrix uses the latter.

Preserved learners and common evaluation. The official supervised perception frontend is frozen. JEPA, CPC, and RSSM learn dynamics from its 784-dimensional visual features without physical labels or coordinate targets. CPC retains conditional future discrimination; RSSM retains a deterministic state, stochastic latent, conditional prior and posterior, reconstruction, and KL regularization (Hafner et al., 2019). Its Native model already trains multistep prior prediction. Source future prediction never reads the future posterior. CoPhyNet retains its supervised source objective.

For JEPA/CPC/RSSM, Native preserves the corresponding base architecture; Structure adds the persistent–current interface; SPRII adds Cross; Random keeps the same Cross objective with randomized relations. The supervised CoPhyNet SPRII instance instead adds both Align and Cross; its Random control randomizes the relation under that combined objective. Collision uses all-object persistent substitution in the reported source run. Native representation dimensions are preserved rather than compressed to a common persistent-only code. Sources are frozen before a common newly trained prediction head receives the complete legal source state and S3 history. For example, structured RSSM exports $U = [P_{64}, h_{128}, z_{32}]$ with 224 dimensions, whereas Native exports its 160-dimensional recurrent state. The supervised split CoPhyNet code is $U = [P_{16}, T_{16}]$. The training-time persistent branch is distinct from the complete state exposed to a head or probe.

Every main head is trained for 100 epochs. The source budget is 100 epochs for CPC and RSSM. In the held-out Collision comparison, all four JEPA arms use 50 source epochs, matching the validation-selected 50-epoch JEPA source budget; control rows with unequal budgets are not used in this comparison. The JEPA comparison has four arms and three joint source/reader seeds on the same 1,994 held-out Collision episodes. These fits extend evaluation of the existing test cohort. All conditions use common queries, targets and visible-object eligibility within each reported population. Balls/Collision use direct query-pose perception; Blocktower uses its cached feature and matching pose-head route. These paths are held fixed within scene.

Operation-specific source paths. CPC matches contrastive candidates by public time, object slot and visible type. RSSM filters the first three query frames with zero persistent input, preventing initialization from bypassing the tested history path; Cross predicts through its prior rollout. The supervised CoPhyNet route retains the original 32-dimensional AB code, split into P_{16} and T_{16}. Align acts on P_{16}; Cross replaces that part while keeping the recipient's remaining code and initial query state. The common frozen-source reader uses the full 32-dimensional representation. These implementations preserve native objectives without imposing a common code.

Baxter tactile grasps

We use the public Baxter tactile hardness bank of Amin et al. (2023) from the dataset release of Amine et al. (2026). Each grasp peak is an 80×16 sequence. The primary crossed bank retains cube and cylinder objects at the three hardness levels shared by both shapes, yielding six known configurations and 170 grasps per configuration. A single permutation of grasp identifiers (Table 8) is reused across configurations and split 110/30/30 into train/validation/confirmation interactions. The release also provides 40-sample windows; this study uses the 80-sample files. The available input consists of grasp-peak tactile histories; pre-contact frames and action sequences are unavailable.

For $\mathcal{R}_H$, a donor shares the hardness level and changes shape; for $\mathcal{R}_S$, it shares shape and changes hardness level. Random uses a uniform derangement among the other five configurations. Every batch contains eight queries from each configuration and exactly one different-grasp donor per query, preserving balanced query and donor marginals.

The split architecture uses a per-sample MLP, a four-layer causal Transformer persistent encoder, 64-dimensional current-context and persistent codes, and prediction at $+1, +20, +40$ samples. All conditions use 10,000 updates, 48 query–donor pairs per batch, and seeds 0/1/2. Six randomly initialized supervised input-readout fits (early 8 samples versus full 80 samples) and nine world-model fits are frozen before the single confirmation evaluation. The source input is a 16-dimensional tactile sample and the causal history contains 40 steps. The reported source recipe enables Align + Cross for both partial-factor relations. Its six-configuration visualization and complete selectivity results appear in Appendix E.3.

OvercookedV2

Environment and base learner. OvercookedV2 is a cooperative cooking environment with coordination-dependent reward (Gessler et al., 2025; Rutherford et al., 2024). We use its native multi-episode AD setting (Jing et al., 2026): the controlled entity is a fixed neural partner whose behavior persists while locations, actions, and outcomes change across episodes. The training bank contains 2,560 histories and 37,376,000 transitions from 20 partners drawn from ten partner-training sources. Nested half- and quarter-history subsets retain 1,280 and 640 histories, or 18,688,000 and 9,344,000 transitions, with 64 and 32 rather than 128 histories per partner. Each history contains 146 recorded 100-step episode prefixes. The layout is `grounded_coord_simple`. Two additional partners from one excluded training source form the held-out development cohort; they are absent from policy training. Relation metadata identifies the partner for sampling without exposing its network parameters to the learner.

Matched adaptation. The AD action cross-entropy already consumes prior interactions. The H condition adds a 32-dimensional persistent history slot; H+VC adds variance/covariance regularization; H+SPRII additionally aligns histories of the same partner. The matched conditions share architecture, initialization and ordered training batches within a seed. The action objective reads the persistent slot, so SPRII uses Align without an additional Cross loss. The primary comparison is H+SPRII versus H+VC at full, half and quarter history budgets. Each pair uses 20,000 updates, effective batch 1,024, microbatch 128, learning rate 3×10^{-4}, and training seeds 4200, 4201 and 4202. The losses are $L_{\mathrm{CE}} + 0.001(25V + C)$ and $L_{\mathrm{CE}} + 0.001(25I + 25V + C)$, respectively. Both retain two 100-step support episodes and a 300-step query. All 18 fitted models use their fixed 20,000-update endpoints. The history-budget intervention subsamples this fixed offline bank; partner training and collection costs lie outside this budget.

Evaluation and replication. Each frozen endpoint is evaluated for 20 episodes with each of the 20 familiar and two held-out development partners: 440 episodes per model, with no optimizer updates. The primary return averages episodes 6–20 within partner, then averages partners within each cohort. Return and probe summaries report the mean and sample SD over the three training seeds. Formation and functional probes use the same frozen endpoints and common sample plan; the Random condition has three probe seeds and no return evaluation. The matched/null/wrong-history donor intervention uses one frozen checkpoint.

History interface. The dynamic path retains the AD observation encoder, action/reward tokens and causal Transformer. Each disjoint support episode is processed separately by the shared backbone; its final valid token projects to a 32-dimensional persistent slot. Attention pooling combines available slots, and a fusion MLP reads that result with the recipient's causal query representation to predict the ego action. H, H+VC and H+SPRII share this interface. Their common action loss already consumes earlier episodes; the tested increment is alignment.

RH20T

RH20T records multimodal robot manipulation (Fang et al., 2023). The operative relation groups different episodes by task identity. It supplies repeatable task context, without assuming known

physical-parameter equivalence. Models predict normalized force/torque and TCP motion from synchronized visual, robot-state and action inputs. The representation and prediction heads are trained jointly; this application uses supervised prediction targets.

Matched conditions. The complete matrix contains 14 conditions and three seeds each. All share prediction windows, normalization, targets and checkpoint step. "Indep" denotes a matched different-episode donor, "SameEp" a causal non-overlapping donor from the query episode, and "Random" balanced random training groups. LowDim is a distinct input/representation route whose controls are compared within that route. Evaluation keeps the query, target, horizon and actions fixed while changing self, matched and shuffled-task context. Task-macro results aggregate 25 held-out tasks.

Visual and low-dimensional sensor embeddings are fused before the causal history encoder. A common prediction trunk feeds latent, six-dimensional force/torque and three-dimensional TCP heads, all trained jointly. Cross changes the persistent input while retaining current context and task conditioning. The 25-task forecasting results are in Appendix F.3; the 24-task hard-negative bank is specified separately in Appendix F.3.

Articulated Swimmer

Articulated Swimmer is a three-link MuJoCo system with three link masses and two joint damping parameters. The task predicts state responses to actions. Each frozen architecture retains its own training-only normalization. Appendix F.5 compares own- and wrong-system persistent updates within each architecture, keeping the source and predictor weights fixed.

Pendulum control

Pendulum tests whether the learned dynamics context supports an online controller when mass and length vary between systems. The ID-selected SPRII recipe uses weight 0.003 and variance target 0.1 with the existing CaDM control interface. It is evaluated intact on ID and four OOD groups, with three training seeds and ten 200-step episodes per group. Return, unchanged upright-success criteria and same-checkpoint physical probes are reported together in Appendix G.5; logged-action donor interventions are not presented as online controller interventions.

NOD-Burgers: one-dimensional field evolution

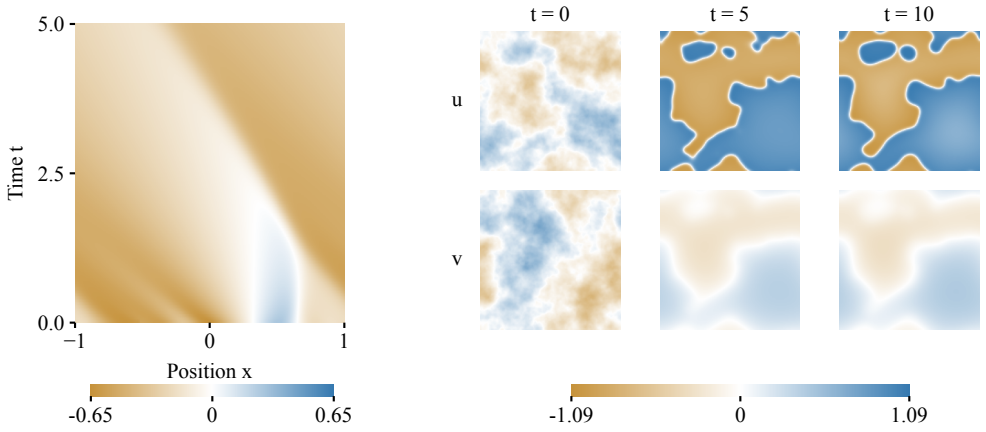
Burgers uses a spatial field rather than a rigid-body scene. Independent trajectories share viscosity ν while differing in initial conditions; the recipient field provides the query state. The released trajectory shown in Figure 9(a) has 401 spatial samples and 101 saved times. Native prediction, viscosity accessibility and donor-use evidence, including three-seed alignment and continuation comparisons, are specified in Appendix C.4.

NOD-FHN: two-dimensional reaction–diffusion

The FitzHugh–Nagumo (FHN/DR2D) environment evolves two coupled fields u, v on a periodic 128×128 grid. Related trajectories share reaction parameters (k, β) with independent Gaussian-random-field initial conditions. The Python port follows the released fourth-order spatial stencil and RK4 update ($\Delta t = 0.001$), saving 101 frames through $t = 10$. Conditioning uses frames $0, 25, 50, 75, 100$ from an independent trajectory; the hierarchical operator predicts the recipient's field evolution. The missing original initial-condition bank was regenerated, so these are reproducible Python simulations, not the authors' exact data.

Training schedules and selection. Both methods use batch size 8 and RMSprop, with initial learning rate $0.5/\sqrt{P}$, where P is the number of trainable parameters, and a factor of 0.95 decay every 500 updates. NOD uses prediction-only stages ending at 20,000, 30,000, 40,000 and 45,000 updates. SPRII uses alignment weight 0.01 with a 2,000-update linear warmup through update 10,000, followed by prediction-only stages ending at 30,000, 40,000 and 50,000. Both recipes reset the learning rate to 10^{-5} after update 40,000 while retaining RMSprop moments and scheduler state. Each continuation restarts its seeded sampling stream.

Seed 42 selects the recipes using mean ID relative-L_2 error across H1/H5/H50 on initial condition 5. Seeds 43/44 repeat the selected stages and endpoints without further checkpoint selection; reporting uses initial conditions 15/24/45. The NOD and SPRII recipes therefore have different update budgets. Appendix H.3 reports the three-source comparison and conditioning-history experiments.



(a) Burgers: recorded field **(b) FHN: field evolution**

Figure 9: **External field environments.** (a) A released Burgers training trajectory at $\nu = 0.01$. (b) A separately simulated FHN trajectory at $k = 0.03, \beta = 0.20$. FHN panels share a signed field scale; Burgers uses a separate scale. These are environment illustrations, not prediction results.

B.5 Probes, fresh readers and fixed consuming routes

A *complete-method comparison* changes the trained source recipe; a *fresh-reader comparison* fits alternative task readers on a frozen source; a *fixed-route intervention* changes context through the same fitted weights. They answer different questions even when the same source model appears in all three. The measurement definitions below complement the task-specific controls in the main text.

Table 6: Terminology used in the main text and appendix.

Term	Meaning
Persistent context	History-derived representation supplied to a predictor; source weights are frozen for the stated diagnostic evaluations.
Probe	Supervised diagnostic fitted to a frozen code; measures accessibility.
Task reader	Predictor fitted on fixed source features; measures realized task value.
Fixed-predictor intervention	Donor replacement with all source and consuming weights fixed.
Native	History-conditioned learner with its original context interface and objective.
Structure	Separate persistent/current interface, without relation objectives.
Random	The instantiated relation objective with a wrong-donor pairing control.

Input evidence and representation accessibility. An estimator trained on the model-visible history checks whether factor evidence is recoverable from the input; a privileged-input estimator is a separate reference. A representation probe fits a supervised factor readout of a frozen representation, using the training systems for fitting. Each result identifies its input: code centroids, actual donor windows, complete source state U, independent-support code S, or learned reader memory M. Probes measure accessibility, while cross-history geometry measures organization; both inform Formation. A memory probe includes the trained reader projection, while a source probe reads the frozen encoder alone.

Fresh task readers and fixed-route use. Null, Persistent, Decode and Oracle fit separate task readers with their stated inputs; Oracle supplies true parameters to a learned reader, not an optimal-risk bound. A fixed Wrong intervention substitutes context through an unchanged reader, keeping source, predictor and any output decoder fixed. The resulting output or risk change measures Use under that substitution. Fresh-reader task errors measure realized Value under the compared inputs and fitting recipe. A separately trained Shuffled/Wrong arm differs from fixed-input substitution.

Frozen-source realization routes. Let h denote the frozen recipient feature, p the frozen persistent code, and $R_0(h)$ the separately fitted, then frozen recipient-only base reader. The 128-unit residual reader computes $v = \mathrm{GELU}(W_v h)$ and $q = W_q h + b_q + W_{\mathrm{proxy}} h$. With elementwise product $\odot$,

$$M_1(h) = R_0(h) + W_o[v \odot \tanh(q)], \tag{11}$$

$$M_2(h, p) = R_0(h) + W_o[v \odot \{\tanh(q + W_p p) - \tanh(q)\}]. \tag{12}$$

The persistent projection and output projection are bias-free, so $M_2(h, 0) = R_0(h)$ by construction (up to floating-point evaluation order). A zero-context penalty therefore cannot alone identify donor specificity. The query-only arm registers the same persistent projection but does not use it or update it by gradients: the architecture is shared, active capacity is not exactly matched. The Oracle route replaces the learned code with a projection of true parameters and uses the same residual family.

The `phys` variants multiply the residual output coordinatewise by a fixed, nonnegative weight vector derived from the prescribed physical sensitivity procedure. They do not add a physical loss or supply true parameters to M_2. Source encoders and the 256-unit base are frozen during residual fitting. Reader fitting uses MSE, AdamW, gradient clipping at 1, and nominal 1,000-step base and route budgets with batch size 128 and learning rate 10^{-3}; development error selects the checkpoint. The SpringWorld M_1/M_2 study in Appendix G.4 uses batch size 256 for base and route fitting, with the same 1,000-update budgets and learning rate. The physical target and evaluation population are held fixed within each pair.

The results of these routes appear in Appendix G.4. We evaluate $M_2(h, sp)$, using s for the post-fitting context amplitude and reserving α for training-pair reliability. The zero-amplitude identity is $M_2(h, 0) = R_0(h)$, not $M_1(h)$.

B.6 Measurement, aggregation and selection

Geometry and factor accessibility. Controlled codes are standardized using training statistics. Within-system distances measure stability across interactions; between-system distances measure separation. The reported between/within ratio is averaged at the source level, not computed from pooled distances. The Balls memory assay uses the inverse within/between ratio and labels it explicitly. Partial factor geometry correlates representation distance with factor distance while controlling the stated other factors; it is distinct from supervised-probe R^2. Probe fitting and selection use training systems. Transient position and velocity targets use regression, while contact uses AUROC.

Prediction targets and scales. The controlled source objective predicts each learner's own observation embedding, so self losses characterize native training. Physical-space comparisons use downstream targets: SpringWorld readers predict train-standardized eight-state increments; D-Clean readers in the state-space and same-donor studies report raw-state MSE on different populations. RH20T uses normalized force/torque MSE and TCP error in cm.

Evaluation populations. We use *sealed test* for the separately held-back evaluations identified as such in the experiment records, *held-out official episodes* for the CoPhy test cohort, and *confirmation* for the frozen relation-design evaluations. Development results retain their own populations and selection rules. SpringWorld's 100-system common-adapter cold-start comparison and its 178-system cold-query horizon curve are different evaluations: their h16 history gains are 36.10% and 35.68%, respectively. The sealed method comparison uses a separate 256-system population.

Table 7 lists the distinct evaluation populations and consuming routes used by the functional analyses.

Source and reader dispersion. For cell error E_{sr}, the mean first averages R_s fitted readers within source, $\bar{E}_s = R_s^{-1} \sum_r E_{sr}$, then averages S sources equally. The displayed source SD is

$$s_{\mathrm{source}} = \sqrt{\frac{1}{S-1} \sum_{s=1}^{S} (\bar{E}_s - \bar{E})^2}, \qquad \bar{E} = \frac{1}{S} \sum_{s=1}^{S} \bar{E}_s.$$

A 3×3 grid therefore represents three independently trained sources, each evaluated with three fitted readers. When source and reader seeds vary jointly, SD describes the complete fitting pipeline. In incomplete grids, the fitted reader counts are reported rather than treating the grid as balanced.

Table 7: **Evaluation populations, fitted-model coverage and consuming routes.**

Analysis	Population	Source, reader and measurement
SpringWorld		
Sealed comparison	256 continuous primary; 36 factorial secondary	3 sources $\times$ 3 readers; matched inputs. The 292-system pool is auxiliary.
Original reader	100-system cold start; 178-system mixture	Separate grids: source 0/reader 0 reference; three-source component mean with reader 0; original 3×3 comparison.
Common adapter	100-system bottleneck; 178-system horizon/probes	FCRL uses each source's own query encoder; the four bottleneck arms share one source's query encoder.
Landscapes	64 development systems	Three sources, reader 0; fixed donor–target interventions.
Controlled D-Clean and PokeWorld		
D-Clean TDS	100-system report	3×3 grid; selected state-space source and common raw-state reader.
D-Clean same-donor	200 validation systems	Common Align + Cross query encoder; 3×3 grid, with 27 Null/Persistent/Oracle and 9 Decode cells.
Poke geometry/probes	400 confirmation; 400 validation systems	Centroid probes and actual donor-window probes are separate measurements.
Poke physical-risk use	96 development systems	Wrong-factor physical MSE; native predictor and state decoder fixed.
Poke native direction	144 validation systems	G1/G2, three sources each; dimensionless projection in each source's native embedding.
Poke fresh-reader value	400 development systems	Separately fitted task readers; Shuffled is a trained arm, not a fixed-input substitution.
Transferred learners, robot data and partners		
Collision test	1,994 held-out episodes	JEPA, CPC and RSSM each have 3 joint source/reader seeds per arm; variation across trained fits is reported as source SD.
Balls extension	2,000 development recipients	The reporting cohort includes source/readout selection examples.
RH20T	25-task forecast; 24-task hard negatives; separate 25-task input validation	Specificity and absolute risk share the hard-negative bank. Input controls use the separate validation bank.
Overcooked	20 familiar and 2 held-out development partners	One layout; 3 training seeds for return/probes at each history budget. Return mean and sample SD are over seeds; donor interventions retain one checkpoint.

Paired effects. Complete-method improvement is $100(\bar{E}_{\text{control,matched}} - \bar{E}_{\text{SPRII,matched}})/\bar{E}_{\text{control,matched}}$. Within-source fresh-reader history gain instead compares separately trained Null and Matched readers. Fixed-weight wrong-minus-matched error measures reliance on the supplied donor. Relative effects use ratios of mean errors, not averages of case-wise percentages. In the same-donor bottleneck discussion, relative differences use Decode as the denominator: Persistent is 22.8% lower in D-Clean and 2.12% higher in SpringWorld. Raw Decode-minus-Persistent differences retain their original units and intervals. Intervals pair conditions within the appropriate unit before averaging fitted seeds and resampling that unit. Analyses specify their system/task/recipient unit, stratification, multiplicity and bootstrap seed.

Table 8: Fixed randomization settings used in the appendix.

Experiment	Operation	Seed
Baxter	Grasp-identifier split permutation	20260822
RH20T additional 25-task bank	Paired task bootstrap	20260923

Checkpoint and configuration selection. CoPhy's cross-scene development and held-out Collision comparisons use the distinct recipes specified in Appendix B.4. Each analysis specifies whether it uses a final-step checkpoint, a validation-selected reader or a released-default reference.

Coverage and selection. The source-0 CoPhy matrix has second-source JEPA fits in all three scenes and CPC Collision, plus three-source RSSM Collision. The held-out Collision comparison includes four arms with three joint source/reader seeds each for JEPA, CPC and RSSM. Balls separately has three-source RSSM (four arms) and CoPhyNet (three arms, reader seed 0) extensions. These are distinct fitted grids; Table 7 and the experiment-specific captions retain their evaluation populations. Overcooked has three training seeds for returns and probes at each history budget; its donor intervention remains a single-checkpoint study (Appendix B.4). Swimmer's three frozen bases span separate architectures.

CoPhy development reporting includes reader-selection examples; CoPhyNet Balls also uses its source-selection cohort. The held-out Collision episode comparison uses a separate population. Recipient-bootstrap intervals describe variation across recipients conditional on fitted source/readers; source SD describes fitted-model variation.

Unless stated otherwise, subsequent analyses use the population definitions, source/reader aggregation rules and fitted-model uncertainty conventions given here; experiment-specific departures are stated with the results. Public datasets and implementations retain their original access and license terms.

C Predictive Performance and Closest Comparisons

This section expands Tables 1–2: the primary task comparisons come first, followed by closest adapted constructions and the released NOD operator. Each comparison retains its own inputs, readout and evaluation population; these task effects motivate the representation analyses in Appendix E.

Primary-comparison average. The abstract's 14.6% is the equal-weight mean of five relative error reductions in Table 1: SpringWorld versus Native JEPA (22.3%), Collision CPC/RSSM/JEPA versus Structure (15.4%, 11.4%, 14.2%), and Balls RSSM versus Structure (9.9%). Calculations use values before rounding; each setting–learner comparison receives one weight. This summarizes these five primary comparisons, not all thirteen settings.

C.1 Cold-start prediction on sealed systems: SpringWorld

Align + Cross reduces cold-start error relative to Native and TDS on the 256 continuous systems in the primary sealed SpringWorld test. All methods receive an independent 96-frame same-system donor and six cold-start queries per system at $h = 16$. The final 10,000-update sources and readers are frozen: three source seeds and three readers per source for each recipe. We average query errors within each system, then across readers within each source, and finally equally across sources. Paired 95% intervals use 10,000 stratified physical-system draws, conditional on these fitted models.

All three source means and all nine paired cells favor SPRII against each comparator. The 292-system pool weights systems rather than strata equally; it does not replace the continuous primary endpoint.

Table 9: **Sealed SpringWorld prediction and paired contrasts.** Results average three sources and three readers per source. Brackets show 95% physical-system intervals conditional on these fitted models. Positive gaps favor Align + Cross; the system-weighted pooled result is auxiliary.

A. Absolute prediction error

Source	Primary: 256 systems	Secondary: 36	Pooled: 292
Native	0.13820 [0.11850, 0.15920]	0.21370	0.14750
TDS	0.13030 [0.11180, 0.15020]	0.19670	0.13840
Align + Cross	**0.10740** [0.09036, 0.12670]	**0.12730**	**0.10990**

B. Paired contrasts within each population

Population	Comparator	MSE gap [95% interval]	Reduction [95% interval]
Primary	Native	0.03084 [0.01735, 0.04429]	22.31% [13.15, 30.69]
Primary	TDS	0.02284 [0.01039, 0.03487]	17.54% [8.40, 25.87]
Secondary	Native	0.08645 [0.02195, 0.16640]	40.45% [16.34, 54.55]
Secondary	TDS	0.06944 [0.01587, 0.13300]	35.30% [12.10, 49.13]
Pooled	Native	0.03769 [0.02285, 0.05302]	25.55% [16.90, 33.32]
Pooled	TDS	0.02859 [0.01560, 0.04176]	20.65% [12.00, 28.45]

C.2 Held-out prediction: Collision and Balls

SPRII lowers error across CPC, RSSM and JEPA, improving on Native, Structure and Random in every fitted seed. Reductions relative to Structure are 15.4%, 11.4% and 14.2%, respectively (Table 10). Three independent donor histories and the first three query frames predict twelve future frames under the S3/query3 task. A frozen donor plan admits 1,994 of 2,000 official normal-test episodes; six lack three distinct legal donors in one object slot.

All four arms cover three joint source/reader seeds. CPC and RSSM use checkpoints after 100 source epochs; every JEPA arm uses checkpoints after 50 source epochs. Readers use a 100-epoch budget. This comparison uses the donor eligibility above and checkpoint selection in Appendix B.

Table 10: **Held-out Collision prediction.** Physical trajectory MSE on 1,994 eligible episodes, mean $\pm$ sample SD across three joint source/reader seeds. Training budgets match within each learner.

	Physical trajectory MSE				Reduction	
Family	Native	Structure	Random	SPRII	vs. Structure	vs. Random
CPC	0.248 ± 0.008	0.223 ± 0.006	0.221 ± 0.006	$\mathbf{0.189} \pm 0.007$	15.4%	14.7%
RSSM	0.268 ± 0.007	0.228 ± 0.005	0.222 ± 0.007	$\mathbf{0.202} \pm 0.007$	11.4%	9.0%
JEPA	0.250 ± 0.004	0.215 ± 0.007	0.265 ± 0.003	$\mathbf{0.184} \pm 0.005$	14.2%	30.4%

Table 63 reports additional CoPhy configuration comparisons.

Held-out Balls4 prediction with RSSM. On the same 1,000 four-ball recipients and eligibility mask, all four RSSM arms use 100 source epochs and 100 reader epochs, with three independent source/reader seeds. The existing checkpoints were evaluated on this population; this extends the existing test evaluation and is not a newly sealed population. Cross lowers mean error by 10.7% against Native, 9.9% against Structure and 13.1% against Random, with the same direction in all three seeds. This result concerns four-ball prediction, not extrapolation to other ball counts.

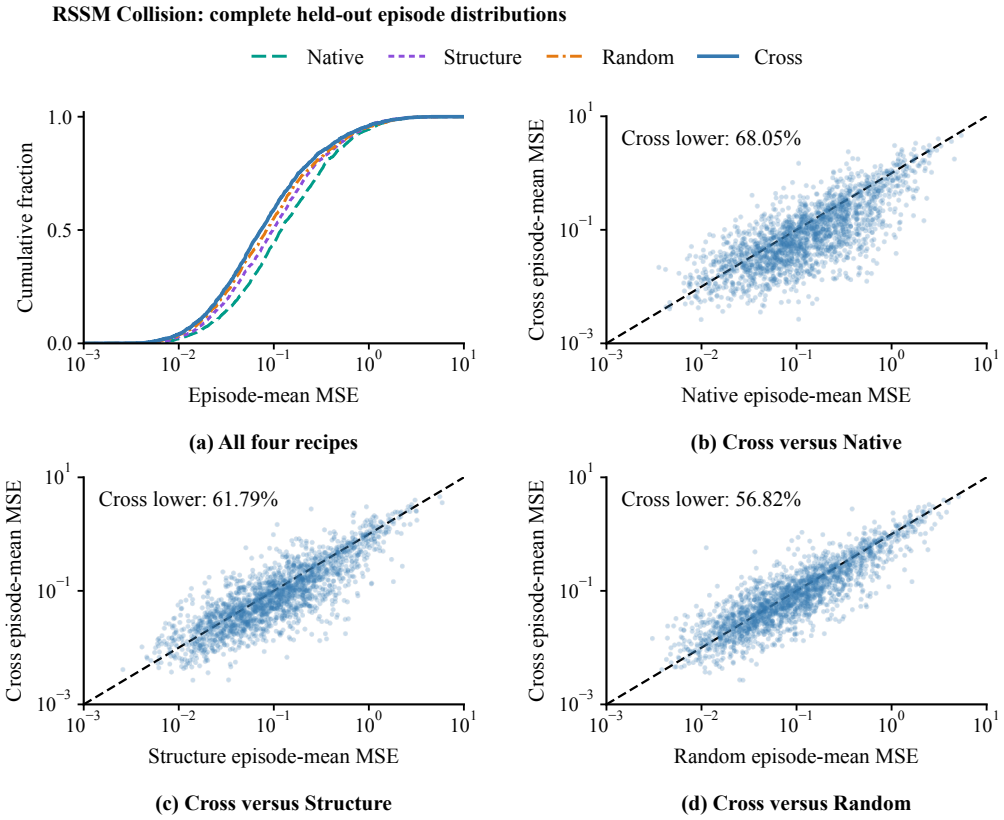


Figure 10: **Cross lowers RSSM error on most held-out episodes.** (a) Error distributions on 1,994 episodes. (b)–(d) Paired comparisons with each control. MSEs are means over three source/reader runs. Points below the diagonal favor Cross; fractions show their share. Axes are logarithmic.

Table 11: **RSSM on 1,000 held-out Balls4 recipients.** MSE: mean ± sample SD over three seeds.

Method	MSE	Source / reader epochs
Native	1.292 ± 0.037	100 / 100
Structure	1.280 ± 0.031	100 / 100
Random	1.327 ± 0.011	100 / 100
SPRII (Cross)	1.154 ± 0.025	100 / 100

C.3 Matched comparisons with adapted constructions

SPRII gives lower matched-reader error than the closest constructions under the task interfaces specified in Appendix B. Tables below group results by reader and evaluation population.

SpringWorld trajectory-decoupled prediction. SPRII lowers matched cold-start error relative to both full and compact visual TDS under the original SpringWorld reader. The NOD-inspired TDS adaptation predicts recipient visual embeddings from a same-system donor, current input and future actions, using $\mathcal{L}_{\text{cross}} + 0.02\mathcal{L}_{\text{SIGReg}}$ and no self or alignment term. It receives the same 10,000 source updates and independent 96-frame histories as SPRII. Compact TDS selects a zero-dimensional history slot ($d = 0$) from $0, \ldots, 6$ on a separate selection-development bank. A common original physical reader evaluates three sources by three readers on 100 development systems.

Pooled Matched reductions are 29.73% and 34.40%, respectively, with the same direction in all three source means. This visual TDS construction uses the trajectory-decoupled principle of Chen et al. (2026).

Fixed-recipe temporal FCRL-style comparison. The fixed temporal FCRL-style adaptation realizes little history gain, whereas SPRII lowers matched error under the common-adapter comparison.

This temporal adaptation of same-function contrastive learning (Gondal et al., 2021) uses two independent histories per system, a shared visual/history encoder and a discarded contrastive projection. Temperature 0.07 is fixed without a sweep. Each of three sources uses 10,000 updates; three fresh readers follow the common-adapter recipe. Each method retains its own frozen query encoder, so the absolute comparison is between complete source recipes.

SPRII lowers Matched error by 58.44% [52.90,63.74]% under this adaptation and budget. Training-only physical probes remain near zero for the contrastive codes on the separate 178-system mixture.

D-Clean state-space TDS. SPRII has lower h32 common-reader error than state-space TDS even though both frozen sources retain strong drag accessibility. State-space TDS predicts true state increments from independent donor history, current state and actions at horizons $1, 4, 16$. A separate source selects $d = 4$ from $0, \ldots, 4$ using native h16 error. Three sources by three readers then evaluate 100 report systems. At h32, common-reader raw-state Matched MSE is 1.507×10^{-4} for SPRII and 1.343×10^{-3} for TDS (88.78% lower); the paired gap is 1.193×10^{-3} $[0.4695, 2.240] \times 10^{-3}$, with five-comparison adjusted interval $[0.3807, 2.678] \times 10^{-3}$. All source means agree in direction. TDS retains drag $R^2 = 0.9712$–0.9767 (SPRII: 0.9965–0.9967) alongside larger task error. Native direct h16 errors of $(2.617$–$2.834) \times 10^{-4}$ and iterative h32 errors of $(1.827$–$1.987) \times 10^{-3}$ use different input constructions. This 100-system comparison does not share the population of the 200-system same-donor experiment in Appendix G.2. The separate released-component comparison below uses the same report population.

D-Clean: common-reader dynamics baselines. SPRII lowers long-horizon error relative to NOD, FCRL, CaDM and the DALI context adaptation (Table 12). CaDM learns a context encoder through forward and backward dynamics prediction (Lee et al., 2020); SPRII reduces its h16 and h32 errors by 32.16% and 33.70%, respectively. Both reductions hold across all three sources. The h32 reduction against FCRL is 88.70%, also with the same direction in every source. The implementation and common-reader protocol are specified in Appendix B.

The same source checkpoints support factor-accessibility and fixed-reader use measurements. Training-fitted damping probes attain $R^2_{\log \gamma} = 0.9739$ for CaDM and 0.9962 for SPRII. Their h32 wrong-minus-matched MSEs are 0.05329 and 0.05305, respectively: both readers use history.

The DALI comparison adapts its context-learning component to the same frozen-code common-reader interface; it is not a native control-return ranking. The selected 160,000-update source uses a separate 100-system selection bank; all three source seeds and their three 20,000-update readers are included on the 100-system report bank. Other sources use 20,000 updates. SPRII lowers h32 mean error by 96.7% relative to this longer-budget DALI source, in the same direction for all three sources. The 20k/80k/160k DALI rows in Table 12 retain the budget dependence; unequal source budgets are not described as equal total computation. At 160k, DALI's damping probe has $R^2 = 0.785$ and its h32 wrong-minus-matched MSE is 0.0398.

Table 12: **D-Clean prediction with a common reader.** Raw-state MSE in units of 10^{-4}, mean $\pm$ source SD after averaging three readers per source (3 sources). Source budgets vary as shown. The 100-system report population was previously evaluated; it is not a new sealed test.

Source	Updates	h1	h4	h16	h32
SPRII	20,000	0.0314 ± 0.002915	0.1166 ± 0.005082	0.5232 ± 0.01173	1.507 ± 0.1355
CaDM	20,000	0.03324 ± 0.001838	0.1474 ± 0.008051	0.7711 ± 0.008115	2.273 ± 0.06284
FCRL	20,000	0.1228 ± 0.01122	0.762 ± 0.08086	4.833 ± 0.492	13.34 ± 1.014
NOD	20,000	0.3894 ± 0.6261	3.814 ± 6.412	23.79 ± 40.18	61.88 ± 104.1
DALI	20,000	0.5579 ± 0.02027	5.307 ± 0.2763	35.49 ± 1.735	89.3 ± 3.886
DALI	80,000	0.3314 ± 0.035	2.962 ± 0.2441	20.54 ± 1.948	53.9 ± 5.067
DALI	160,000	0.3201 ± 0.03572	2.732 ± 0.3115	17.26 ± 1.991	45.92 ± 5.362

Source dispersion and native controls. For SPRII/NOD/FCRL/CaDM, h32 source SD is 0.135/104.076/1.014/0.063 in 10^{-4} raw-state MSE units. Damping-probe R^2 is 0.996/0.643/0.852/0.974 and fixed-reader wrong-minus-matched MSE is 0.05305/0.03556/0.04891/0.05329.

Table 13: **Prediction with adapted baselines and context controls.** A/B report Spring standardized MSE; C reports D-Clean raw-state MSE, each averaged equally over three sources and three readers. Fixed Wrong uses the Matched reader. D uses native TDS decoders rather than C's common reader; its direct and rollout errors are separate diagnostics.

A. Spring: original reader

Reader	Source	Fresh Null	Matched	Fixed Wrong
Original	Align + Cross	0.1775	0.1121	0.2439
Original	TDS	0.1716	0.1596	0.1787
Original	Compact TDS ($d = 0$)	0.1709	0.1709	0.1709

B. Spring: common adapter

Reader	Source	Fresh Null	Matched	Fixed Wrong
Common adapter	Align + Cross	0.1778	0.1136	—
Common adapter	FCRL-style	0.2767	0.2735	—

C. D-Clean: common-reader h32 context controls

Source	Matched	Fresh Null	Fixed Wrong	Fixed Zero
SPRII	0.0001507	0.01821	0.05320	0.03105
State-space TDS	0.001343	0.01821	0.05257	0.03143

Other source comparisons and budgets are in Table 12.

D. D-Clean: TDS native-decoder diagnostics

Measurement	Source 0	Source 1	Source 2
Direct h16	0.0002706	0.0002834	0.0002617
Free rollout h32	0.001948	0.001827	0.001987

C.4 Prediction and continuation on a released operator: Burgers

The fixed SPRII endpoint has lower mean native prediction error than released-default NOD across ID, viscous OOD, and inviscid OOD. We use the released NOD implementation and training configuration (Chen et al., 2026), with source seeds 1234, 5678 and 9012. The field environment and available viscosity grid are specified in Appendix B.4 and Figure 9(a).

Training and selection. SPRII retains the native prediction objective during relation training, then continues prediction with $\lambda_{\text{align}} = 0$. With alignment disabled, a ten-update matched implementation check reproduced the unchanged NOD path in predictions, gradients, parameters and optimizer state. Both source searches span 200 epochs and select checkpoints by ID development loss. The SPRII pre-continuation checkpoints receive another 25, 113 and 127 epochs, respectively, at 45 updates per epoch; the final 25 epochs use learning rate 5×10^{-5}. All reported SPRII values use these fixed final endpoints. Continued NOD starts from its own selected checkpoints and receives the same additional updates. This control matches continuation updates, not total ancestral training and search budgets.

Native prediction. SPRII reduces error by 1.075%, 6.554% and 2.921% across the three distributions (Table 14). Native MSE follows the released evaluator's unweighted mean of batch MSE over the full trajectory, with 45 ID, 500 viscous-OOD and 100 inviscid-OOD examples. These populations were used in earlier comparisons; the recipe was selected on development data. Source SD and paired-source intervals accompany the table.

Relative to continued NOD, SPRII lowers ID error by 4.06%, with 1.06%/0.20% higher viscous/inviscid OOD errors. Relative to Random continuation, it lowers ID and viscous-OOD errors, while inviscid error is nearly tied. The complete three-source controls appear in the same table.

Organization and fixed-predictor use. The primary-checkpoint measurements in Table 15A use training-standardized latent geometry, training-fitted viscosity probes and a common history bank. SPRII has a higher between/within ratio and viscosity probe R^2 than released-default NOD. At

Table 14: **Burgers prediction and continuation controls.** Native trajectory MSE in units of 10^{-3}, mean $\pm$ sample SD across three sources. SPRII includes prediction-only continuation; controls match its added updates. Released-default NOD has no added updates. Dashes mark unreported SD.

Track	ID	Viscous OOD	Inviscid OOD
Primary comparison			
Released-default NOD	9.096 ± 0.544	8.012 ± 0.815	11.86 ± 0.628
SPRII fixed endpoint	$\mathbf{8.999} \pm 0.798$	$\mathbf{7.487} \pm 0.183$	$\mathbf{11.51} \pm 0.360$
Matched additional-update controls			
Continued NOD	9.379 ± 0.340	7.408 ± 0.130	11.49 ± 0.215
Random continuation	9.538 (–)	8.900 (–)	11.50 (–)

Paired-source 95% t intervals for SPRII minus released-default NOD ($n = 3$, df=2; MSE $\times 10^{-3}$): ID $[-2.472, 2.277]$, viscous OOD $[-2.982, 1.932]$, and inviscid OOD $[-2.783, 2.091]$.

direct horizon $H = 50$, the mean wrong-donor MSE changes are close to zero for both models. The additional-update analysis in panel B uses a distinct 11-system, relative-L_2 donor assay: both models incur positive wrong-donor penalties, and continued NOD has stronger geometry and viscosity accessibility. These measurements distinguish viscosity accessibility from fixed-predictor donor responses.

Table 15: **Burgers organization, viscosity accessibility and donor use.** Entries are mean $\pm$ SD across three sources. B/W is the latent between/within squared-distance ratio. A uses 45 ID recipients and a direct MSE donor assay; B uses training geometry and a separate 11-system relative-L_2 assay. Positive penalties indicate wrong-donor harm; 95% intervals resample paired sources and systems.

A. Primary comparison: released NOD and SPRII

	ID B/W	ID R^2	$H = 50$ MSE ($\times 10^{-3}$)	
Method			**Matched**	**Wrong−matched**
NOD	22.3 ± 19.09	0.651 ± 0.458	12.89 ± 0.992	0.014 ± 0.391
SPRII	$\mathbf{31.64} \pm 4.506$	$\mathbf{0.762} \pm 0.141$	$\mathbf{12.19} \pm 1.944$	$\mathbf{-0.054} \pm 0.597$

B. Additional-update comparison

Method	Train B/W	ID R^2	Viscous-OOD R^2	$H = 50$ relative-L_2 penalty [95% interval]
Continued NOD	42.07 ± 14.74	0.924 ± 0.006	0.975 ± 0.004	0.1331 $[0.06729, 0.2127]$
SPRII	24.58 ± 8.62	0.762 ± 0.141	0.895 ± 0.116	0.1275 $[0.06538, 0.2041]$

Test-time-adaptive operator references. In the Burgers reference runs reported here, GEPS and CoDA use 50 support-code optimization steps per query, while NOD and the SPRII endpoint use none. Table 16 reports these references under the released native metric. CoDA has lower mean error in all three distributions; GEPS has lower ID error and higher OOD means, with substantial source dispersion.

Table 16: **Burgers prediction with test-time adaptation.** Native full-trajectory MSE in units of 10^{-3}, mean $\pm$ sample SD across three sources. The reported references use 50 support-code updates per query; zero-step results are in Table 14.

Method	ID	Viscous OOD	Inviscid OOD	Code steps
GEPS	5.60 ± 7.05	12.37 ± 14.21	13.84 ± 17.41	50
CoDA	3.38 ± 0.57	6.48 ± 1.42	5.89 ± 1.55	50

C.5 Relation supervision on a frozen CoDA host

An amortized CoDA instance adds a history encoder to each of three fixed CoDA source decoders. The four arms in Table 17 share a teacher-fit initialization, a fixed encoder budget and final-checkpoint selection. Each source seed retains its own decoder and train-only teacher-code cache. Seeds 1234, 5678 and 9012 retain their 5,000-update decoders. All arms use the same 101-frame support histories

and fixed released evaluation populations: 45 ID, 500 viscous-OOD and 100 inviscid-OOD pairs. The relation-trained arm uses Cross.

Table 17: **Objectives on the frozen CoDA decoder.**

Arm	Objective addition	Role
Plain amortized CoDA	Self prediction and common regularization	History-encoder baseline.
Matched Cross	A second prediction using the related donor	Relation-supervised arm.
Random Cross	The same second prediction using a wrong-system donor	Pairing control with matched donor marginals.
Duplicate Self	A second self-prediction forward/backward pass	Matched computation and reconstruction-loss scale.

A shared 500-update teacher fit precedes 1,000 encoder updates per arm. Within each seed, initialization and sampled histories are identical across arms. Plain uses one prediction branch; the other arms use two, so equal update counts do not imply equal training FLOPs. The primary endpoint uses the inferred context with no support-code optimization (code-0). The secondary test-time-adaptation endpoint (code-50) initializes the code from the same encoder and applies fifty support-code optimization steps; the encoder and decoder remain frozen. Original CoDA with zero initialization and fifty support optimization steps is the external reference in Table 16. The prespecified code-0 criterion compares Matched Cross with both Plain and Duplicate Self in each distribution: lower mean error and the same direction in a majority of source seeds. Random Cross measures sensitivity to the relation assignment. Paired contrasts use the same decoder seed, evaluation inputs and adaptation budget. Native MSE averages batch-8 errors equally and includes the initial frame. Viscosity ridge probes fit 360 training histories, with regularization selected by five-fold training cross-validation.

Matched Cross meets this code-0 criterion in all three distributions: relative to Plain, mean error falls by 6.70%, 5.17% and 6.50%; relative to Duplicate Self, it falls by 4.98%, 6.74% and 7.39%, respectively. Each comparison favors Matched Cross in all three seeds. Its smaller advantages over Random Cross (1.74%, 0.38% and 1.46%) have directions 2/3, 1/3 and 2/3 across seeds. With only three source runs, all nine paired t intervals include zero (Table 19).

After fifty support-code updates, Matched Cross has lower mean error than Duplicate Self in all three distributions, but lower mean error than Plain only on viscous OOD; Random Cross has slightly lower means than Matched Cross in all three distributions. All nine paired intervals include zero (Table 20). Each arm's mean query error is higher at code-50 than at code-0 in every distribution. Matched Cross retains the highest mean viscosity-probe R^2 (Table 21), separating parameter accessibility from forecasting value under this fixed adaptation budget. The gains concern amortized inference.

Table 18: **Amortized prediction with a frozen CoDA decoder.** Native full-trajectory MSE in units of 10^{-3}, mean $\pm$ sample SD across three sources. Code-0 is the primary endpoint.

Distribution	Arm	Code-0	Code-50
ID	Plain	3.4834 ± 0.5669	$\mathbf{3.5123} \pm 0.5353$
ID	Matched Cross	$\mathbf{3.2500} \pm 0.7286$	3.5566 ± 0.4909
ID	Random Cross	3.3076 ± 0.7055	3.5164 ± 0.4933
ID	Duplicate Self	3.4203 ± 0.7573	3.5899 ± 0.4809
Viscous OOD	Plain	6.3900 ± 1.4357	7.1890 ± 1.6605
Viscous OOD	Matched Cross	$\mathbf{6.0598} \pm 1.1847$	7.0845 ± 1.4797
Viscous OOD	Random Cross	6.0832 ± 1.2241	$\mathbf{7.0310} \pm 1.5672$
Viscous OOD	Duplicate Self	6.4976 ± 1.5413	7.4010 ± 1.6684
Inviscid OOD	Plain	5.8833 ± 1.6998	$\mathbf{6.7663} \pm 2.2725$
Inviscid OOD	Matched Cross	$\mathbf{5.5006} \pm 1.4956$	6.8105 ± 2.1187
Inviscid OOD	Random Cross	5.5822 ± 1.5558	6.8046 ± 2.2922
Inviscid OOD	Duplicate Self	5.9394 ± 1.8774	7.0893 ± 2.5458

Table 19: **Paired CoDA contrasts at code-0.** Differences are comparator minus Matched Cross in units of 10^{-3}; positive favors Matched Cross. Intervals are paired-source 95% t intervals ($n = 3$); n_+ counts positive source differences.

Distribution	Comparator	Seed 1	Seed 2	Seed 3	Mean [95% CI]	n_+
ID	Plain	0.1270	0.4228	0.1502	$0.2333\,[-0.1752, 0.6418]$	3
ID	Random Cross	-0.0113	0.0710	0.1129	$0.0576\,[-0.0994, 0.2145]$	2
ID	Duplicate Self	0.3036	0.1917	0.0155	$0.1703\,[-0.1904, 0.5310]$	3
Viscous OOD	Plain	0.1989	0.7125	0.0793	$0.3302\,[-0.5055, 1.1659]$	3
Viscous OOD	Random Cross	-0.0904	0.1676	-0.0071	$0.0234\,[-0.3038, 0.3506]$	1
Viscous OOD	Duplicate Self	0.4304	0.8332	0.0499	$0.4378\,[-0.5351, 1.4108]$	3
Inviscid OOD	Plain	0.2188	0.7455	0.1836	$0.3826\,[-0.3993, 1.1646]$	3
Inviscid OOD	Random Cross	-0.0237	0.2427	0.0257	$0.0816\,[-0.2704, 0.4335]$	2
Inviscid OOD	Duplicate Self	0.2938	0.9736	0.0488	$0.4387\,[-0.7515, 1.6290]$	3

Table 20: **Paired CoDA contrasts at code-50.** Differences are comparator minus Matched Cross in units of 10^{-3}; positive favors Matched Cross. Intervals are paired-source 95% t intervals ($n = 3$); n_+ counts positive source differences.

Distribution	Comparator	Seed 1	Seed 2	Seed 3	Mean [95% CI]	n_+
ID	Plain	0.0046	-0.0930	-0.0446	$-0.0443\,[-0.1655, 0.0768]$	1
ID	Random Cross	-0.0361	-0.0165	-0.0681	$-0.0402\,[-0.1049, 0.0244]$	0
ID	Duplicate Self	0.0232	0.0760	0.0006	$0.0333\,[-0.0629, 0.1294]$	3
Viscous OOD	Plain	0.1935	0.2277	-0.1076	$0.1045\,[-0.3537, 0.5628]$	2
Viscous OOD	Random Cross	-0.0976	0.1018	-0.1646	$-0.0534\,[-0.3977, 0.2908]$	1
Viscous OOD	Duplicate Self	0.3286	0.5325	0.0882	$0.3165\,[-0.2360, 0.8689]$	3
Inviscid OOD	Plain	0.0396	0.0626	-0.2347	$-0.0442\,[-0.4551, 0.3668]$	2
Inviscid OOD	Random Cross	-0.0369	0.1850	-0.1656	$-0.0059\,[-0.4464, 0.4347]$	1
Inviscid OOD	Duplicate Self	0.0376	0.8296	-0.0308	$0.2788\,[-0.9092, 1.4669]$	2

Table 21: **CoDA viscosity accessibility and decoder invariance.** Probe R^2 is mean $\pm$ sample SD across three sources on 45 ID histories. Decoder change is the maximum absolute change during encoder fitting, over all three decoders; all remain unchanged.

Arm	Code-0 viscosity R^2	Code-50 viscosity R^2	Max. decoder change
Plain	0.1325 ± 0.1388	0.2765 ± 0.2184	0
Matched Cross	$\mathbf{0.1752} \pm 0.1545$	$\mathbf{0.3010} \pm 0.1946$	0
Random Cross	0.1291 ± 0.1324	0.2845 ± 0.1945	0
Duplicate Self	0.1527 ± 0.1604	0.2878 ± 0.2334	0

Table 22: **CoDA fitting and evaluation costs.** Times are seconds, averaged over three sources: per fit for training and per query for evaluation. Code-0 includes encoder inference and forecasting; Code-50 rows exclude encoder inference. Evaluation times are amortized execution costs, excluding data transfer, decoder construction and scoring. Shared decoder and teacher costs are reported separately.

Quantity	Plain	Matched Cross	Random Cross	Duplicate Self
Trainable encoder parameters	89,282	89,282	89,282	89,282
Encoder fitting time	1268	2489	2518	2519
Code-0 encoder + forecast	0.03416	0.03500	0.03510	0.03353
Code-50 adaptation time	6.84	6.75	6.57	6.47
Code-50 forecasting time	0.03490	0.03604	0.03407	0.03370
Shared decoder fitting time			5372	
Shared teacher-code adaptation			2286	
Shared teacher-fit time			1.44	

D SpringWorld Case Study

This section collects the questions needed to interpret the SpringWorld instance: what history alone provides, how Align and Cross contribute, and what training resources and objective weights accompany those effects. The existing comparisons use the visual input and training protocol in Appendix B.4. Development component and sealed prediction results use separate populations.

D.1 History-conditioned baselines

Native and Structure both receive history. Introducing the separate channel alone does not reproduce the joint method's benefit: the development component means are 0.170 for Native, 0.172 for Structure and 0.110 for Align + Cross (Table 34). Appendix C.1 gives the primary comparison against Native and trajectory-decoupled prediction on the separate sealed systems.

D.2 Contributions of Align and Cross

Align accounts for most of the observed organization and task improvement in the component study. The joint recipe lowers development error to 0.110, compared with 0.123 for Align and 0.168 for Cross; its source-wise between/within ratio is 2.360, compared with 2.045 and 0.252, respectively. Table 34 gives the full component comparison and separate task and geometry populations.

D.3 Computational cost

Both sources use 10,000 optimizer updates and their frozen-context readers use 10,000 updates. Align and Cross are added within source updates; this SpringWorld instance has no continuation stage. Table 23 separates optimizer budgets, parameter counts and measured running time.

Table 23: **SpringWorld training cost per fit.** Update times use a common RTX 6000D benchmark. Full-fit times average three original runs under different hardware and runtime conditions and are not directly comparable to the update benchmark.

Quantity	Native	Align + Cross
Source optimizer updates	10,000	10,000
Additional continuation updates	0	0
Reader updates per fitted reader	10,000	10,000
Trainable source parameters	4,509,120	4,525,632
Compute-only time per source update (s)	0.1756	0.1948
Original full-fit wall time (s)	34,270	34,324

D.4 Budgets and configuration selection

Table 24 separates the source budget, reader budget and selection rule. Three fitted readers on one source are not three independent source runs. The initial comparisons fix the relation weights at $\lambda_{\mathrm{p}} = 1$ and $\lambda_{\mathrm{x}} = 0.1$; the sensitivity study below is a separate development analysis.

Table 24: **SpringWorld budgets and configuration selection.** Updates are per fit.

Comparison	Source	Reader	Configuration and selection
Native / Structure / Align / Cross / joint	10,000	10,000	Fixed component weights; final source step. Component table uses reader 0.
Sealed Native / TDS / joint	10,000	10,000	Three sources $\times$ three readers; sources and readers frozen before the primary test.
Full / compact TDS	10,000	10,000	$d \in \{0, \ldots, 6\}$ for compact TDS; separate selection-development bank.
Temporal FCRL-style adaptation	10,000	10,000	Fixed temperature 0.07; three sources $\times$ three common-adapter readers.

The TDS and FCRL-style inputs and matched-reader restrictions are specified in Appendix C.3. Configuration counts include only the distinct recipes used in the reported comparisons and their declared selection stages, excluding seed replicas. Native, Structure, Align, Cross, the joint recipe, full TDS and temporal FCRL each use one fixed source configuration. Compact TDS declares seven candidate dimensions, $d \in \{0, \ldots, 6\}$, for its selection stage.

D.5 Sensitivity to relation weights

The one-at-a-time development analysis varies $\lambda_{\mathrm{p}} \in \{0.25, 1, 4\}$ at $\lambda_{\mathrm{x}} = 0.1$, and $\lambda_{\mathrm{x}} \in \{0.025, 0.1, 0.4\}$ at $\lambda_{\mathrm{p}} = 1$. The shared default gives five distinct settings, with three source seeds per setting. Architecture, data, update count and reader protocol are fixed. Prediction and training-fitted factor probes use the development population; the sealed primary test is excluded from configuration selection. All fifteen sources and forty-five readers are included. The prediction measure is cold-start $q = 0$, $H = 16$ error on 100 development systems (600 cases); the probes are evaluated on 178 development systems.

Table 25: **Sensitivity to Align and Cross weights.** Mean ± SD over three sources; MSE first averages three readers per source. Probes predict $\log m$, $\log \gamma$ and $\log k$.

λ_{p}	λ_{x}	**Dev. MSE**	**Mass** R^2	**Drag** R^2	**Stiffness** R^2
0.25	0.1	0.1203 ± 0.0025	0.2837 ± 0.0494	0.1441 ± 0.1210	-0.0204 ± 0.0218
1	0.1	0.1100 ± 0.0081	0.3630 ± 0.0408	0.1598 ± 0.0220	0.1430 ± 0.0284
4	0.1	0.1609 ± 0.0360	0.3498 ± 0.0989	0.2694 ± 0.1131	0.3166 ± 0.2892
1	0.025	0.1049 ± 0.0112	0.4162 ± 0.0846	0.1645 ± 0.0447	0.3578 ± 0.2626
1	0.4	0.1274 ± 0.0027	0.3751 ± 0.0322	0.1331 ± 0.0277	0.2403 ± 0.1694

Relative to the default $(1, 0.1)$, changing Align to 0.25 or 4, or Cross to 0.4, increases mean cold-start error by 9.38%, 46.23% and 15.83%, respectively, with the same direction in all three sources. Reducing Cross to 0.025 lowers the mean by 4.69%, with lower error in two of three sources. Probe scores and prediction error do not vary in lockstep: for example, $\lambda_{\mathrm{p}} = 4$ raises mean drag and stiffness R^2 while increasing prediction error.

D.6 Gradient relationships

Every 100 optimizer steps, the diagnostic computes the cosine between $\nabla \mathcal{L}_{\mathrm{align}}$ and $\nabla \mathcal{L}_{\mathrm{cross}}$ on the shared persistent encoder g_p, using the same diagnostic batch and parameter subset. It records both gradient norms and zero-norm cases without altering the training state. Cosine summaries exclude zero-norm observations and retain negative values. Early, middle and late phases are the three equal thirds of training. These local gradient measurements describe objective interaction; task performance is measured separately in Table 25.

Table 26: **Distribution of Align–Cross gradient cosines.** Quantiles and negative frequencies use valid observations pooled across three sources; zero-norm frequencies use all observations.

$(\lambda_{\mathrm{p}}, \lambda_{\mathrm{x}})$	**Valid** n	**Mean**	**Q1**	**Median**	**Q3**	**Negative (%)**	**Zero norm (%)**
$(0.25, 0.1)$	300	0.0263	−0.0152	0.0177	0.0720	35.0	0.0
$(1, 0.1)$	300	0.0214	−0.0387	0.0181	0.0770	40.7	0.0
$(4, 0.1)$	300	0.0246	−0.0355	0.0173	0.0842	40.3	0.0
$(1, 0.025)$	300	0.0308	−0.0194	0.0214	0.0765	34.3	0.0
$(1, 0.4)$	300	0.0379	−0.0383	0.0317	0.1028	37.3	0.0

Mean gradient cosines range from 0.0214 to 0.0379, while 34.3–40.7% of observations are negative. No diagnostic observation has a zero gradient norm. Mean gradient cosines remain close to zero across source seeds and early, middle, and late training phases. Thus the small positive mean does not imply uniformly aligned objectives.

Relation-reliability effects are examined directly in Appendix E.1 and analytically in Appendix I.2.

E FORMATION: RELATION DESIGN AND REPRESENTATION ORGANIZATION

PokeWorld, SpringWorld and Baxter test relation design, objective components and tactile selectivity. Fixed-route Use is measured separately in Appendix F.

E.1 RELATION RELIABILITY, SHARED FACTORS, AND OBJECTIVES: POKEWORLD

Takeaway. More reliable relations strengthen organization; changing the pairing rule redirects frozen-code accessibility.

The experiments vary pair reliability at fixed R8, shared factors and objective composition. Within each study, sources 0/1/2 share architecture and budget; checkpoints and endpoints are fixed after validation and evaluated on 400 confirmation systems. Figure 11 shows the two formation maps.

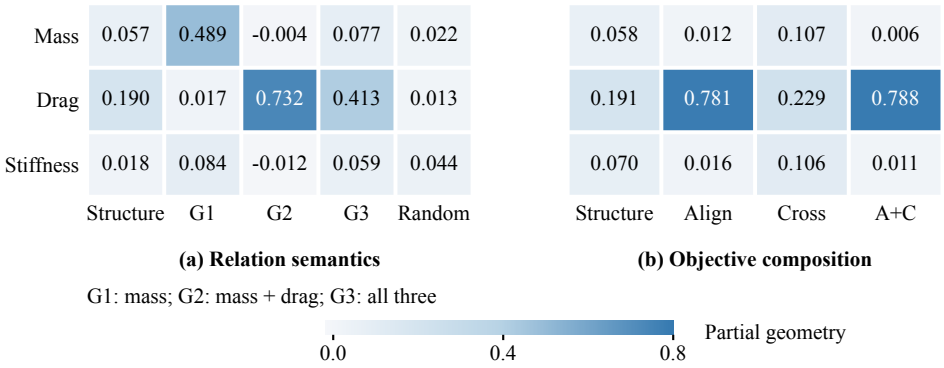


Figure 11: **Relation semantics and objective composition organize different factors.** Mean partial geometry across three sources on 400 confirmation systems. (a) Shared factors $G_1 = m$, $G_2 = (m, \gamma)$ and $G_3 = (m, \gamma, k)$; (b) objective components. Blocks use separate Structure references and a common color scale. Source SDs are in Tables 31 and 32.

Input evidence and interaction diversity. Rendered histories support factor readout, with higher accuracy for drag than for mass and stiffness. A separate privileged-dynamics estimator quantifies recoverability when numerical state is available. Replacing eligible same-rollout donors with independent rollouts raises drag readout from 0.2647 to 0.3469 while native h16 self loss changes from 0.3641 to 0.3869. R2/R4/R8 vary history diversity using two/four/eight independent rollouts per training system before relation randomization; α varies pair quality at fixed R8.

Table 27: **Factor readout from rendered histories and privileged dynamics.** Rendered-history entries show means and source SD.

Input readout (R^2)	Mass	Drag	Stiffness
Rendered history	0.270 ± 0.026	0.574 ± 0.009	0.149 ± 0.034
Privileged dynamics	0.762	0.937	0.760

Shared factors and frozen accessibility. Changing from G_1 to G_2 shifts accessibility from mass toward drag. Training-only ridge and MLP probes of frozen code centroids both favor G_1 for mass and G_2 for drag in every source replica on 400 validation systems. The spectral construction in Figure 26 shows how a wider sharing relation can change the selected predictive direction without lowering the original factor's score.

Table 28: **Linear and nonlinear readouts show the G_1/G_2 accessibility shift.** Validation probes of frozen centroids. Error contrasts ($G_2 - G_1$), paired-system 95% intervals; positive favors G_1.

Probe	Target	G_1 R^2	G_2 R^2	MSE contrast	95% interval
Ridge	$\log m$	0.5271	0.0709	0.1492	$[0.1231, 0.1758]$
Ridge	γ	0.3075	0.9249	-0.7704	$[-0.8572, -0.6869]$
MLP	$\log m$	0.5272	0.0273	0.1635	$[0.1303, 0.1974]$
MLP	γ	0.1291	0.9644	-1.042	$[-1.143, -0.9438]$

Visualizing the frozen representation. Figure 12 displays all three replicas under both relations. Its factor coloring complements the probes; apparent spacing or orientation across independently projected sources is not a quantitative comparison.

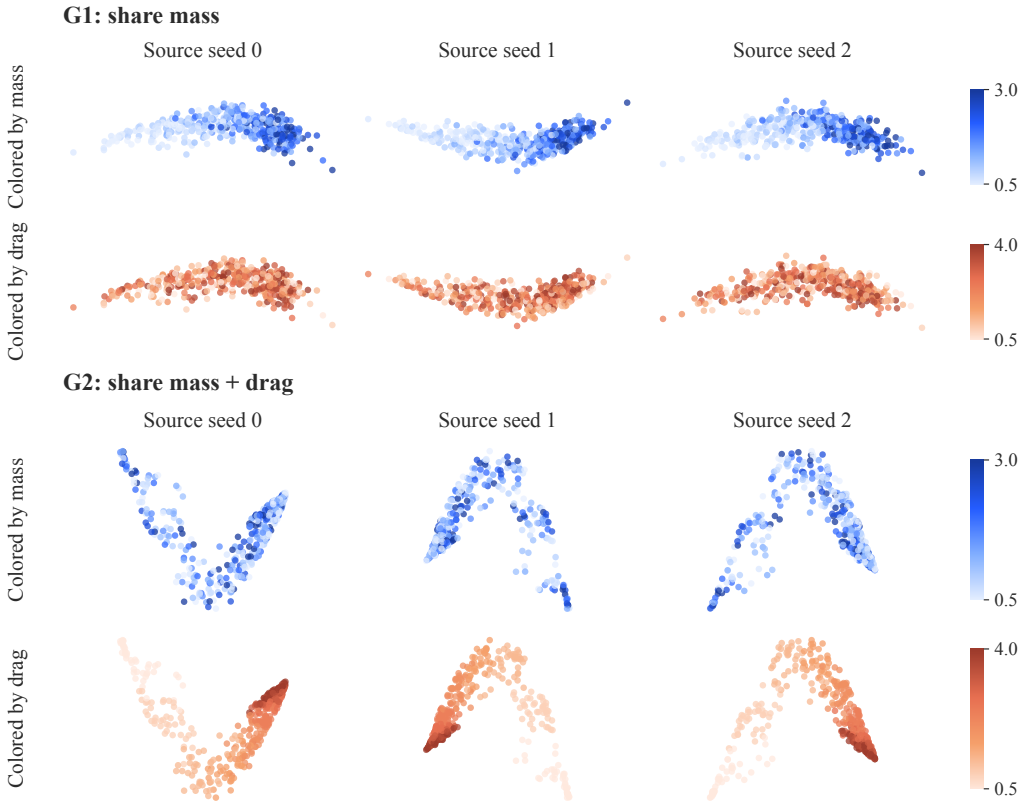


Figure 12: **Persistent-context atlas under G_1 and G_2.** Each panel shows the same 400 validation systems. Each frozen source has an independent PCA projection; mass and drag rows recolor identical coordinates using common physical-value ranges. Axes are not aligned across sources. The atlas is illustrative; quantitative comparisons use the geometry and probes in Appendix E.1.

Persistent-context atlas. The atlas uses the same six frozen step-20,000 G_1/G_2 checkpoints and 400-system validation bank as the factor probes. Using the fixed rollout/anchor set, rollout-averaged codes are standardized with training statistics before forming system centroids; each source then has an independent PCA projection. Physical labels and confirmation/test data are not used to construct the projection. Because axes are not aligned across sources, quantitative claims rely on the probes and geometry rather than visual orientation.

Objective controls and native references. The factorial keeps the two-view batch, predictor, dimensions and budget fixed. The relational InfoNCE alternative uses normalized persistent codes, one positive and 94 negatives per anchor, with all three sources in the $\tau \in \{0.07, 0.10, 0.20\} \times \lambda \in$

$\{0.1, 1\}$ validation search. Confirmation geometry is 0.730 ± 0.019 and ratio 1.736 ± 0.104. Native embedding losses retain each learner's own target space. Table 32 gives the full objective factorial.

The acquisition studies use separate D-Clean and PokeWorld banks: Poke's raw 24-step drag estimator scores 0.921, versus 0.320 ± 0.025 for monolithic JEPA and 0.481 ± 0.092 for Align + Cross, with 7.1% lower h16 object-state error on the original acquisition bank.

Actual-input versus privileged input estimates remain in Table 27; D-Clean acquisition uses its own one-shot sealed population. Native self prediction has no donor route.

Table 29: **D-Clean results on 200 one-shot sealed-test systems.** Means and sample SD over three sources. Errors use each source's own embedding target. Correct uses a same-system donor from another rollout; Gap is shuffled-minus-correct error. Native has no donor route.

Method	Drag $R^2 \uparrow$	Self h16 $\downarrow$	Correct $\downarrow$	Gap $\uparrow$
Native	0.9932 ± 0.0007	0.0789 ± 0.0356	n.a.	n.a.
Structure	0.9935 ± 0.0007	0.0736 ± 0.0404	0.5512 ± 0.0681	0.0797 ± 0.0079
Same-rollout alignment	0.4434 ± 0.0367	0.0520 ± 0.0078	0.0542 ± 0.0090	0.00635 ± 0.00104
Align	0.9984 ± 0.0001	0.0140 ± 0.0040	0.0141 ± 0.0041	0.0948 ± 0.0037
Align + Cross	0.9986 ± 0.0002	0.0155 ± 0.0073	0.0155 ± 0.0072	0.0941 ± 0.0031
Align–Random	0.5299 ± 0.0400	0.0563 ± 0.0100	0.0591 ± 0.0100	0.00034 ± 0.00009
Supervised γ	0.9985 ± 0.0002	0.0119 ± 0.0024	0.0124 ± 0.0024	0.0969 ± 0.0044

Table 30: **PokeWorld geometry under history diversity and relation fidelity.** Means and source SD over three seeds. A: validation geometry with train-standardized codes; partial drag correlation controls mass and stiffness. B: validation and frozen-confirmation results.

A. Independent-history budget

Relation	Budget	Within $\downarrow$	Between $\uparrow$	Ratio $\uparrow$	Partial $\rho_\gamma \uparrow$
Correct	R2	78.35 ± 4.67	29.38 ± 1.78	0.375 ± 0.017	0.095 ± 0.022
Correct	R4	55.66 ± 5.17	43.32 ± 2.82	0.781 ± 0.062	0.361 ± 0.036
Correct	R8	28.09 ± 1.93	104.30 ± 2.01	3.729 ± 0.313	0.773 ± 0.009
Random	R2	70.80 ± 8.74	17.28 ± 2.03	0.244 ± 0.002	-0.001 ± 0.017
Random	R4	83.68 ± 16.91	20.39 ± 3.64	0.245 ± 0.008	-0.010 ± 0.016
Random	R8	126.90 ± 0.89	34.18 ± 0.39	0.269 ± 0.004	-0.006 ± 0.004

B. Relation fidelity

	Validation		Confirmation	
α	Ratio $\uparrow$	Drag $\uparrow$	Ratio $\uparrow$	Drag $\uparrow$
0	0.272 ± 0.006	0.061 ± 0.031	0.260 ± 0.006	0.031 ± 0.026
0.5	0.976 ± 0.051	0.243 ± 0.056	0.880 ± 0.047	0.264 ± 0.063
1	3.749 ± 0.058	0.781 ± 0.027	4.391 ± 0.156	0.796 ± 0.023

Table 31: **Factor-selective geometry under relation refinement.** Means and sample SD over three sources. Exact factor tuples are split-disjoint; individual factor levels are shared.

	Validation			Confirmation		
Condition	Mass	Drag	Stiffness	Mass	Drag	Stiffness
Structure	0.069 ± 0.008	0.194 ± 0.007	0.037 ± 0.012	0.057 ± 0.010	0.190 ± 0.002	0.018 ± 0.008
G_1	0.419 ± 0.006	0.013 ± 0.009	0.103 ± 0.004	0.489 ± 0.009	0.017 ± 0.006	0.084 ± 0.003
G_2	0.015 ± 0.000	0.720 ± 0.009	-0.013 ± 0.001	-0.004 ± 0.001	0.732 ± 0.012	-0.012 ± 0.001
G_3	0.064 ± 0.013	0.427 ± 0.087	0.067 ± 0.035	0.077 ± 0.035	0.413 ± 0.082	0.059 ± 0.023
Random	0.030 ± 0.007	0.036 ± 0.014	0.054 ± 0.008	0.022 ± 0.009	0.013 ± 0.007	0.044 ± 0.006

Donor support and candidate composition. G_1 and G_2 use the same frozen donor-bank file. Each recipient has three candidate donor systems and four rollouts per candidate, giving twelve supported donor histories under either relation. At each update, the sampler selects one of the three systems

Table 32: **Objective components on 400 confirmation systems.** Means and sample SD over three sources. Factor columns give partial geometry; Ratio is between/within separation. Native h16 errors use each source's own embedding target.

Condition	Mass ↑	Drag ↑	Stiff. ↑	Ratio ↑	h16 ↓
Structure	0.058 ± 0.005	0.191 ± 0.005	0.070 ± 0.011	0.299 ± 0.006	0.404 ± 0.033
Align	0.012 ± 0.009	0.781 ± 0.004	0.016 ± 0.005	4.203 ± 0.289	0.386 ± 0.019
Cross	0.107 ± 0.009	0.229 ± 0.009	0.106 ± 0.002	0.362 ± 0.009	0.373 ± 0.005
Align + Cross	0.006 ± 0.004	0.788 ± 0.001	0.011 ± 0.002	4.411 ± 0.142	0.381 ± 0.010

uniformly, then selects one of its four rollouts uniformly. This fixed training support is distinct from the full set of semantically eligible systems (Table 33).

G_1 candidate construction uses fixed drag-index offsets $(1, 3, 7)$ and a different-stiffness system at each offset. G_2 selects three candidates from the same-mass, same-drag set with different stiffness. The observed G_1/G_2 accessibility contrast therefore compares equal-sized training supports with different candidate composition and preselection. A literal reduction of G_1 to G_2's training support size leaves three candidates unchanged. Candidate composition and donor-exposure frequencies follow the pairing rule; equal support does not separate them from shared-factor semantics.

Table 33: **Semantic eligibility and the fixed training support.** Counts are per recipient. Eligible systems satisfy the relation; training uses donors from only three preselected systems.

Split	Relation	Eligible systems	Selected systems	Rollouts/system	Supported histories
Training	G_1	173–174	3	4	12
Training	G_2	19	3	4	12
Validation	G_1	35–36	3	4	12
Validation	G_2	3	3	4	12

E.2 Objective components and reference learners: SpringWorld

Alignment-containing recipes improve cold-start prediction and cross-history organization. Task error uses one separately fitted reader per source recipe (reader seed 0) on 100 systems and 600 cases. Geometry uses frozen P_{64} before reader fitting on 178 systems with four histories each.

Align accounts for most of the original cold-start gain. Adding Cross to Align lowers error in all three sources, by 23.04%, 8.27% and 0.10%. Geometry uses train-standardized codes and partial Spearman factor-distance correlations, controlling the other two factors (k/m controls drag). Larger ratios indicate greater between-system separation relative to within-system variability.

Supervised reference encoders use factor-related supervision and serve as references rather than competitors. The question for the predictive learners is whether they organize this information with relation assignments but without regressing the numerical factor labels.

Supervised reference encoders. Table 35 compares GRU, Transformer, TCN and DeepSets using one source and reader per recipe. The supervised TCN has the lowest development error in this reference grid. The three-source objective comparison is in Table 34.

Table 34 reports task error and source geometry for the two cohorts.

The pooled Cross addition to Align is 10.37%; adding Align to Cross gives 34.10%. Align + Cross has greater source separation than Structure in all three geometry replicas.

Table 34: **Align accounts for most of the SpringWorld gain; Cross lowers error further.** Three-source means, separate cohorts: A, cold-start error on 100 development systems (one reader per source); B, frozen-source geometry on 178 systems. Ratio averages source-wise between/within ratios.

A. Three-source component mean (reader 0)

	Native	Structure	Align	Cross	Align + Cross
Mean	0.1703	0.1716	0.1232	0.1675	0.1104

B. Frozen-source geometry

Recipe	Within	Between	Ratio	ρ_m	ρ_γ	ρ_k	$\rho_{k/m}$
Structure	131.6	31.64	0.240	0.001	0.005	-0.014	-0.025
Align	47.25	90.9	2.045	0.361	0.224	0.132	0.414
Cross	115.7	29.22	0.252	0.200	0.184	0.056	0.066
Align + Cross	40.58	93.63	2.360	0.354	0.193	0.141	0.421

Table 35: **SpringWorld development references and physical readout.** A: single-source standardized MSE on 100 cold-start and 178 query-mixture systems; supervised references use their own training recipes. B: training-fitted ridge R^2 for log factors on 100 validation systems. These are separate from the sealed and multi-reader evaluations.

A. Single-source reference (source 0, reader 0)

Source	Cold start	Query mixture
Controlled JEPA objectives		
Native	0.1716	0.8511
Structure	0.1711	0.8468
Align	0.1195	0.8033
Cross	0.1646	0.8443
Align + Cross	**0.0920**	**0.7892**
Other reference learners		
CPC	0.2218	0.9858
RSSM	0.1653	0.7882
Supervised split	0.1841	0.8864
Supervised reference encoders		
GRU	0.0934	0.6328
Transformer	0.0887	0.6631
TCN	0.0593	0.5948
DeepSets	0.0892	0.6597

B. Frozen-source physical probes

Source	m	γ	k	k/m
Native	0.0076	-0.0234	-0.0265	-0.0085
Align + Cross	0.3806	0.1529	0.2023	0.4945

E.3 Relation-selective tactile organization: Baxter

Hardness and shape relations organize real tactile histories selectively. The bank crosses cubes/cylinders with three hardness levels; confirmation holds out grasps from known configurations.

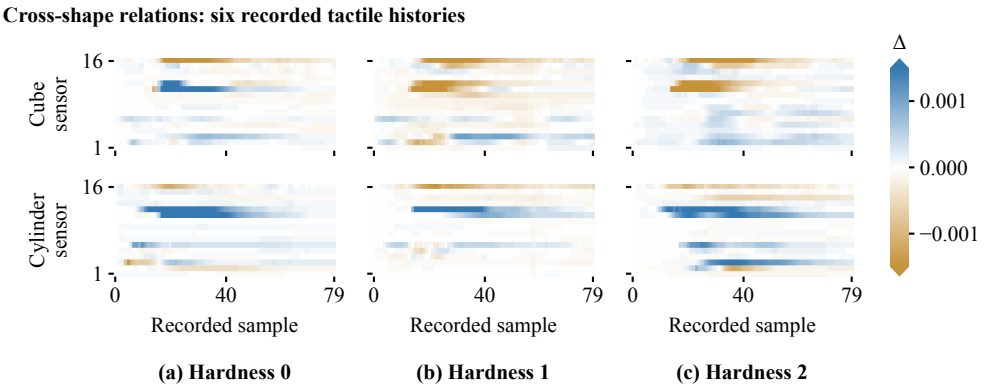


Figure 13: **Shared hardness connects tactile histories across object shapes.** Columns share hardness levels 0–2; rows differ in shape. Sensor changes are relative to the initial sample, with a common symmetric scale saturated at the pooled 97th percentile of absolute change.

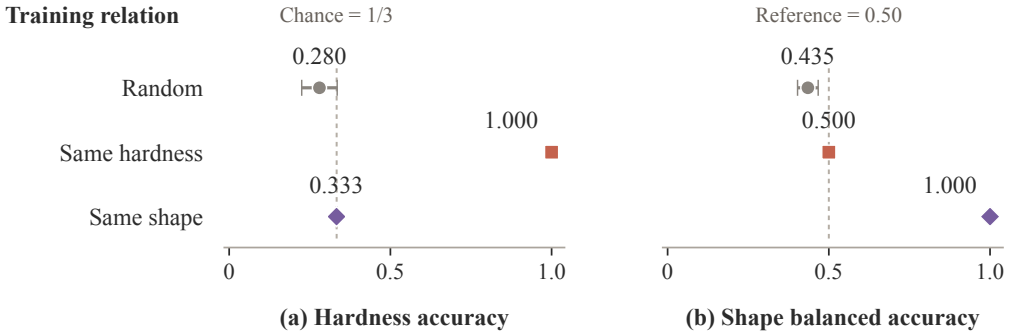


Figure 14: **Relation choice selects which tactile property becomes accessible.** Means and sample SD over three sources on held-out grasps from six known configurations. (a) Cross-shape hardness accuracy; (b) shape balanced accuracy with one hardness level held out. Dashed lines mark the respective chance references, $1/3$ and 0.50.

The input recoverability test reaches hardness accuracy 0.954 ± 0.003 using eight samples and 1.000 ± 0.000 using the full peak; shape is already perfect at both lengths. Thus, the input already supports both classifications.

The key comparison is between the two relation-trained models: hardness accuracy is 1.000 under $\mathcal{R}_H$ versus 0.333 under $\mathcal{R}_S$; shape balanced accuracy is 1.000 under $\mathcal{R}_S$ versus 0.500 under $\mathcal{R}_H$. The corresponding contrasts $C_H = 0.667$ and $C_S = 0.500$ are positive in every source on validation and confirmation. This reversal shows that the pairing rule selects which property is accessible within the six known configurations; Random provides an additional unstructured-pairing reference.

E.4 Structured relation errors

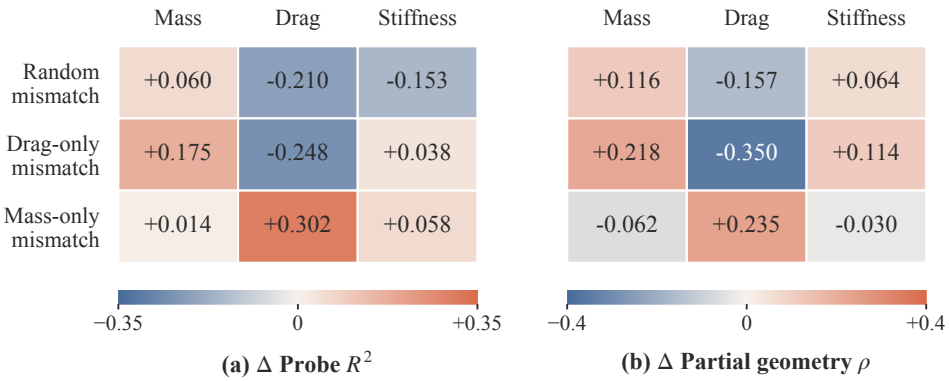
This three-source PokeWorld comparison separates correct-pair frequency from the structure of pairing errors. The conditions are correct pairing; random mismatch with $\alpha = 0.5$; drag-only mismatch with $\alpha = 0.5$; and mass-only mismatch with $\alpha = 0.5$. Structured errors preserve the other two physical factors. All four conditions use a common training bank with legal exact single-factor mismatches, and share initialization, recipient draws, correct/error masks, source budgets and evaluation protocol within seed. All twelve Align-only sources ($\lambda_p = 1$, $\lambda_x = 0$) were trained from

scratch for 20,000 updates on this distinct 2,000-system bank, with source seeds 0–2. Probe fitting uses 200 training systems, and evaluation uses 200 common validation systems with four rollouts and four history anchors each. All results below are development-only. The finite-bank mismatch kernel selects a distinct legal donor; it differs from independent factor resampling in Appendix I.2.

Partial geometry and training-fitted probe R^2 quantify each factor's organization and accessibility. The between/within ratio measures system separation. Differences from correct pairing are computed within source. The corollary characterizes alignment pressure for a fixed representation; the factor-specific response of a trained encoder is measured by these comparisons.

Table 36: **Factor measurements under structured relation errors.** Means and sample SD across three sources. Probe targets and partial correlations follow Appendix E.1's PokeWorld protocol.

Pairing	**Factor**	**Probe R^2**	**Partial geometry ρ**
Correct	Mass	0.1630 ± 0.0138	0.0796 ± 0.0105
Correct	Drag	0.6087 ± 0.0881	0.4665 ± 0.0461
Correct	Stiffness	0.0589 ± 0.0432	0.0168 ± 0.0186
Random mismatch	Mass	0.2229 ± 0.0274	0.1954 ± 0.0013
Random mismatch	Drag	0.3987 ± 0.0305	0.3092 ± 0.0043
Random mismatch	Stiffness	-0.0938 ± 0.1124	0.0806 ± 0.0024
Drag-only mismatch	Mass	0.3377 ± 0.0457	0.2978 ± 0.0152
Drag-only mismatch	Drag	0.3608 ± 0.0682	0.1162 ± 0.0224
Drag-only mismatch	Stiffness	0.0970 ± 0.0451	0.1312 ± 0.0320
Mass-only mismatch	Mass	0.1766 ± 0.1025	0.0180 ± 0.0064
Mass-only mismatch	Drag	0.9105 ± 0.0435	0.7013 ± 0.0186
Mass-only mismatch	Stiffness	0.1166 ± 0.0127	-0.0129 ± 0.0026



(a) Δ Probe R^2 **(b) Δ Partial geometry ρ**

Figure 15: **Structured mismatches redistribute factor organization.** Mean paired-source changes from correct pairing in (a) probe R^2 and (b) partial geometry ρ; positive indicates an increase. All mismatch rules use $\alpha = 0.5$ and three Align-only sources on 200 development validation systems. Mass-only mismatch lowers mass geometry but not its mean probe score (source SDs: Table 36).

Drag-only mismatch changes drag geometry from 0.4665 to 0.1162 and mass geometry from 0.0796 to 0.2978. Mass-only mismatch changes mass geometry to 0.0180 and drag geometry to 0.7013. All four directions hold in each of the three source seeds. Accessibility is less uniform: mass-only mismatch does not lower mass probe R^2 (0.1766 versus 0.1630 for correct pairing). These results support factor-directed organization under structured errors. Correct pairing does not minimize native latent MSE at any evaluated horizon; the prediction panel reports each frozen encoder's own target space and scale, and does not measure common physical-state error or control return.

Table 37: **Pairing quality, system separation and native prediction.** Coverage is the fraction of recipients with a legal donor; factor columns give match rates, and separation is the between/within ratio. Native latent MSE uses each encoder's own target space, with target variance in brackets. Errors and ratios show means and sample SD across three sources.

Pairing	Coverage (%)	Mass (%)	Drag (%)	Stiffness (%)	Separation
Correct	100	100	100	100	1.0601 ± 0.1093
Random mismatch	100	54.97	54.99	51.76	0.9694 ± 0.0236
Drag-only mismatch	100	100	50	100	1.1618 ± 0.1712
Mass-only mismatch	100	50	100	100	3.8995 ± 1.3415

Pairing	$H = 1$	$H = 4$	$H = 16$
Correct	0.0719 ± 0.0071 [0.6226]	0.1103 ± 0.0054 [0.6267]	0.2312 ± 0.0040 [0.6285]
Random mismatch	0.0108 ± 0.0008 [0.7416]	0.0506 ± 0.0010 [0.7439]	0.2126 ± 0.0015 [0.7457]
Drag-only mismatch	0.0265 ± 0.0034 [0.7061]	0.0688 ± 0.0045 [0.7075]	0.2216 ± 0.0038 [0.7083]
Mass-only mismatch	0.0223 ± 0.0004 [0.6841]	0.0675 ± 0.0043 [0.6885]	0.2157 ± 0.0041 [0.6966]

E.5 Relations inferred from learned histories

A Structure teacher trained without relation supervision encodes one 24-frame history per training rollout. Each source seed defines a frozen graph over 8,000 rollouts from the 2,000-system bank. After per-coordinate standardization on training codes, each rollout receives three Euclidean nearest neighbors, excluding itself. Physical labels and system grouping are not used to construct or select the graph. RandomHistory draws three neighbors from the same candidate universe. The graphs are fixed before source training; physical labels are used only for subsequent diagnostics.

Inferred and RandomHistory each use three newly trained Align-only sources ($\lambda_p = 1$, $\lambda_x = 0$), with 20,000 updates per source. True reuses three G_3 sources from the same bank with the same source hyperparameters.[1] All nine models use the common evaluation protocol: probes are fitted on 200 training systems, and evaluation uses 200 validation systems with 3,200 windows per model. No sealed data enter this analysis. The relation-fidelity curve in Appendix E.1 uses a different bank and serves as a descriptive reference, not an interpolation rule for these graphs.

Table 38: **Agreement and coverage of history-inferred relations.** Match rates and coverage are percentages, averaged over three source-specific graphs. Donor usage is donor-rollout entropy normalized by $\log 8000$. True uses exact same-system relations.

Relation	Tuple p	Mass	Drag	Stiffness	Coverage	Donor usage
True	100	100	100	100	100	1.000
Inferred	0.0639	11.15	13.30	3.64	100	0.974
RandomHistory	0.0431	10.04	10.01	3.50	100	0.980

For a shared-space score T, recovery denotes the fraction of the True–RandomHistory gap attained by Inferred: $(T_{\text{Inferred}} - T_{\text{RandomHistory}})/(T_{\text{True}} - T_{\text{RandomHistory}})$. Ratios use source means and are not clipped. We report them only where the True reference exceeds RandomHistory in every paired-source resample. Native latent errors are retained as diagnostics, not common physical-task errors or recovery scores.

Inferred relations modestly enrich physical matches, but recovery relative to the True relation varies across factors and measurements.

[1] The reused G_3 checkpoints and newly trained graph-based sources were trained under different software/hardware environments.

Table 39: **Representation and native prediction with inferred relations.** Three-source means and sample SD; target variance is a mean only. Each source defines its own latent prediction target.

A. Factor accessibility and geometry

Relation	Factor	Probe R^2	Partial geometry ρ
True	Mass	0.1700 ± 0.0154	0.0737 ± 0.0092
True	Drag	0.6556 ± 0.0791	0.4807 ± 0.0877
True	Stiffness	0.0670 ± 0.0020	0.0314 ± 0.0453
Inferred	Mass	0.0538 ± 0.0423	0.0360 ± 0.0069
Inferred	Drag	0.1563 ± 0.0316	0.0061 ± 0.0111
Inferred	Stiffness	0.0314 ± 0.0111	0.0666 ± 0.0028
RandomHistory	Mass	0.0911 ± 0.0318	0.0256 ± 0.0117
RandomHistory	Drag	0.1065 ± 0.0870	0.0342 ± 0.0043
RandomHistory	Stiffness	-0.0235 ± 0.0202	0.0323 ± 0.0255

B. System separation and native prediction

Relation	Between/within	Native $H = 16$	Target variance
True	1.0683 ± 0.2664	0.2327 ± 0.0099	0.6249
Inferred	0.2577 ± 0.0013	0.2357 ± 0.0051	0.7482
RandomHistory	0.2499 ± 0.0117	0.2268 ± 0.0051	0.7436

Table 40: **Inferred-relation changes and recovery relative to True.** Δ is Inferred minus RandomHistory. Recovery intervals are 95% paired-source bootstrap intervals over three seeds; they do not measure cross-task stability. Dashes denote inapplicable ratios.

Measurement	Factor	Δ	Recovery (%)
Probe R^2	Mass	-0.0373	$-47.3\,[-135.5, 1.0]$
Probe R^2	Drag	$+0.0498$	$9.1\,[-2.7, 30.5]$
Probe R^2	Stiffness	$+0.0549$	$60.7\,[55.3, 70.0]$
Geometry ρ	Mass	$+0.0104$	$21.6\,[-27.6, 53.4]$
Geometry ρ	Drag	-0.0281	$-6.3\,[-8.6, -4.2]$
Geometry ρ	Stiffness	$+0.0343$	—[a]
Between/within ratio	—	$+0.0077$	$0.9\,[-0.5, 2.4]$
Native h16 error	—	$+0.0090$	—[b]

[a] The True–RandomHistory stiffness-geometry gap crosses zero under source resampling; no consistent True advantage defines a recovery scale.

[b] The native models use different latent target spaces.

F USE: FIXED-ROUTE CONTEXT INTERVENTIONS

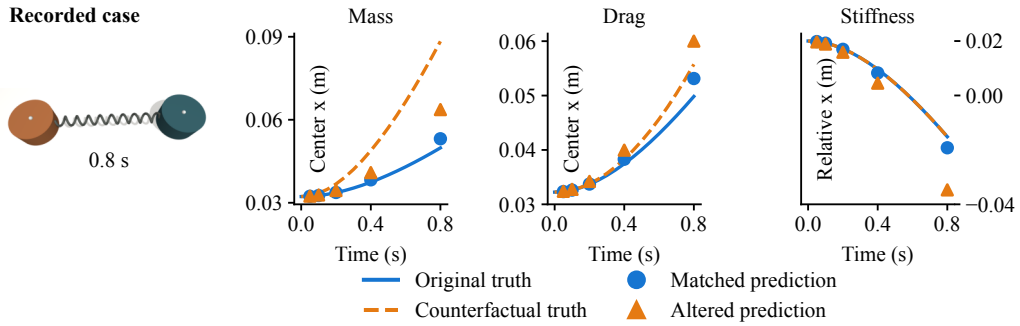
This section expands Figure 4: SpringWorld provides physical donor landscapes, PokeWorld tests factor-specific direction, and RH20T and Balls examine recorded task context and actual reader memory. Swimmer tests recipient-specific updates through frozen predictor weights. Probes measure accessibility; fixed-route interventions measure dependence on supplied context.

F.1 PHYSICAL DONOR LANDSCAPES: SPRINGWORLD

Changing donor physics redirects the fixed reader toward the corresponding target dynamics. The assay uses 64 continuous development systems, three frozen sources and reader seed 0. For each factor, five donor levels and five simulator target levels share initial states and actions; the two other physical parameters remain fixed. A fixed rule selects the query without using model error. All 5×5 grids are complete; h16 is primary.

Valley depth is mean off-diagonal error minus mean diagonal error, averaging 20 and five cells respectively. Models use the original training-only target normalization. Systems then sources receive equal weight; intervals use 10,000 paired-system draws conditional on these fitted routes.

Align + Cross has positive valley depth for all three factors in every source. The balanced factorial main effect of Align, $(\text{Align} - \text{Structure} + \text{Align} + \text{Cross} - \text{Cross})/2$, is 0.1653, 0.02480



(a) Physical substitutions at a fixed recipient state

Recipe	Mass	Drag	Stiffness
Structure	11.79 [5.73, 19.00]	3.10 [1.56, 4.85]	0.16 [-0.14, 0.55]
Align	192.24 [140.24, 247.06]	31.24 [21.98, 41.52]	23.83 [15.86, 33.53]
Cross	31.97 [18.37, 48.07]	8.86 [5.08, 13.44]	0.33 [-0.66, 1.42]
A+C	182.05 [132.00, 234.29]	30.33 [21.58, 39.97]	23.95 [15.63, 33.83]

(b) Population matching advantage: mean [95% CI], MSE × 10^{-3}

Figure 16: **Physical donor substitutions redirect predictions and produce error valleys.** (a) One case selected by the fixed, model-independent rule described above: lines show simulator truth and markers show predictions at horizons 1/2/4/8/16. (b) Valley depths for four separately fitted recipes on 64 systems, with paired-system 95% intervals. Align + Cross uses the pooled intervals in Table 41.

and 0.02365 for mass, drag and stiffness, with respective reported intervals [0.1209,0.2118], [0.01720,0.03286] and [0.01573,0.03306]; the corresponding Cross main effects 0.00499, 0.00243 and 0.00015 have intervals spanning zero. Each contrast compares complete source/reader recipes; donor substitutions within each recipe keep its fitted reader fixed.

Output-coordinate decomposition. The equal-mass center/relative transform decomposes the frozen reader's physical output response. Simulator relative shares are 0.7143/0.7149/1.0000 for mass/drag/stiffness; reader shares span 0.610–0.700/0.650–0.689/0.637–0.696 across sources. The stiffness relative-minus-center statistic is 1.55×10^{-3}, quantifying the distribution of the response across physical output coordinates.

The center/relative transform acts on the position and velocity coordinates $(x_1, y_1, x_2, y_2, v_{x1}, v_{y1}, v_{x2}, v_{y2})$.

Table 41: **Align + Cross yields positive donor-matching valleys in every source.** Depth is off-diagonal minus diagonal h16 error on 64 development systems. Comparisons use three frozen sources and one reader per source. Paired-system 95% intervals condition on fitted models.

Factor	Source	Valley depth	95% interval	ρ
Mass	Pooled	0.1820	[0.1320, 0.2343]	0.591
	Seed 0	0.1950	[0.1379, 0.2558]	0.606
	Seed 1	0.1612	[0.1131, 0.2134]	0.592
	Seed 2	0.1899	[0.1364, 0.2462]	0.601
Drag	Pooled	0.0303	[0.0216, 0.0400]	0.585
	Seed 0	0.0334	[0.0238, 0.0439]	0.600
	Seed 1	0.0335	[0.0229, 0.0458]	0.578
	Seed 2	0.0240	[0.0159, 0.0327]	0.542
Stiffness	Pooled	0.0240	[0.0156, 0.0338]	0.672
	Seed 0	0.0285	[0.0184, 0.0407]	0.776
	Seed 1	0.0229	[0.0143, 0.0328]	0.401
	Seed 2	0.0205	[0.0124, 0.0299]	0.628

Table 41 (continued). Recipe comparison.

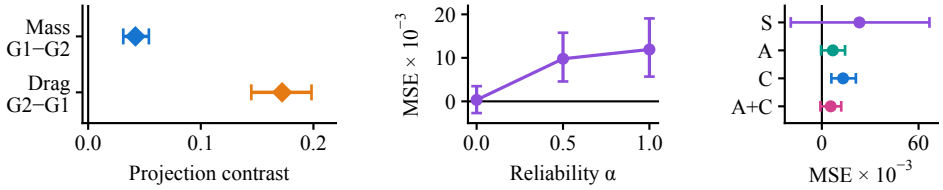
Recipe	Mass: depth [95% CI]	Drag: depth [95% CI]	Stiffness: depth [95% CI]
Structure	0.01179 [0.00573, 0.01900]	0.00310 [0.00156, 0.00485]	0.00016 [−0.00014, 0.00055]
Align	0.1922 [0.1402, 0.2471]	0.03124 [0.02198, 0.04152]	0.02383 [0.01586, 0.03353]
Cross	0.03197 [0.01837, 0.04807]	0.00886 [0.00508, 0.01344]	0.00033 [−0.00066, 0.00142]
Align + Cross	0.182 [0.132, 0.2343]	0.03033 [0.02148, 0.03983]	0.02395 [0.01539, 0.03398]

The landscape figure uses the upper pooled intervals; this panel uses the four-recipe intervals.

F.2 Factor-specific responses and direction: PokeWorld

Takeaway. Shared-factor relations redirect frozen predictive responses toward the corresponding mass or drag counterfactual direction.

Physical-risk and native-embedding assays measure this response in different spaces. The 96-system physical-risk assay in Figure 17 holds the native predictor and training-only ridge state decoder fixed. Matched histories come from an independent recipient interaction; a wrong-factor donor changes exactly one physical parameter. Three deterministically selected windows per system retain current observations, actions, target and h16. All three source seeds use common training-target normalization; Δ_f is wrong-factor minus matched four-coordinate object-state MSE.



(a) Native direction · 144 systems **(b) Physical use · 96 systems**

Relation	Mass	Drag	Stiffness
G1	5.27 [-0.79, 12.63]	3.36 [-1.30, 8.32]	2.84 [-1.60, 8.29]
G2	1.35 [-1.11, 4.30]	2.75 [-5.07, 10.89]	0.23 [-1.33, 1.73]
G3	0.94 [-1.69, 3.66]	7.20 [1.67, 13.54]	2.59 [-0.34, 5.99]

(c) Complete physical-risk contrasts · mean [95% CI] × 10^{-3}

S: Structure; A: Align; C: Cross.

Figure 17: **Shared-factor relations redirect predictive responses.** (a) Mean G_1-minus-G_2 mass and G_2-minus-G_1 drag projections on 144 systems. (b)–(c) Wrong-factor minus matched h16 object-state MSE on a separate 96-system bank. Pointwise 95% intervals resample systems with three sources fixed (and state decoders fixed in (b)–(c)). Yellow marks an interval above zero. Native projections and physical MSE have distinct units.

The principal direction assay evaluates step-20,000 Align G_1/G_2 checkpoints (three sources per relation) on 144 eligible validation systems with three queries each. It uses each source's native 128-dimensional observation embedding without fitting a new encoder, reader or decoder. For factor f and relation G, let $\Delta\hat{z}_{f,s}$ be the wrong-donor minus matched prediction displacement and $\Delta z^{\mathrm{cf}}_{f,s}$ the paired counterfactual target displacement in the native embedding space. We define

$$P(f,G) = \frac{\sum_s \langle \Delta\hat{z}_{f,s}, \Delta z^{\mathrm{cf}}_{f,s} \rangle}{\sum_s \|\Delta z^{\mathrm{cf}}_{f,s}\|_2^2},$$

pooling within a physical system and then giving source seeds equal weight. The contrasts are

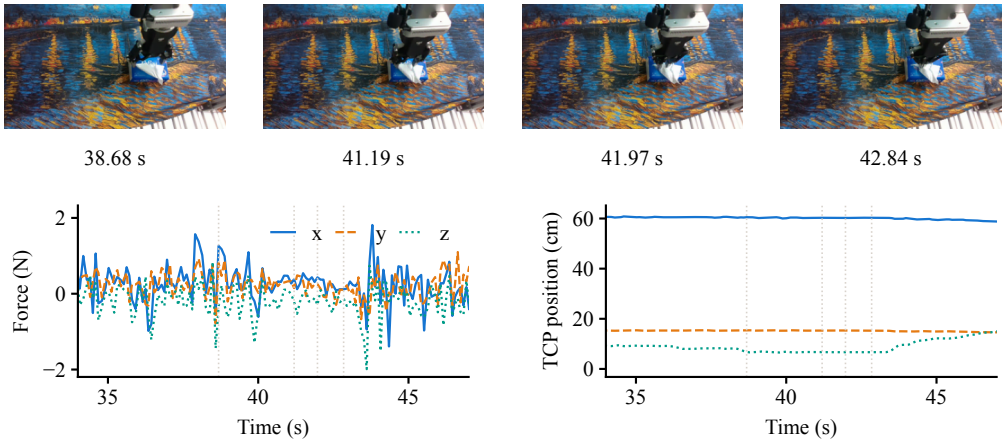
$$\Delta_m = P(m,G_1) - P(m,G_2) = 0.04197, \qquad 95\%\ \mathrm{CI} = [0.03118, 0.05390],$$
$$\Delta_\gamma = P(\gamma,G_2) - P(\gamma,G_1) = 0.1722, \qquad 95\%\ \mathrm{CI} = [0.1448, 0.1983].$$

Reported intervals use paired physical-system bootstrap. Their sum is 0.2142 $[0.1846, 0.2432]$. The already fitted state-decoder version reports the same directions at a smaller scale: $\Delta_m = 0.004986$ and $\Delta_\gamma = 0.02059$. The primary statistic is a dimensionless projection in each model's native 128-D embedding. Both contrasts show relation-specific functional redirection on this development bank; the state-decoder assay expresses that response in state coordinates.

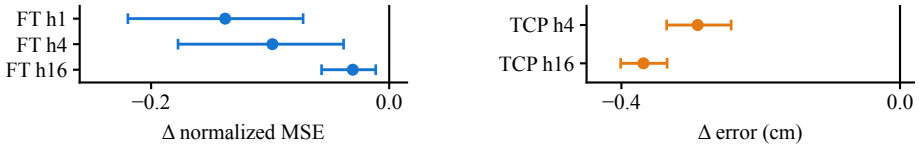
F.3 Task-specific donor use in recorded interactions: RH20T

RH20T relations join different episodes sharing task metadata. Three evaluations test different aspects of the resulting context: a 25-task forecasting cohort compares matched-donor prediction; a 24-task hard-negative bank compares donor specificity and absolute risk; and a separate 25-task validation bank tests action use and donor-count sensitivity.

Matched-donor forecasting. Cross lowers all five endpoint means relative to Structure under matched-donor evaluation. The models jointly predict force/torque and TCP from synchronized visual, state and action histories. Table 42 reports all 14 conditions and three fitted seeds.



(a) Recorded interaction and synchronized measurements



(b) Matched forecasting: Cross − Structure (25 held-out tasks)

Figure 18: **Cross reduces all five matched-donor forecasting endpoint means.** (a) RH20T frames and synchronized force/TCP observations. (b) Cross-Indep minus Structure on 25 held-out tasks; negative favors Cross. Force/torque uses normalized MSE and TCP uses cm. Task-paired 95% intervals use three-source averages. Absolute risks on the separate 24-task bank are in Table 43.

Table 42: **Self-context and matched-donor forecasting on RH20T.** Equal-weight means over 25 held-out tasks and three sources. Force/torque uses normalized MSE; TCP uses cm. LD denotes LowDim; A+C denotes Align + Cross. Dashes indicate unavailable matched inputs. Shading highlights the Structure/Cross comparison.

A. Self-context

	Force/torque MSE			TCP (cm)	
Condition	**h1**	**h4**	**h16**	**h4**	**h16**
Native	0.6346	1.117	1.497	1.962	4.063
Structure	**0.6199**	**1.099**	**1.498**	**1.944**	**3.965**
Align–Indep	0.6633	1.132	1.503	2.262	4.429
Align–Random	0.6232	1.106	1.510	2.045	4.257
Cross–SameEp	0.6145	1.090	1.495	1.963	3.965
Cross–Indep	**0.6230**	**1.093**	**1.497**	**1.901**	**3.902**
Cross–Random	0.6307	1.096	1.495	1.948	3.950
A+C–SameEp	0.6407	1.120	1.500	2.098	4.371
A+C–Indep	0.6602	1.130	1.505	2.268	4.435
A+C–Random	0.6225	1.101	1.507	2.036	4.258
LD–Native	0.5909	1.089	1.489	1.730	3.887
LD–Cross–Indep	0.5760	1.069	1.488	1.598	3.748
LD–A+C–Indep	0.6162	1.118	1.504	2.032	4.302
LD–A+C–Random	0.6090	1.122	1.505	1.866	4.191

B. Matched donor

	Force/torque MSE			TCP (cm)	
Condition	**h1**	**h4**	**h16**	**h4**	**h16**
Native	–	–	–	–	–
Structure	**0.7991**	**1.226**	**1.551**	**2.429**	**4.952**
Align–Indep	0.6667	1.135	1.504	2.536	4.644
Align–Random	0.6255	1.108	1.510	2.062	4.274
Cross–SameEp	0.6709	1.140	1.527	2.254	4.745
Cross–Indep	**0.6612**	**1.128**	**1.521**	**2.139**	**4.584**
Cross–Random	0.6665	1.127	1.516	2.158	4.604
A+C–SameEp	0.6498	1.127	1.506	2.407	4.578
A+C–Indep	0.6633	1.133	1.506	2.573	4.670
A+C–Random	0.6250	1.103	1.507	2.045	4.269
LD–Native	–	–	–	–	–
LD–Cross–Indep	0.6039	1.131	1.531	1.893	4.471
LD–A+C–Indep	0.6187	1.121	1.506	2.078	4.339
LD–A+C–Random	0.6108	1.124	1.506	1.905	4.230

Donor specificity and absolute risk. The 24-task bank contains 10,681 windows with matched, action-metadata-matched/different-task, calibration-matched/different-task and random donor eligibility. Align + Cross and Monolithic Q+D use the same evaluation windows and donor assignments, 48-pair batch, 20,000-step checkpoint and three seeds; parameter counts differ by 0.309%. Define

$$S_f = E_f(\text{hard}) - E_f(\text{matched}), \qquad D_{\text{spec}} = S_{\text{A+C}} - S_{\text{Mono}}. \tag{13}$$

Align + Cross has greater FT donor specificity, while Monolithic has lower matched error. Table 43 reports both comparisons with paired intervals; the TCP matched-error interval favors Monolithic.

Table 43: **Donor specificity and absolute risk on the 24-task bank.** Means weight tasks equally after averaging three sources, with task-paired 95% intervals. Specificity is hard-minus-matched error; contrasts are A+C minus Monolithic. Positive risk differences favor Monolithic. FT h4 is primary; TCP h16 is secondary.

A. Donor specificity

Endpoint	$S_{\mathrm{A+C}}$	S_{Mono}	D_{spec}
FT h4 MSE	0.00260 [0.00107, 0.00429]	0.00030 [−0.00050, 0.00134]	**0.00230** [0.00073, 0.00404]
TCP h16 (cm)	0.0891 [0.0209, 0.1614]	0.0176 [−0.0025, 0.0414]	0.0715 [−0.0042, 0.1480]

B. Absolute matched and hard-donor risk

Endpoint	Model	Matched	95% interval	Hard	95% interval
FT h4	Align+Cross	0.3343	[0.2501, 0.4262]	0.3369	[0.2518, 0.4300]
FT h4	Monolithic	0.3205	[0.2390, 0.4104]	0.3208	[0.2391, 0.4105]
TCP h16 (cm)	Align+Cross	4.899	[4.355, 5.482]	4.989	[4.446, 5.561]
TCP h16 (cm)	Monolithic	4.349	[3.925, 4.786]	4.367	[3.948, 4.800]

C. A+C minus Monolithic risk

Endpoint	Donor	Difference	95% paired interval
FT h4	matched	0.0138	[−0.0002, 0.0345]
FT h4	hard	0.0161	[0.0028, 0.0363]
TCP h16 (cm)	matched	0.5503	[0.3699, 0.7485]
TCP h16 (cm)	hard	0.6218	[0.4302, 0.8282]

Action use and donor-count controls. The additional 25-task validation bank holds checkpoints fixed while zeroing actions, replacing donors, or averaging $K = 1, 2, 4$ histories. Task-paired differences first average three seeds, then use 10,000 task draws. Align + Cross has positive donor penalties for FT and TCP; Cross has positive FT but unresolved TCP penalties. The Align + Cross TCP donor effect decreases with donor count, while Random retains a small positive FT effect. Ablation establishes action use. Table 44 lists action, donor and history-count effects.

Table 44: **Action use and donor-count sensitivity on a separate 25-task bank.** A: action-zero minus observed, and shuffled-minus-correct donor effects. B: random-minus-correct effects at fixed donor count K and fixed weights. Pointwise 95% task-bootstrap intervals follow averaging three sources. FT uses MSE; TCP uses cm.

A. Action and donor interventions

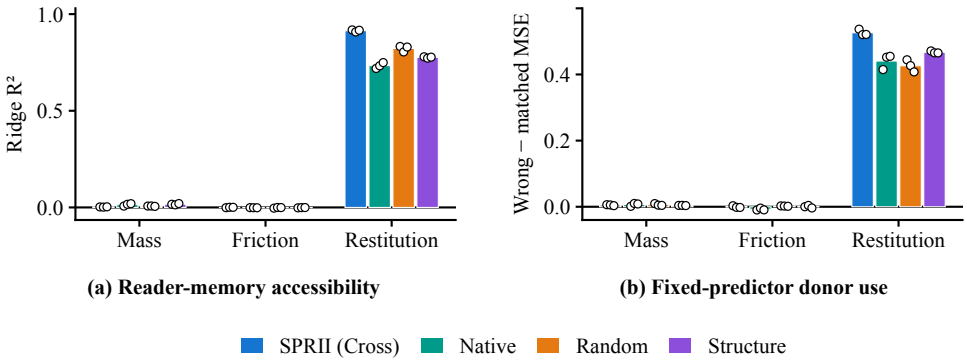
Intervention/model	Metric	Difference	95% task interval
A+C: action zero	FT h4	0.002074	[0.000740, 0.003650]
A+C: action zero	TCP h4	0.06439	[0.04507, 0.08274]
A+C: action zero	TCP h16	0.1399	[0.07687, 0.2025]
Align + Cross	FT h4	0.009692	[0.004828, 0.01624]
Align + Cross	TCP h4	0.4705	[0.2801, 0.6948]
Align + Cross	TCP h16	0.3385	[0.1768, 0.5304]
Cross	FT h4	0.01711	[0.009042, 0.02876]
Cross	TCP h4	0.004189	[−0.02059, 0.02922]
Cross	TCP h16	0.01571	[−0.0472, 0.07335]

Table 44 (continued). B. Donor-count sensitivity at fixed weights

Model	K	FT h4 MSE	TCP h4 (cm)	TCP h16 (cm)
Align + Cross	1	0.01019 [0.00476, 0.01786]	0.4686 [0.2741, 0.6968]	0.3368 [0.1719, 0.5327]
Align + Cross	2	0.00874 [0.00325, 0.01648]	0.3652 [0.1675, 0.6004]	0.2614 [0.09066, 0.4618]
Align + Cross	4	0.00794 [0.00240, 0.01566]	0.2978 [0.08930, 0.5383]	0.2191 [0.04070, 0.4249]
Cross	1	0.01606 [0.00836, 0.02703]	0.00488 [−0.01979, 0.03002]	0.01573 [−0.04830, 0.07455]
Cross	2	0.01593 [0.00634, 0.02960]	0.00691 [−0.01343, 0.02892]	0.01599 [−0.02912, 0.05778]
Cross	4	0.01602 [0.00531, 0.03114]	0.00869 [−0.01117, 0.02953]	0.01166 [−0.02378, 0.04752]
Random	1	0.00071 [0.00017, 0.00133]	0.00162 [−0.00353, 0.00672]	0.00113 [−0.01151, 0.01315]
Random	2	0.00067 [0.00017, 0.00126]	0.00110 [−0.00355, 0.00570]	−0.00018 [−0.01234, 0.01100]
Random	4	0.00066 [0.00016, 0.00122]	0.00078 [−0.00368, 0.00521]	−0.00091 [−0.01283, 0.00794]

F.4 Physical information in reader memory: Balls

RSSM reader memory selectively exposes restitution, and fixed-predictor substitutions show corresponding donor use. The measured memory includes the trained support projection and is distinct from complete source state U and independent-support code S. Twelve RSSM fits (four recipes by three joint source/reader seeds) undergo one-factor donor substitutions at fixed reader weights.



(a) Reader-memory accessibility **(b) Fixed-predictor donor use**

Figure 19: **Restitution dominates reader-memory accessibility and donor effects.** (a) Training-fitted ridge R^2 on RSSM reader memory. (b) Wrong-one-factor minus matched MSE: 2,000 recipients, fixed weights. Bars average three source/reader fits; symbols mark fits. SPRII uses Cross.

For the third SPRII (Cross) replica, wrong restitution changes MSE from 1.1823 to 1.7033; mass and friction changes are 0.00331 and −0.00209. Its within/between ratio is 1.059: within-system code variation remains slightly larger than between-system variation. S3/S5/S8 supply three/five/eight distinct eligible support histories. Repeat supplies three copies of one history; results average over the three original histories. All inputs pass through the same fixed S3-trained reader.

JEPA and CoPhyNet factor substitutions also favor restitution; their cohorts and replication counts are given with the corresponding results.

Actual-memory channels and fixed-reader substitutions. The CoPhyNet probes below measure actual reader memory M; source U and independent-support S are separate channels. Development-bank coverage is specified in Appendix B.6.

Table 45: **Distinct histories lower error through the fixed S3-trained RSSM reader.** MSE averages three fits; Repeat averages repetitions of each individual history. Reader weights remain fixed.

Recipe	Repeat	S3	S5	S8
SPRII (Cross)	1.29	1.171	1.156	1.147
Native	1.532	1.286	1.241	1.206
Random	1.497	1.356	1.333	1.316
Structure	1.517	1.288	1.246	1.218

Table 46: **Reader-memory probes, RSSM geometry and separate donor assays.** A: training-fitted CoPhyNet memory probes, including the learned projection. B: RSSM same-physics/wrong-physics code-distance ratios. C: separate single-source evaluations, reporting wrong-factor minus matched MSE. These panels measure distinct channels and cohorts.

A. CoPhyNet reader-memory probes (R^2)

Recipe	Factor	Source 1	Source 2	Source 3
SPRII (A+C)	Mass	0.0509	0.0546	0.0678
	Friction	0.0027	−0.0020	0.0002
	Restitution	0.8297	0.8266	0.8210
Native	Mass	0.0283	0.0271	0.0411
	Friction	−0.0014	0.0024	0.0009
	Restitution	0.7581	0.7599	0.7684
Random	Mass	0.0306	0.0262	0.0295
	Friction	−0.0025	−0.0043	0.0007
	Restitution	0.7611	0.7681	0.7598

B. RSSM within/between geometry

Recipe	Source 1	Source 2	Source 3
SPRII (Cross)	1.182	1.373	1.059
Native	3.904	1.769	2.493
Random	2.370	2.429	2.334
Structure	2.479	2.415	2.496

C. Separate donor assay at 100 source epochs

Learner	Matched MSE	Δm	Δ friction	Δ restitution
JEPA	1.240	+0.000490	+0.001173	0.5140
CoPhyNet	1.006	+0.004422	−0.001568	0.5184

Source U, independent support S and actual reader memory M are distinct channels. A+C denotes Align + Cross. Table 45 gives history-count means.

F.5 Recipient-specific updates through frozen weights: Swimmer

Recipient-specific persistent updates reduce error for three frozen predictors. The 512 prospective systems are disjoint from training and selection; averages cover three adapter checkpoints, then 36 query/candidate rows per system. Four thousand paired-system draws quantify uncertainty.

Table 47: **Recipient-specific updates improve all three frozen Swimmer predictors.** Gain is wrong-system minus own-system update error in 10^{-5} units. Intervals are paired-system 95% bootstrap intervals on 512 new systems.

Frozen base	Gain	95% interval	Positive systems
Persistent-JEPA	5.329	[3.930, 6.724]	65.04%
Masked-GRU (base seed 64101)	6.930	[5.415, 8.621]	67.97%
Masked-GRU (base seed 64103)	6.661	[4.878, 8.486]	68.36%

Each base retains its architecture and training normalization. The within-base comparison measures the value of its recipient-specific update through fixed predictor weights.

Task benefit under different targets and fitted readouts is examined in Appendix G.

G VALUE: CONDITIONAL TASK BENEFIT AND READER REALIZATION

This section expands Figure 5: SpringWorld tests horizon and reader effects, D-Clean contrasts direct context with decoded parameters, and PokeWorld separates aggregate from component-level benefit. Population and replication details follow Appendix B.6.

G.1 PREDICTION HORIZON AND RECIPIENT EVIDENCE: SPRINGWORLD

History benefit grows toward longer horizons for both cold and moving queries in the 178-system common-adapter grid. Both query types have zero observed recipient transitions and differ in initial motion. The comparison spans four strata, three sources and three readers per source.

Table 48: **History benefit increases with prediction horizon.** Error reduction (%) for separately fitted Persistent versus Null readers; pointwise stratified system-bootstrap 95% intervals.

Horizon	Cold: gain [95% interval]	Moving: gain [95% interval]
1	$1.95\,[-8.47, 11.01]$	$1.15\,[-0.83, 3.01]$
2	$13.39\,[3.93, 21.49]$	$1.98\,[-0.22, 4.02]$
4	$23.19\,[14.37, 30.35]$	$3.23\,[0.91, 5.77]$
8	$35.20\,[26.74, 41.92]$	$11.62\,[7.17, 16.57]$
16	$35.68\,[28.22, 42.27]$	$29.73\,[22.68, 36.45]$

The 36.10% cold-start and 7.11% full-mixture summaries use different populations of 100 and 178 systems, respectively. The temporal-reader study varies recipient evidence under a separate protocol.

Recipient-prefix manipulation. A separate temporal reader uses $r \in \{0, 1, 2, 4, 8\}$. At h16, r0/r8 errors are 0.4693/0.4672 for Null, 0.3344/0.3334 for Persistent and 0.1646/0.1562 for Oracle. The observed endpoint changes are small under this temporal-reader protocol.

The recipient-prefix reader uses 100 validation systems and three sources by three readers. Its endpoint intervals use 4,000 system draws and generally include zero.

G.2 DIRECT CONTEXT VERSUS PHYSICAL BOTTLENECKS

Takeaway. Direct context outperforms a decoded-parameter interface in D-Clean; the same comparison favors decoded parameters in SpringWorld.

Four fresh readers share the frozen Align + Cross query encoder: Null, Persistent, Decode and Oracle. Persistent and Decode receive exactly the same independent donor history; the train-fitted parameter decoder is frozen before reader fitting. Each arm uses the same 10,000-step horizon-conditioned recipe, three sources and three readers, with 32 windows from each of 200 validation systems. This population is separate from the 100-system TDS comparison.

At h32, Persistent MSE is 5.90×10^{-4} versus 7.64×10^{-4} for Decode and 2.23×10^{-4} for Oracle. The paired Decode-minus-Persistent gap is 1.74×10^{-4} (95% interval: $[0.682, 3.07] \times 10^{-4}$), positive in all nine cells. Independent validation of the frozen decoder gives $R^2 = 0.9967, 0.9973, 0.9970$ on systems 1000–1199. Accurate parameter estimation therefore need not supply a predictively equivalent learned interface.

SpringWorld's opposite ordering. The common-adapter comparison projects the same donor code either directly or through training-only ridge estimates of three log-parameters. All arms share query encoder, donor identities, sample order and target normalization; Decode/Oracle add only the 256-parameter input projection to the common reader. On the 100-system cold-start bank, Decode-minus-Persistent is -0.002362 $[-0.004573, -0.0004762]$: explicit physical parameters preserve

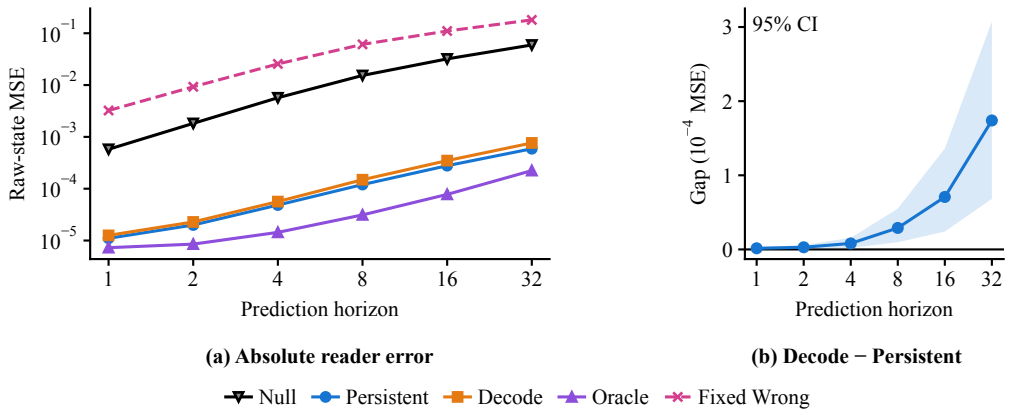


Figure 20: **Direct context outperforms the physical bottleneck across D-Clean horizons.** (a) Raw-state MSE (log scale); Fixed Wrong substitutes donors through the Persistent reader. (b) Decode-minus-Persistent MSE with pointwise system-bootstrap 95% intervals, conditional on fitted sources and readers. The primary endpoint is h32.

and slightly improve realized value here. Persistent reduces Null error by 36.10%, and Persistent error is 2.12% higher than Decode error, using the same Decode-denominator convention as the D-Clean comparison. Table 50 gives actual-donor probes on 178 systems and separate full-mixture results.

Table 49: **Direct-context and bottleneck readers use the same D-Clean donors.** (A) Raw-state MSE; (B) pointwise paired-system 95% intervals conditional on the fitted source–reader grid; (C) independent validation of the frozen drag decoder. The primary endpoint is h32.

A. Absolute reader and intervention error

Horizon	Null	Persistent	Decode	Oracle	Fixed Wrong
1	0.000577	0.000011	0.000013	0.000007	0.003227
2	0.001821	0.000020	0.000023	0.000009	0.009263
4	0.005689	0.000048	0.000056	0.000014	0.025560
8	0.015320	0.000120	0.000149	0.000031	0.061030
16	0.031970	0.000278	0.000349	0.000078	0.110300
32	0.059360	0.000590	0.000764	0.000223	0.180900

B. Paired contrasts and intervention uncertainty

Horizon	Fixed Wrong 95% CI	D−P (10^{-4})	D−P 95% CI (10^{-4})
1	[0.002261, 0.004323]	0.0154	[0.0059, 0.0281]
2	[0.006822, 0.01199]	0.0291	[0.0076, 0.0586]
4	[0.01949, 0.03224]	0.0809	[0.0218, 0.1565]
8	[0.04752, 0.07559]	0.2898	[0.0965, 0.5504]
16	[0.08612, 0.1362]	0.7078	[0.2409, 1.3620]
32	[0.1433, 0.2213]	1.7390	[0.6824, 3.0740]

C. Independent frozen-decoder validation

Sampling seed	Validation R^2	γ MSE	95% MSE interval
0	0.9967	0.003653	[0.0027, 0.0052]
1	0.9973	0.002957	[0.0022, 0.0043]
2	0.9970	0.003344	[0.0025, 0.0046]

D−P is Decode minus Persistent. Fixed Wrong changes the Persistent input, without fitting a fifth reader. The grid contains 27 Null/Persistent/Oracle fits and nine Decode fits; no per-arm source SD is reconstructed.

Table 50: **SpringWorld reader error and donor-factor accessibility.** (A) Common-adapter MSE across three sources and three readers. (B) Training-fitted log-parameter probes on 712 donor histories from 178 validation systems, weighted equally by system.

A. Reader risk: cold-start and full-mixture populations

Population	Null	Persistent	Decode	Oracle
Cold start: 100 systems / 600 cases	0.17784	0.11364	0.11128	0.06188
Full mixture: 178 systems	0.87168	0.80973	0.80873	0.76594

B. Actual-donor accessibility: 178 systems

Source recipe	Seed	R^2_m	R^2_γ	R^2_k	$R^2_{k/m}$
Align + Cross	0	0.3918	0.1754	0.2050	0.4842
	1	0.3112	0.1537	0.1421	0.3774
	2	0.3569	0.1618	0.2345	0.5115
FCRL-style temporal	0	−0.0000	−0.0019	−0.0180	−0.0091
	1	0.0107	−0.0073	−0.0349	−0.0154
	2	0.0030	0.0013	−0.0134	−0.0056

Cold-start Decode-minus-Persistent is -0.002362 $[-0.004573, -0.0004762]$. The k/m estimate derives from fitted mass/stiffness coordinates. B's probe bank differs from A's 100-system cold-start risk population.

G.3 Aggregate and component-level benefit: PokeWorld

Persistent context improves object-position and velocity prediction, with benefits that vary across output components. The R8 source study fits separate Null, Persistent, Shuffled and Oracle readers with the same architecture and training schedule. All 36 combinations use three sources by three reader seeds, 10,000 updates, and the reused 400-system development bank with 16 windows per system. Shuffled fits a separate reader using wrong-context inputs.

At the prespecified h16 endpoint, object-position and velocity errors fall, while finger-position error rises. The aggregate Null-minus-Persistent interval spans zero; Persistent improves on the separately trained Shuffled reader.

Actual donor-window drag readout is strong ($R^2 = 0.747$–0.783), compared with 0.131–0.164 for query-only inputs. Even with true physical parameters, the fitted Oracle reader improves aggregate h16 error by only 0.200% in this reader family (Persistent: 0.057%), while finger-position error rises.

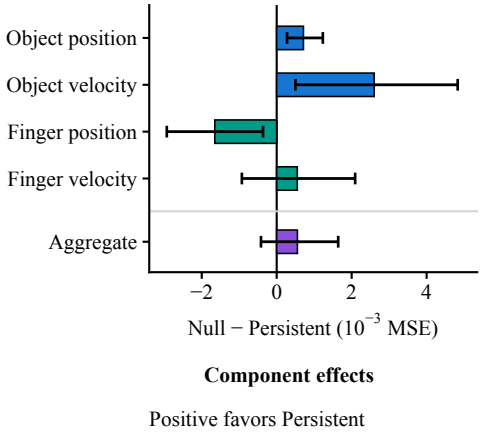


Figure 21: **Persistent context improves object-state prediction, with mixed effects on finger state.** Null-minus-Persistent MSE on the development bank; positive favors Persistent. Pointwise paired-system 95% intervals condition on the fitted source–reader grid; the aggregate interval adjusts for three contrasts. Components are not additive.

Factor-targeted utility. The 36-reader G_1/G_2 development grid pairs task gains with actual-donor physical probes. Its utility-versus-simulator-sensitivity slope is 0.002810 $[-0.004443, 0.009203]$, with positive slopes in each source. G_1 retains higher mass readout and G_2 higher drag readout in 5,913 actual donor windows. Table 52 gives the slopes and donor probes. Additional population checks are grouped in Appendix G.4.

Aggregate, component and factor-targeted value. For targeted utility, $\Delta U = (E_N - E_P)_{G_1} - (E_N - E_P)_{G_2}$ is regressed against $d = z_{\text{train}}(S_m) - z_{\text{train}}(S_\gamma)$. Simulator sensitivities use central

Table 51: **Aggregate PokeWorld gains are small despite component-level benefits.** Separate reader fits on 400 development systems; aggregate 95% intervals adjust for three contrasts. Oracle uses true parameters.

A. Aggregate h16 MSE

Arm	MSE
Null	0.9629
Persistent	0.9624
Shuffled	0.9637
Oracle	0.961

B. Prespecified aggregate contrasts

Contrast	Difference	Adjusted 95% interval
Null–Persistent	0.000553	$[-0.000419, 0.001641]$
Null–Oracle	0.001927	$[0.000427, 0.003615]$
Shuffled–Persistent	0.001367	$[0.000274, 0.002498]$

C. Oracle component reference: Null–Oracle

Component	Position	Velocity
Object	0.002809	0.006168
Finger	-0.003009	0.001740

$\pm 1\%$ perturbations, fixed query/actions and training factor/target scales. Actual-donor probes cover 5,913 distinct windows, equally weighting systems, with ridge coefficient 1.

Table 52: **PokeWorld factor-targeted utility and donor accessibility.** (A) The G_1-minus-G_2 history-benefit slope against simulator sensitivity has a system-bootstrap interval spanning zero. (B) Source-averaged probe R^2 on actual donor windows. The development grid includes initial fits.

A. Continuous slope of ΔU against sensitivity dominance

Source seed	0	1	2
Expanded-grid slope	0.003838	0.003307	0.001284

Mean: 0.002810; 95% system-bootstrap interval: $[-0.004443, 0.009203]$

B. Train-fitted donor-code probe, validation R^2

Recipe	$\log m$	$\log \gamma$	$\log k$	$\log(k/m)$
G_1	0.20700	0.12460	0.08187	0.24440
G_2	0.07075	0.78890	0.07491	0.14640

G.4 Realizing value through frozen-source readouts

Changing the consuming route recovers additional value from frozen SpringWorld context, while the PokeWorld comparisons show that this recovery is not automatic.

Reader realization across frozen sources. Separately fitted, development-selected M_1/M_2 readers favor M_2 for SpringWorld Align + Cross in all nine standard and physically weighted pairs, but favor M_1 in all five PokeWorld G_1/G_2 pairs (Figure 22(a); Table 53). Align + Cross uses three sources by three readers; Structure uses one source and three readers, and Poke G_1/G_2 use source 2 with two/three readers. The routes share their architecture but not exactly their active capacity; Appendix B.5 defines the M_1/M_2 formulas.

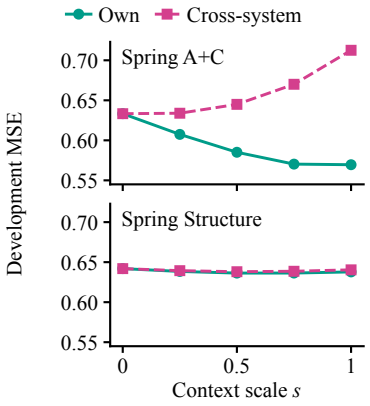
Table 53: **Reader recovery differs across frozen source families.** Descriptive mean errors and gains $E(M_1) - E(M_2)$; positive favors M_2. Weighted gains use the separate physically weighted route. These fitted-run means do not estimate uncertainty.

Family	Sources × readers	M_1	M_2	Gain	Weighted gain
Align + Cross	3×3	0.6185	0.5695	0.04901	0.05059
Structure	1×3	0.6316	0.6378	-0.006213	-0.006971
Poke G_1	1×2	0.8099	0.8127	-0.002805	-0.001495
Poke G_2	1×3	0.7919	0.7987	-0.006841	-0.006443

All nine Align + Cross pairs favor M_2 in both route variants; all five Poke pairs favor M_1. Figure 22 shows within-fit percentages, while this table retains absolute differences.

Source	Seeds × readers	Standard	Weighted
Spring A+C	s0/1/2 × 3	+7.94%	+8.22%
Spring Structure	s0 × 3	-0.99%	-1.11%
Poke G1	s2 × 2	-0.35%	-0.18%
Poke G2	s2 × 3	-0.86%	-0.81%

(a) Mean within-fit error reduction



(b) Fixed M2 response

Figure 22: **Reader recovery depends on source family and donor context.** (a) Mean within-fit error reduction, $100[E(M_1) - E(M_2)]/E(M_1)$; positive favors M_2. Reader percentages are averaged within source, then across sources. (b) SpringWorld raw MSE under own- and cross-system donors through fixed M_2 readers; $s = 0$ gives the recipient-only base R_0.

Matched-architecture context control. Matched context lowers error relative to mismatched context in separately fitted M_2 readers with identical architecture. The control reuses the three-source by three-reader grid, with each source and recipient-only base R_0 frozen. Mismatched readers use a fixed different-system donor map during training, selection and evaluation, with unchanged recipient inputs and targets. They receive 1,000 AdamW updates with batch size 256 and learning rate 10^{-3}, selecting checkpoints every 100 updates on the original development population. All readers are evaluated on the same records; frozen-source probes appear in Table 50.

Matched M_2 lowers mean error by 8.7% on the full development mixture and 33.7% at cold-start h16 relative to mismatched M_2, with the same ordering in all nine pairs at both endpoints.

Mismatched M_2 improves on M_1 in none of the nine pairs. Figure 23 shows the paired contrasts and both source-paired and hierarchical intervals; Table 54 retains the raw errors.

Table 54: **Matched context lowers error with identical M_2 architecture.** Task MSE, mean ± sample SD across three sources, each averaging three readers. Both populations are development data: 178 systems for the full mixture and 100 for cold-start h16.

Endpoint	M_1	M_2 matched	M_2 mismatched
Full development	0.6185 ± 0.009762	0.5695 ± 0.01796	0.6240 ± 0.007561
Cold start, h16	0.1799 ± 0.006272	0.1216 ± 0.01436	0.1835 ± 0.008084

Formation does not guarantee reader value. Increasing relation reliability strengthens organization and drag accessibility, but mean M_2 gains remain negative at the 5,000-update h16 endpoint for every α; the $\alpha = 1$ interval lies entirely below zero (Table 55). A larger fitting budget reduces absolute error. Each $\alpha \in \{0, 0.5, 1\}$ uses three frozen 20,000-update PokeWorld sources and three readers per source. Unlike post-fitting scale s, α changes source training relations. Each paired contrast compares readers within a fixed source; query encoders vary across α.

The reader predicts train-standardized eight-state displacements on 400 systems from the existing validation bank. The recipient-only base R_0 and each route fit for 1,000 or 5,000 updates, with batch size 128 and learning rate 0.001, using the same 5,913 windows and development selection rule. Budgets share the same frozen sources, cached observations and evaluation geometry. Table 55 reports paired source/system bootstrap intervals. At 5,000 updates, all three wrong-minus-matched intervals include zero. Zero-context and wrong-donor interventions measure channel dependence and content specificity, respectively.

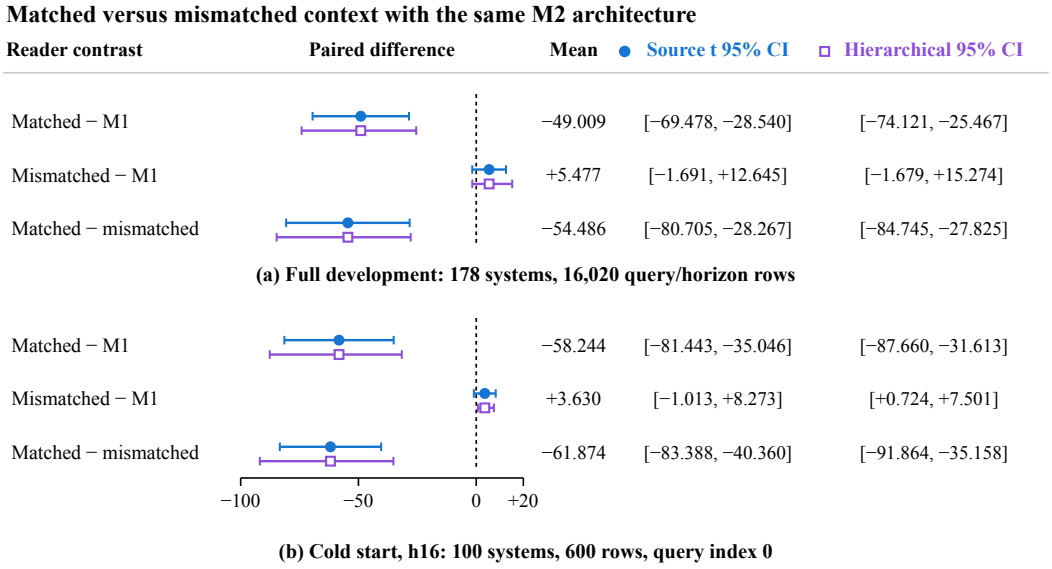


Figure 23: **Paired error differences for the matched-architecture control.** (a) Full development mixture; (b) cold-start h16. Negative values favor the first reader (10^{-3} MSE). Source-paired 95% t intervals use three source means; hierarchical 95% intervals resample sources, readers and systems. Both condition on development-selected fits; matched and mismatched M_2 are fitted separately.

Table 55: **Stronger formation does not ensure greater reader value.** h16 gain is $E(M_1) - E(M_2)$ in 10^{-3} normalized MSE; positive favors M_2. Budgets share frozen sources and development systems, averaging three readers per source. B/W uses training-standardized latent statistics.

A. Organization, accessibility and pooled reader value

α	B/W	$R^2_{\log \gamma}$	1k gain	5k gain	5k 95% interval
0	0.320	0.101	-2.512	-1.890	$[-8.366, +3.116]$
0.5	1.042	0.254	$+0.061$	-2.942	$[-8.255, +1.948]$
1	3.796	0.797	-4.469	-6.291	$[-12.010, -0.328]$

B. Source-specific reader value

α	Reader updates	Source 0	Source 1	Source 2
0	1000	-3.463	-3.017	-1.057
0.5	1000	$+1.222$	-0.015	-1.023
1	1000	-3.310	-4.186	-5.910
0	5000	-4.739	-1.617	$+0.686$
0.5	5000	-2.123	-3.436	-3.267
1	5000	-5.288	-11.270	-2.314

Intervals use 5,000 paired source/system bootstrap draws and are descriptive with three source seeds.

Donor-specific responses through fixed readers. SpringWorld donor substitutions keep the fitted source and reader weights fixed while varying post-fitting context scale s. Own-system context lowers Align + Cross error as its scale increases, whereas whole-system cross donors raise it (Figure 22(b); Table 56). This difference supports content-specific use through the fixed reader. The single-source Structure comparison has smaller, nonmonotone own-context effects. At $s = 0$, M_2 equals the recipient-only base R_0, not M_1; standard and physically weighted routes are evaluated separately.

Table 56: **Donor identity changes the response to context scale.** Development MSE as persistent inputs are multiplied by s, with donor substitutions through fixed readers. At $s = 0$, the route equals the recipient-only base R_0, not M_1.

A. Standard M_2; displayed in Figure 22

Source	Donor	$s = 0$	0.25	0.5	0.75	1
Align + Cross	Own	0.6333	0.6074	0.5851	0.5703	0.5695
	Cross-system	0.6333	0.6339	0.6450	0.6700	0.7125
Structure	Own	0.6419	0.6384	0.6363	0.6364	0.6378
	Cross-system	0.6419	0.6394	0.6381	0.6386	0.6404

B. Physically weighted $M_{2,\mathrm{phys}}$

Source	Donor	$s = 0$	0.25	0.5	0.75	1
Align + Cross	Own	0.6333	0.6090	0.5859	0.5684	0.5660
	Cross-system	0.6333	0.6337	0.6434	0.6671	0.7102
Structure	Own	0.6419	0.6387	0.6365	0.6361	0.6372
	Cross-system	0.6419	0.6394	0.6376	0.6376	0.6388

Route-selection and fixed-intervention evaluators remain separate.

Additional PokeWorld reader checks. On 100 new systems with 16 windows each, six frozen source exports and twelve paired readers give utility slopes $+0.009202, -0.1686, +0.1314$ that vary in sign. A separate single-source object-only check uses four-dimensional targets on 400 development systems and independently selected readers: the R8 $M_2 - M_1$ interval includes zero, while G_2 has higher M_2 mean error (Table 57).

Table 57: **Object-only PokeWorld reader comparisons.** h16 object-state MSE on 400 development systems for one source–reader pair. M_1 and M_2 are selected independently; the paired interval is conditional on these fits and evaluation systems.

Source	M_2	M_1	Recipient-only
R8	0.9011	0.9026	0.9118
G_2	0.9132	0.9085	0.9080

R8 $M_2 - M_1$: -0.001497 $[-0.011855, 0.008400]$.

G.5 Closed-loop control with an ID-selected recipe: Pendulum

The ID-selected SPRII recipe uses relation weight 0.003 and variance target 0.1. Mass and length are sampled independently from discrete sets: both use $\{0.75, 0.80, \ldots, 1.25\}$ in ID. For OOD c0–c3, the respective (mass, length) sampling sets are $\{0.2, 0.4\} \times \{1.6, 1.8\}$, $\{0.5, 0.7\} \times \{1.3, 1.5\}$, $\{1.3, 1.5\} \times \{0.5, 0.7\}$, and $\{1.6, 1.8\} \times \{0.2, 0.4\}$. Three training seeds each contribute ten 200-step episodes per group. Table 58 reports the return and success rate from those same trajectories. Success means that all of the final 100 post-action states lie within 60° of upright; this criterion is unchanged.

Table 58: **Pendulum control with one ID-selected SPRII recipe.** Mean return ± sample SD across three seeds, with success percentage in parentheses; higher is better. All OOD groups use this recipe.

Group	CaDM	Random relation	SPRII (ID-selected)
ID	-326.5 ± 46.5 (96.7%)	-321.4 ± 33.2 (100.0%)	-323.9 ± 26.2 (100.0%)
OOD_c0	-376.6 ± 63.4 (93.3%)	-346.8 ± 30.6 (100.0%)	-322.2 ± 29.4 (100.0%)
OOD_c1	-387.0 ± 11.9 (93.3%)	-400.1 ± 24.6 (93.3%)	-381.0 ± 13.6 (100.0%)
OOD_c2	-355.5 ± 33.9 (90.0%)	-479.6 ± 126.7 (76.7%)	-370.2 ± 51.8 (90.0%)
OOD_c3	-1100 ± 273.5 (20.0%)	-1185 ± 131.2 (6.7%)	-1060 ± 384.4 (30.0%)

SPRII has higher mean return than CaDM in four of five groups, but lower return in OOD c2; the direction is not shared by all seeds in every group. The return and success endpoints remain separate. All five groups use the same ID-selected SPRII recipe; no per-group recipe selection is performed.

Same-checkpoint ID probes give mass R^2 of 0.027, 0.012 and 0.050 for CaDM, Random and SPRII, respectively, and length R^2 of 0.670, 0.683 and 0.669. The corresponding common-bank own-donor H10 MSEs are 0.0181, 0.0226 and 0.0190; wrong-minus-matched errors are 0.1132, 0.0956 and 0.1018. These logged-action, fixed-model interventions test context use separately from online control. The weak mass probes do not establish recovery of all physical parameters.

H Generality Across Learners, Partners, and Fields

CoPhy varies learner and scene; Overcooked separates behavioral readout from cooperation outcomes; and FHN varies the number of independent histories for field prediction. The suite map in Appendix A.3 links each application to its implementation. The corresponding Formation, Use and Value analyses appear in Appendices E, F and G.

H.1 Learner and scene transfer: CoPhy

The CoPhy comparisons show both gains and adverse effects across learner families and scenes. We distinguish the source-0 development matrix, additional-source replication, and sensitivity to the fitted reader. The three-seed held-out Collision comparison is reported in Appendix C.2.

Breadth across learners and scenes. The source-0 matrix contains 45 evaluated learner–scene–recipe cells across JEPA, CPC, RSSM and supervised CoPhyNet in Balls, Collision and Blocktower. Figure 24 gives relative error reductions against each available control, and Table 59 gives absolute errors. CoPhyNet has Native and Random controls; its Structure arm was not evaluated.

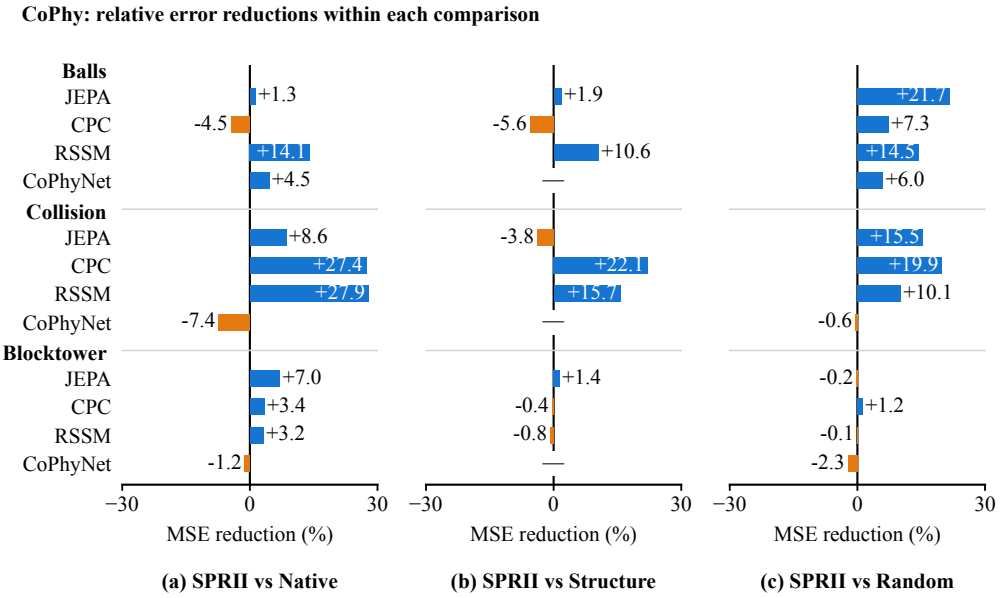


Figure 24: **CoPhy gains depend on learner, scene and comparator.** (a)–(c) Percentage MSE reduction against Native, Structure and Random: $100(\text{control} - \text{SPRII})/\text{control}$; positive favors SPRII. These are single-source development comparisons. CoPhyNet Structure was not evaluated in this matrix; held-out Collision results appear separately in Table 10.

Table 59: **CoPhy prediction error by learner and scene.** Source-0 development MSE under S3/query3 and 100 source epochs / 100 reader epochs. Dashes mark unevaluated Structure arms; bold marks the lowest MSE per row.

Scene	Family	Native	Structure	SPRII	Random
Cross: self-supervised dynamics objectives					
Balls	JEPA	1.2563	1.2630	**1.2396**	1.5827
	CPC	1.4399	**1.4239**	1.5042	1.6228
	RSSM	1.3105	1.2582	**1.1254**	1.3155
Collision	JEPA	0.2512	**0.2212**	0.2297	0.2717
	CPC	0.2448	0.2283	**0.1778**	0.2220
	RSSM	0.2731	0.2338	**0.1970**	0.2192
Blocktower	JEPA	0.0931	0.0878	0.0865	**0.0863**
	CPC	0.1030	**0.0991**	0.0995	0.1007
	RSSM	0.0974	**0.0935**	0.0943	0.0942
Align + Cross: supervised CoPhyNet objective					
Balls	CoPhyNet	1.0534	–	**1.0056**	1.0698
Collision	CoPhyNet	**0.2033**	–	0.2184	0.2171
Blocktower	CoPhyNet	0.0871	–	0.0882	**0.0862**

Source information and consumed memory. Source U, independent support S and trained reader memory M are separate measurements (Appendices B.3 and B.5). The complete source-U grid covers all 45 source roles after 100 source epochs, seed 0; it uses train-fitted ridge probes before the supervised reader projection. In Balls, SPRII restitution R^2_U is 0.7875/0.7672/0.8133/0.6170 for JEPA/CPC/RSSM/CoPhyNet, while mass/friction accessibility stays weak. Accessibility and task error dissociate here as well: CPC Balls has $R^2_U = 0.7672$ versus Native 0.5959 but MSE 1.5042 versus 1.4399. Collision mass access is lower for SPRII than Native in JEPA (0.3683 versus 0.3799) and CoPhyNet (0.1757 versus 0.2004).

Replication across sources and in Balls. Additional sources preserve the comparison grid: JEPA Blocktower has higher error than Random at both sources, while JEPA Collision changes ordering against Structure. Appendix B.6 specifies source coverage and development selection.

Table 60: **CoPhy prediction and accessibility across source seeds.** Development results include reader-selection examples. S denotes independent support and M fitted reader memory. Probes target restitution (Balls), mass (Collision) and vertical gravity (Blocktower). Dashes: unevaluated cells.

	Prediction MSE			Support R^2_S			Reader memory R^2_M		
Arm	**Src 0**	**Src 1**	**Src 2**	**Src 0**	**Src 1**	**Src 2**	**Src 0**	**Src 1**	**Src 2**
JEPA / Balls									
Native	1.256	1.319	—	0.7977	0.7981	—	0.7865	0.7778	—
Structure	1.263	1.294	—	0.7855	0.7866	—	0.7832	0.7788	—
Cross	1.240	1.305	—	0.9162	0.9185	—	0.9051	0.9116	—
Random	1.583	1.644	—	0.6777	0.6918	—	0.6884	0.6823	—
JEPA / Collision									
Native	0.2512	0.2444	—	0.4431	0.4366	—	0.5657	0.5619	—
Structure	0.2212	0.2238	—	0.5976	0.6246	—	0.6659	0.6706	—
Cross	0.2297	0.2154	—	0.6422	0.6718	—	0.6451	0.6713	—
Random	0.2717	0.2645	—	0.5005	0.5250	—	0.5286	0.5502	—
JEPA / Blocktower									
Native	0.0931	0.0895	—	0.1028	0.0902	—	0.0430	0.0467	—
Structure	0.0878	0.0876	—	0.1252	0.1301	—	0.0714	0.0879	—
Cross	0.0865	0.0876	—	0.1761	0.1564	—	0.1242	0.0860	—
Random	0.0863	0.0853	—	0.1673	0.1678	—	0.1256	0.1154	—
CPC / Collision									
Native	0.2448	0.2517	—	0.4372	0.4213	—	0.5604	0.5484	—
Structure	0.2283	0.2323	—	0.6126	0.5929	—	0.6476	0.6367	—
Cross	0.1778	0.1841	—	0.7495	0.7411	—	0.7541	0.7419	—
Random	0.2220	0.2212	—	0.6657	0.6588	—	0.6717	0.6711	—
RSSM / Collision									
Native	0.2731	0.2674	0.2729	0.334	0.240	0.344	0.497	0.494	0.477
Structure	0.2338	0.2332	0.2273	0.564	0.565	0.570	0.627	0.642	0.634
Cross	0.1970	0.1997	0.2112	0.738	0.723	0.733	0.736	0.719	0.724
Random	0.2192	0.2199	0.2284	0.633	0.643	0.634	0.678	0.669	0.664

RSSM Collision: three-source summary

Arm	**MSE: mean ± SD**	**Support R^2_S**	**Reader memory R^2_M**
Native	0.2711 ± 0.0032	0.306	0.490
Structure	0.2314 ± 0.0036	0.566	0.635
Cross	0.2026 ± 0.0075	0.731	0.727
Random	0.2225 ± 0.0051	0.636	0.671

The three-source Balls extension uses 2,000 development recipients with exact pairing. Its relation-trained arm improves Matched error against Native/Random/Structure by 8.95%/13.68%/9.13% for RSSM, and against Native/Random by 3.91%/3.19% for CoPhyNet. All five recipient-bootstrap family-adjusted intervals are positive. Table 61 reports Null, Matched and Wrong inputs and all five contrasts; these intervals condition on the fitted models and exclude selection uncertainty. Actual-memory use is analyzed in Appendix F.4; the source-U probes above measure a different channel.

Reader and configuration sensitivity. With frozen sources, the CoPhyNet reader recipe changes comparison sign and magnitude across scenes. At 50 source epochs, the factorial in Table 62 varies initialization, dropout and delivery; the endpoint at 100 source epochs below uses a separate budget.

For Native, SPRII and Random, respectively, the every-step reader at 100 source epochs yields MSEs of 1.037, 0.9562 and 1.125 on Balls, and 0.2085, 0.2228 and 0.2188 on Collision.

Table 63 complements this reader analysis with operation and configuration comparisons, using the evaluation populations and fitted-model coverage stated in each row.

Table 61: **All five Balls contrasts favor relation training.** Three-source means on 2,000 development recipients. Recipient-bootstrap 95% intervals are Bonferroni-adjusted over five contrasts, conditional on fitted models and excluding selection uncertainty.

A. Absolute risks under recorded context conditions				
Family	**Arm**	**Matched**	**Null**	**Wrong**
RSSM	SPRII (C)	1.171	1.892	2.296
	Native	1.286	2.485	2.293
	Random	1.356	2.004	2.300
	Structure	1.288	1.937	2.335
CoPhyNet	SPRII (A+C)	1.044	1.791	2.228
	Native	1.087	1.773	2.244
	Random	1.079	1.759	2.216

B. Comparator minus SPRII, paired before source averaging			
Family	**Comparator**	**Difference**	**Adjusted 95% interval**
RSSM	Native	0.1151	$[0.0902, 0.1395]$
	Random	0.1856	$[0.1607, 0.2096]$
	Structure	0.1177	$[0.0927, 0.1433]$
CoPhyNet	Native	0.0424	$[0.0276, 0.0574]$
	Random	0.0344	$[0.0193, 0.0497]$

Table 62: **CoPhyNet gains depend on the reader recipe.** SPRII versus Native error reduction (%) for one frozen source on validation data (50 source epochs / 100 reader epochs). A and B denote fixed random-weight initializations. History enters only at initialization unless marked every step.

Reader recipe	**Balls**	**Collision**	**Blocktower**
Initialization B, dropout 0.1, initial only	-5.448	-4.089	-1.224
Initialization A, dropout 0	$+7.506$	-2.104	-1.477
Initialization A, dropout 0.1	$+6.074$	-4.426	$+2.090$
Initialization B, dropout 0	$+0.617$	-7.591	$+0.621$
Initialization B, dropout 0.1, every step	$+6.646$	-2.925	$+1.134$

Table 63: **CoPhy operation and configuration comparisons.** Errors are lower-is-better MSE. Development rows use one fitted source/reader seed; paired intervals condition on the reported models and evaluation units.

Configuration	Coverage	Numerical comparison	Finding / scope
JEPA Collision: selected 50-epoch configuration	1,994 test episodes; 3 seeds per arm	SPRII 0.184 ± 0.005; Structure 0.215 ± 0.007	14.2% lower than Structure (all arms use 50 source epochs); all four arms in Table 10.
CoPhyNet Collision: Cross	4,000 development recipients; seed 0	Cross 0.2032; Align+Cross 0.2107	Near Native; paired interval against Native includes zero.
CPC Balls: Align	2,000 development recipients; seed 0	Align 1.393; Cross 1.415; Structure 1.388	Lower than Cross, higher than Structure.
CoPhyNet Blocktower	8,088 development recipients; seed 0	SPRII 0.088180; Native 0.087114; separately evaluated Structure 0.087114; Random 0.086230. SPRII−Random: 0.001951 $[0.000668, 0.003184]$	Paired interval versus the separate Structure control includes zero; Random has lower mean error.

Each row uses its stated population and fitted-model coverage. The CoPhyNet Collision Align+Cross reference uses $\lambda_{\mathrm{p}} = \lambda_{\mathrm{x}} = 0.1$; the CPC Balls Cross reference uses $\lambda_{\mathrm{x}} = 0.3$. Both use 50 source epochs, 100 reader epochs and init-only history delivery.

H.2 Recurring-partner behavior and cooperation: Overcooked

Relative to H+VC, relation alignment lowers behavioral-probe error across the three history budgets. At full history, familiar-partner returns also improve in each of three training seeds. All comparisons use one layout and fixed 20,000-update endpoints; Appendix B.4 gives the shared evaluation protocol.

Table 64: **Full-history alignment improves familiar-partner returns in all three seeds.** Paired Align − H+VC differences, mean ± sample SD across three training seeds. Lower probe MSE and higher return are better. Familiar and held-out development partners are reported separately.

History	Familiar probe MSE	Familiar return	Held-out dev. return
25%	-0.00567 ± 0.00810	1.489 ± 2.335	-2.222 ± 15.21
50%	-0.01344 ± 0.00728	0.389 ± 1.110	18.44 ± 20.66
100%	$\mathbf{-0.02120} \pm 0.00610$	$\mathbf{1.889} \pm 2.175$	$\mathbf{-1.111} \pm 11.44$

For full history, the familiar return differences are $+4.400$, $+0.667$ and $+0.600$ across seeds, whereas held-out development differences are $+9.333$, $+0.667$ and -13.333. The latter population therefore shows mixed return effects. At 25% and 50% history, seeds also differ in familiar-return signs.

Table 65 compares behavioral readout, representation distances and cooperation returns at the same checkpoints. At full history, Align lowers probe MSE relative to Random in all three seeds (paired difference -0.02583 ± 0.00342), and the Align–H+VC probe difference is -0.02120 ± 0.00610. The Align mean, 0.05395, remains above the constant-mean reference, 0.04333; held-out development probe means also remain above their reference, 0.07062.

Table 65: **Overcooked behavioral readout and cooperation returns.** Mean ± sample SD across three training seeds. A denotes Align; VC denotes H+VC. The distance ratio is different-/same-partner distance on the familiar behavioral-signature bank, computed per seed before averaging. Random has no return evaluation.

A. Behavioral readout and representation

History	Recipe	Familiar MSE ↓	Held-out dev. MSE ↓	Distance ratio
25%	VC	0.06788 ± 0.00667	0.13880 ± 0.04230	1.343 ± 0.061
25%	A	0.06220 ± 0.00974	0.09733 ± 0.01384	1.917 ± 0.071
50%	VC	0.06934 ± 0.00811	0.14090 ± 0.02324	1.408 ± 0.019
50%	A	0.05590 ± 0.00736	0.08966 ± 0.01240	2.329 ± 0.326
100%	VC	0.07516 ± 0.00791	0.12400 ± 0.02857	1.458 ± 0.036
100%	A	0.05395 ± 0.01106	0.10630 ± 0.01046	2.269 ± 0.206
100%	Random	0.07978 ± 0.01423	0.14840 ± 0.03203	1.460 ± 0.066

B. Cooperation returns

History	Recipe	Familiar partners	Held-out dev. partners
25%	VC	15.78 ± 1.262	-1.333 ± 4.372
25%	A	17.27 ± 1.768	-3.556 ± 14.850
50%	VC	15.20 ± 0.200	-19.56 ± 2.341
50%	A	15.59 ± 0.916	-1.111 ± 18.410
100%	VC	14.04 ± 1.500	-19.11 ± 8.572
100%	A	15.93 ± 0.742	-20.22 ± 3.672

H.3 History-count scaling in reaction-diffusion fields: FHN

Additional independent histories reduce h50 field-prediction error in both methods, with larger relative reductions for SPRII on all three splits. From $K = 1$ to $K = 4$, SPRII error decreases by 16.9%, 18.0% and 10.9% on ID, OOD-Intra and OOD-Extra, compared with 8.4%, 9.2% and 7.1% for NOD. The released NOD-Hier architecture runs on the regenerated Python simulation bank described in Appendix B.4. Initial condition 5 selects configurations; reporting uses 15, 24 and 45, with three fixed sources 42/43/44. Source 42 is the development seed; 43/44 repeat the selected recipe. NOD

uses 45,000 updates and SPRII 50,000; these history-count comparisons therefore do not isolate alignment at equal training compute. The training schedules are specified in Appendix B.4.

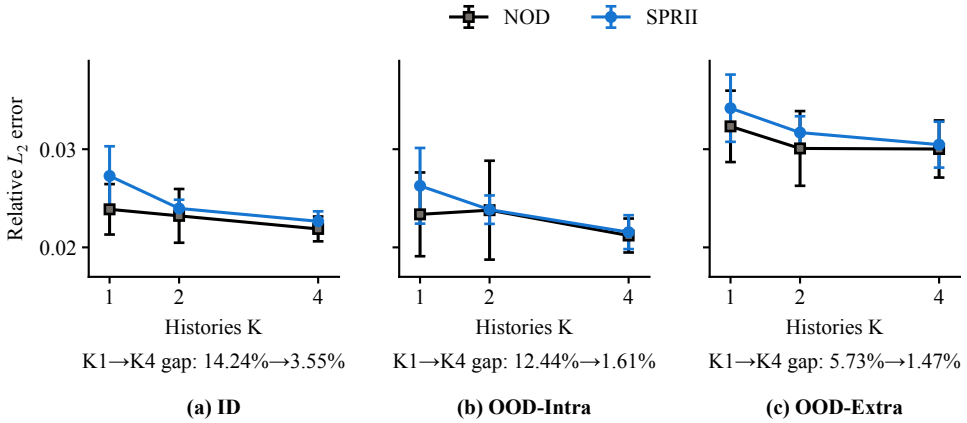


Figure 25: **Additional histories narrow the FHN endpoint gap on all three splits.** H50 relative-L_2 error: (a) ID, (b) OOD-Intra, (c) OOD-Extra. Points and whiskers show mean ± sample SD across three sources. Annotations give $100(\text{SPRII} - \text{NOD})/\text{NOD}$ at $K = 1, 4$.

Table 66: **Additional histories narrow the FHN prediction gap.** H50 relative-L_2 error averaged across three sources; lower is better. Both methods receive the same K independent histories, with different training budgets as described above.

	$K = 1 \to 4$ **reduction**		**NOD error**		**SPRII error**	
Split	**SPRII**	**NOD**	$K = 1$	$K = 4$	$K = 1$	$K = 4$
ID	16.9%	8.4%	0.02388	0.02188	0.02728	0.02266
OOD-Intra	18.0%	9.2%	0.02336	0.02121	0.02627	0.02155
OOD-Extra	10.9%	7.1%	0.03233	0.03002	0.03418	0.03046

Consequently, the gaps to NOD on ID, OOD-Intra and OOD-Extra narrow from 14.24%, 12.44% and 5.73% at $K = 1$ to 3.55%, 1.61% and 1.47% at $K = 4$. NOD retains slightly lower absolute mean error at $K = 4$. The probe table below shows accessibility by factor, split and source.

FHN factor accessibility. The bank contains nine training, four OOD-Intra and twelve OOD-Extra systems; $K = 1$ uses initial condition 36. All six models share 13,500 report-case keys.

Table 67: **FHN accessibility varies by factor, split and source.** Training-fitted probe R^2 with one history ($K = 1$).

A. k

Split	**Method**	**Src 42**	**Src 43**	**Src 44**
ID	NOD	0.9698	0.9301	0.9836
ID	SPRII	0.9444	0.3775	0.8952
Intra	NOD	0.9269	0.8098	0.9377
Intra	SPRII	0.8602	0.5513	0.6923
Extra	NOD	0.9351	0.9139	0.9284
Extra	SPRII	0.8587	0.0949	0.8730

B. β

Split	**Method**	**Src 42**	**Src 43**	**Src 44**
ID	NOD	0.9627	0.9496	0.9394
ID	SPRII	0.9766	0.9449	0.9097
Intra	NOD	0.9877	0.9829	0.9765
Intra	SPRII	0.9600	0.9269	0.8990
Extra	NOD	0.9521	0.9409	0.9299
Extra	SPRII	0.9848	0.9518	0.9154

The additional $K = 2, 4$ probes use their four-recipient population.

A separate single-source comparison (source 42) evaluates SPRII against extra-50k NOD on report initial conditions 15/24/45. SPRII has slightly lower mean error only on ID at $K = 4, H = 50$ (relative L_2: 0.02013 versus 0.02152; the paired-system interval includes zero), while extra-50k NOD is lower in the other eight split/horizon cells.

I THEORETICAL CONNECTIONS

We analyze four reference models connecting relations to persistent-context learning. Pairing determines which variations agreement penalizes; predictive relevance selects among stable directions; fixed-route interventions measure context dependence; and conditional risk characterizes the value available to a downstream reader. Each proposition states its required assumptions.

I.1 SETUP AND LIFECYCLE

Setup. Let G be the grouping variable specified by a relation, H an interaction history, and $Z = \phi(H) \in \mathbb{R}^d$ a fixed representation with finite second moments. A group may denote a whole system, selected attributes, a partner, or a recorded task. Write $\mu_G = \mathbb{E}[Z \mid G]$, $\Sigma_{\text{between}}(G) = \text{Cov}(\mu_G)$, and $\Sigma_{\text{within}}(G) = \mathbb{E}[\text{Cov}(Z \mid G)]$. The sampling law is $\Pi_G(A, B) = P(B)Q_G(A \mid B)$, as in Eq. (5). Conditional independence is imposed explicitly for the iid results below; the general pair law permits dependent interaction histories.

Remark: available information, reliance and realized value. The lifecycle distinguishes information present in a context, reliance by a specified predictor, and task value realized by that predictor. These are operationally assessed with geometry and probes, fixed-predictor substitutions, and task readers, respectively; failure of a particular probe is not evidence that the information is absent. For a fixed joint law of recipient inputs X, donor code D and target Y, conditional independence of Y and D given X implies zero *Bayes-optimal incremental value* $\Delta_\ell^\star$ in Proposition 4. This statement concerns additional target information beyond X, not the difference between arbitrary finite fitted readers: their excess-risk terms can differ even when $\Delta_\ell^\star = 0$. Conversely, accessible information need not be relevant beyond X, and positive available value need not be realized by a fitted reader. Donor sensitivity establishes reliance under the specified substitution, while the residual cross term in Appendix I.3 determines whether that reliance reduces risk. A newly fitted reader's benefit concerns its own use of context; it does not establish use by the original predictor.

I.2 FORMATION: PAIRING, RELIABILITY AND PREDICTIVE SELECTION

Pairing structure.

Proposition 1 (structure exposed by pairing). For a pair law with finite second moments, means μ_A, μ_B, marginal covariances Σ_A, Σ_B, and cross covariance Σ_{AB},

$$\mathbb{E}_{\Pi_G}\|Z^A - Z^B\|_2^2 = \text{tr}\,\Sigma_A + \text{tr}\,\Sigma_B - 2\,\text{tr}\,\Sigma_{AB} + \|\mu_A - \mu_B\|_2^2. \tag{14}$$

If a matched pair is instead specified by drawing G from its population law and then drawing Z^A, Z^B conditionally iid from $P(Z \mid G)$, then

$$\text{Cov}(Z^A, Z^B) = \Sigma_{\text{between}}(G), \qquad \mathbb{E}\|Z^A - Z^B\|_2^2 = 2\,\text{tr}\,\Sigma_{\text{within}}(G). \tag{15}$$

Proof. Expand the squared difference and use $\mathbb{E}\|Z^A\|^2 = \text{tr}\,\Sigma_A + \|\mu_A\|^2$ and $\mathbb{E}[(Z^A)^\top Z^B] = \text{tr}\,\Sigma_{AB} + \mu_A^\top \mu_B$. This proves Eq. (14). For the ideal matched pair, write $Z^i = \mu_G + \epsilon_i$. Conditional centering gives $\mathbb{E}[\epsilon_i \mid G] = 0$, and conditional independence gives $\mathbb{E}[\epsilon_A \epsilon_B^\top \mid G] = 0$. Thus the cross covariance is $\text{Cov}(\mu_G)$, while the covariance of $\epsilon_A - \epsilon_B$ given G is $2\,\text{Cov}(Z \mid G)$. Taking its trace and averaging proves Eq. (15). The same centering argument also yields the total-covariance identity $\text{Cov}(Z) = \Sigma_{\text{between}}(G) + \Sigma_{\text{within}}(G)$. □

What changes under an actual sampler? Conditional centering gives, without the iid premise,

$$\Sigma_{AB} = \text{Cov}(\mathbb{E}[Z^A \mid G], \mathbb{E}[Z^B \mid G]) + \mathbb{E}[\text{Cov}(Z^A, Z^B \mid G)].$$

The two mixed mean–residual terms vanish by iterated expectation. Overlap, shared episode dynamics, balancing, and forced mismatches can leave the second term nonzero or alter the conditional means. The actual pairing law therefore determines which dependence alignment preserves. Under ideal pairing, alignment penalizes within-group differences and leaves group-mean variation as a candidate for retention. A constant code also has zero alignment loss; predictive and non-collapse terms provide the additional constraints needed to learn informative codes.

Reliability and independent repetitions. Reliability changes whether shared variation is exempt from the disagreement penalty. Let Q_c select correct pairs and Q_r random pairs with the same donor and recipient marginals. For $\alpha \in [0,1]$, the mixture $Q_\alpha = \alpha Q_c + (1-\alpha)Q_r$ implies $\mathbb{E}_{Q_\alpha}\ell = \alpha\mathbb{E}_{Q_c}\ell + (1-\alpha)\mathbb{E}_{Q_r}\ell$ for any integrable pair loss at fixed parameters.

Proposition 2 (alignment pressure under corrupted relations). Let $H = L\theta + N$, where θ has covariance Σ_θ and independent zero-mean noise N has covariance Σ_N. Correct pairs share θ and independently resample N; random pairs independently resample both. For a fixed encoder $Z = WH$,

$$\mathbb{E}_{Q_\alpha}\|W(H^A - H^B)\|_2^2 = 2\,\mathrm{tr}(W\Sigma_N W^\top) + 2(1-\alpha)\,\mathrm{tr}(WL\Sigma_\theta L^\top W^\top). \tag{16}$$

Proof. Under Q_c, $H^A - H^B = N^A - N^B$ has zero mean and covariance $2\Sigma_N$. Under Q_r, independence gives covariance $2L\Sigma_\theta L^\top + 2\Sigma_N$ for the difference. In each case, the expected squared norm after applying W is the trace of the transformed covariance. Mixing these two expressions proves the result. □

Correct pairing leaves persistent variation outside this penalty; random pairing contracts it. The coefficient $1-\alpha$ describes the objective in this reference model; Figure 3(a) reports the learned response. If only alignment assignments change, per-view variance/covariance regularizers remain common. If Cross consumes the corrupted assignment, its task loss changes too. The calculation therefore isolates alignment pressure; the monotone geometry in Figure 3(a) is an empirical response of separately trained nonlinear models.

Corollary (factor-specific alignment pressure). In Proposition 2, let $\theta = (m, \gamma, k)$ have independent components, $L = [L_m, L_\gamma, L_k]$, and $\mathrm{Var}(\gamma) = \sigma_\gamma^2$. Correct pairs share all three factors. In the corrupted component, retain m, k and independently redraw only γ from its marginal; independently redraw the observation noise in both components. For a fixed W,

$$\mathbb{E}\|W(H^A - H^B)\|^2 = 2\,\mathrm{tr}(W\Sigma_N W^\top) + 2(1-\alpha)\sigma_\gamma^2\,\mathrm{tr}(WL_\gamma L_\gamma^\top W^\top). \tag{17}$$

Proof. The correct-pair difference is $N^A - N^B$. In the corrupted component it is $L_\gamma(\gamma^A - \gamma^B) + N^A - N^B$; independence and centering remove cross terms and give second moment $2\sigma_\gamma^2 L_\gamma L_\gamma^\top + 2\Sigma_N$. Taking the trace after applying W and mixing the two components yields Eq. (17). □

The additional penalty acts along L_γ; the shared m, k variations cancel from the pair difference. This fixed-encoder statement isolates the pressure exerted by alignment. It does not imply that retraining a nonlinear model leaves all other factors' geometry or probe accuracy unchanged: predictive losses, capacity and variance constraints also select the learned representation. Factor-selective degradation is therefore an empirical hypothesis for Appendix E.4. For a finite bank whose sampler *forces* unequal drag values, independent redrawing is not the sampling law. Replace $2\sigma_\gamma^2$ by the actual $\mathbb{E}_Q[(\gamma^A - \gamma^B)^2]$; for uniform distinct draws from n bank entries, this is $2n\sigma^2_{\gamma,\mathrm{bank}}/(n-1)$ when bank variance uses denominator n. More general balancing and unequal supports use the actual pair second moment in Proposition 1. The qualitative cancellation of exactly shared factors remains, while the coefficient must match the sampler.

Why separate realizations can add evidence. For conditionally iid codes $Z_1, \ldots, Z_R \mid G$ with mean μ_G and covariance $\Sigma(G)$, the average $\bar{Z}_R = R^{-1}\sum_r Z_r$ has conditional covariance $\Sigma(G)/R$: independence removes the off-diagonal covariance terms in the sum. Consequently, $\mathbb{E}\|\bar{Z}_R - \mu_G\|^2 = \mathrm{tr}\,\Sigma_{\mathrm{within}}(G)/R$. For dependent repetitions, the remaining cross-covariance terms determine the reduction. This estimation reference describes the information available from repeated histories; SPRII learns its encoder from those histories. The R2/R4/R8 bank comparison in Table 30 tests the learning effect of providing more interaction realizations.

Semantic refinement and predictive selection. A relation specifies which variation may persist across a pair; the training objective determines which eligible factor the representation emphasizes. We first analyze a fixed code, then give a learning example in which the dominant factor changes.

Semantic refinement for a fixed representation. Let G_2 refine G_1, so that G_1 is a measurable function of G_2. For the same fixed Z under the same population law, total covariance gives

$$\Sigma^Z_{\text{between}}(G_2) - \Sigma^Z_{\text{between}}(G_1) = \mathbb{E}[\text{Cov}(\mathbb{E}[Z \mid G_2] \mid G_1)] \succeq 0.$$

To see this, write $M_i = \mathbb{E}[Z \mid G_i]$. The tower property gives $\mathbb{E}[M_2 \mid G_1] = M_1$; total covariance applied to M_2 yields the identity. Positive semidefiniteness follows because the last term's quadratic form is $\mathbb{E}[\text{Var}(v^\top M_2 \mid G_1)] \geq 0$ for every v. Since the total covariance of Z is unchanged, within-group covariance decreases by the same matrix.

Thus $G_1 = m$, $G_2 = (m, \gamma)$, and $G_3 = (m, \gamma, k)$ successively make drag and stiffness eligible as group-stable variation. This identity compares *one representation under different groupings*. Experimental donor pools may include forced mismatches and have non-nested supports; their effects enter through the actual pair law.

Predictive selection under relation constraints. The preceding identities keep the representation fixed while changing its grouping. We now allow the representation to change and examine the joint influence of predictive reward and relation disagreement in a joint spectral selection problem.

A constrained predictive reference. Let U collect the inputs supplied outside the history code, such as current state, actions, and horizon. To model history information beyond U, assume finite second moments and consider the residuals $\widetilde{H} = H - \mathbb{E}[H \mid U]$ and $\widetilde{Y} = Y - \mathbb{E}[Y \mid U]$, with a centered linear model $\widetilde{Y} = K\widetilde{H} + \eta$ and $\mathbb{E}[\eta\widetilde{H}^\top] = 0$. Assume positive-definite history covariance on the modeled subspace and whiten that subspace. Below, H, Y denote these transformed variables, with $\text{Cov}(H) = I_p$ and $Y = KH + \eta$ (absorbing the whitening into K). Residualization and whitening define this analytical reference model.

Proposition 3 (joint predictive and relation selection). For $Z = WH$, $W \in \mathbb{R}^{d\times p}$ with $1 \leq d < p$, impose $WW^\top = I_d$ and let $\beta \geq 0$. Consider

$$\mathcal{J}_G(W) = \min_C \mathbb{E}\|Y - CWH\|_2^2 + \beta\mathbb{E}_{\Pi_G}\|W(H^A - H^B)\|_2^2. \tag{18}$$

Write $D_G = \mathbb{E}_{\Pi_G}[(H^A - H^B)(H^A - H^B)^\top]$ and $M_G = K^\top K - \beta D_G$. The optimal decoder is $C^\star = KW^\top$, and the optimal row spaces of W are top-d eigenspaces of M_G:

$$W^\star_G \in \arg\max_{WW^\top = I_d} \text{tr}(WM_GW^\top). \tag{19}$$

The selected subspace is unique if $\lambda_d(M_G) > \lambda_{d+1}(M_G)$; its orthonormal basis need not be.

Proof. Since $\text{Cov}(WH) = I_d$ and $\text{Cov}(Y, WH) = KW^\top$, completing the square gives

$$\begin{aligned}\mathbb{E}\|Y - CWH\|_2^2 &= \text{tr}\,\text{Cov}(Y) - 2\,\text{tr}(CWK^\top) + \text{tr}(CC^\top)\\ &= \text{tr}\,\text{Cov}(Y) - \|KW^\top\|_F^2 + \|C - KW^\top\|_F^2.\end{aligned}$$

Minimization sets $C = KW^\top$ and leaves predictive reward $\text{tr}(WK^\top KW^\top)$. The alignment term equals $\text{tr}(WD_GW^\top)$, including when the pair difference has a nonzero mean. Removing the constant target variance therefore yields Eq. (19).

To characterize the maximum, diagonalize the symmetric matrix $M_G = V\,\text{diag}(\lambda_1, \ldots, \lambda_p)V^\top$ with decreasing eigenvalues. Since $W^\top W$ is a rank-d orthogonal projector, $a_i = \|Wv_i\|_2^2$ satisfies $0 \leq a_i \leq 1$ and $\sum_i a_i = d$. Hence

$$\text{tr}(WM_GW^\top) = \sum_{i=1}^p \lambda_i a_i \leq \sum_{i=1}^d \lambda_i.$$

Choosing the first d eigenvectors as rows of W attains the bound. A strict eigengap requires projection onto precisely that subspace; ties permit alternative choices within the tied eigenspace. □

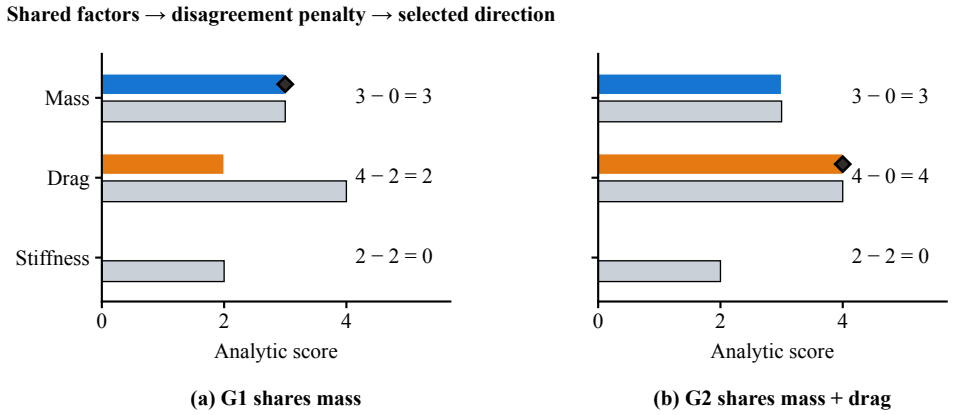


Figure 26: **A broader sharing relation can select a different predictive direction.** In the Proposition 3 example, broadening (a) G_1 to (b) G_2 changes the selected direction (diamond) from mass to drag. Gray bars show predictive scores; colored bars subtract relation disagreement.

A realizable three-factor crossing. Let $H = (M, \Gamma, K_0)$ have independent standard-normal coordinates, use $d = 1$, $\beta = 1$, and set $K^\top K = \mathrm{diag}(3, 4, 2)$. Under $G_1 = M$, pairs share mass and independently resample the other two coordinates. Under $G_2 = (M, \Gamma)$, they share mass and drag. Both laws have the same standard-normal marginals. A shared coordinate cancels in the pair difference; two independent standard normals have difference variance 2. Therefore

$$\begin{aligned} D_{G_1} &= \mathrm{diag}(0, 2, 2), \quad M_{G_1} = \mathrm{diag}(3, 2, 0), \\ D_{G_2} &= \mathrm{diag}(0, 0, 2), \quad M_{G_2} = \mathrm{diag}(3, 4, 0). \end{aligned} \tag{20}$$

Proposition 3 selects $W^\star_{G_1} = e_m^\top$ but $W^\star_{G_2} = e_\gamma^\top$. Mass remains shared, yet removing drag's penalty makes drag the leading direction. Broader semantic eligibility is therefore compatible with a different, non-nested learned subspace. This reproduces the qualitative factor crossover in Figure 3(b) and Appendix E.1: the preferred direction changes while the original factor remains shared.

The argument also presupposes available evidence: for a fixed encoder, a factor independent of its input remains independent of its code. The actual-input estimators in Table 27 therefore check for factor evidence in histories before asking which factors the representation emphasizes.

I.3 Use: fixed-predictor reliance

A fixed-weight intervention holds the predictor and recipient inputs fixed and changes supplied persistent context. It tests functional use without refitting readers for each donor condition.

Risk change under a donor substitution. Fix a predictor f and a joint evaluation law for (X, Y, D_m, D_s), where D_m is matched and D_s is the specified replacement. Assume finite second moments for the target and both predictions. Write $\hat{Y}_m = f(X, D_m)$, $\hat{Y}_s = f(X, D_s)$, and let $\Delta_{\mathrm{spec}} = \mathbb{E}\|Y - \hat{Y}_s\|^2 - \mathbb{E}\|Y - \hat{Y}_m\|^2$. Expanding $Y - \hat{Y}_s = (Y - \hat{Y}_m) + (\hat{Y}_m - \hat{Y}_s)$ yields

$$\Delta_{\mathrm{spec}} = \mathbb{E}\|\hat{Y}_s - \hat{Y}_m\|^2 + 2\mathbb{E}\langle Y - \hat{Y}_m, \hat{Y}_m - \hat{Y}_s\rangle. \tag{21}$$

The risk change includes both prediction sensitivity and its interaction with the original residual. If the matched prediction is $\mathbb{E}[Y \mid X, D_m]$ and the replacement adds no conditional target information, so $\mathbb{E}[Y \mid X, D_m, D_s] = \mathbb{E}[Y \mid X, D_m]$, conditioning makes the cross term vanish. Then specificity equals the mean-squared prediction change. The general decomposition applies to any trained predictor with finite risk.

Positive specificity establishes lower risk with the matched donor under the specified replacement law. Near-zero specificity can arise from cancellation of the two terms despite different predictions. The donor and recipient protocols in RH20T, SpringWorld and Swimmer specify that law. Zero-context replacement measures channel dependence, while wrong-donor replacement measures content specificity. Absolute-risk comparisons and fresh-reader evaluations quantify task benefit.

I.4 VALUE: CONDITIONAL RISK AND READER REALIZATION

The same relation-conditioned interface supports numerical prediction and categorical action learning, but their native losses measure different quantities. We apply standard conditional-risk and information-theoretic identities (Cover & Thomas, 2006; Gneiting & Raftery, 2007) to separate available task value from its realization by a trained reader.

Common setup. Fix the encoders, target construction, and pairing law $\Pi_G(A, B) = P(B)Q_G(A \mid B)$. They induce a joint law of the permitted recipient inputs $X = X_B$, donor code $D = D_A$, and target $Y = Y_B$. Write $R_\ell(f) = \mathbb{E}\ell(f, Y)$. Risk is evaluated under this fixed joint law, allowing statistical dependence between donor and recipient. For Cross it is the recipient-conditioned risk in Eq. (3); the same analysis applies to a native history-conditioned reader in an Align-only instance.

Proposition 4 (task value and reader error). Let $\Delta^\star_\ell$ be the optimal recipient-only risk minus the optimal risk with (X, D), allowing unrestricted measurable readers.

(a) Numerical targets. For square-integrable Y and squared loss, the optimal readers are $m_0 = \mathbb{E}[Y \mid X]$ and $m_D = \mathbb{E}[Y \mid X, D]$, and

$$\Delta^\star_{\mathrm{sq}} = \mathbb{E}\|Y - m_0\|^2 - \mathbb{E}\|Y - m_D\|^2 = \mathbb{E}\|m_D - m_0\|^2 \geq 0. \tag{22}$$

(b) Categorical targets. For Y in a finite alphabet and log loss $\ell(p, Y) = -\log p(Y)$, write $p_0 = P(Y = \cdot \mid X)$ and $p_D = P(Y = \cdot \mid X, D)$. These distributions attain the respective optimal risks $H(Y \mid X)$ and $H(Y \mid X, D)$, so

$$\Delta^\star_{\log} = H(Y \mid X) - H(Y \mid X, D) = I(Y; D \mid X) = \mathbb{E}\,\mathrm{KL}(p_D \| p_0) \geq 0. \tag{23}$$

Logarithms are natural. We assume conditional distributions exist; X and D may be continuous.

For either loss, two learned readers f_0 and f_D with finite risk obey

$$R_\ell(f_0) - R_\ell(f_D) = \Delta^\star_\ell + \mathcal{E}^\ell_0 - \mathcal{E}^\ell_D, \tag{24}$$

where the excess risks are

$$\mathcal{E}^{\mathrm{sq}}_0 = \mathbb{E}\|f_0 - m_0\|^2, \quad \mathcal{E}^{\mathrm{sq}}_D = \mathbb{E}\|f_D - m_D\|^2,$$
$$\mathcal{E}^{\log}_0 = \mathbb{E}\,\mathrm{KL}(p_0 \| \widehat{p}_0), \quad \mathcal{E}^{\log}_D = \mathbb{E}\,\mathrm{KL}(p_D \| \widehat{p}_D).$$

Here the learned log-loss readers are probability vectors $\widehat{p}_0, \widehat{p}_D$. The KL comparisons use the same conditioning inputs; finite risk excludes assigning zero probability to positive-probability events.

Proof. For squared loss, $r = Y - m_D$ has zero conditional mean given (X, D) and is orthogonal to $v = m_D - m_0$. Expanding $Y - m_0 = r + v$ proves Eq. (22). More generally, for a reader f with its corresponding conditional mean m,

$$\mathbb{E}\|Y - f\|^2 = \mathbb{E}\|Y - m\|^2 + \mathbb{E}\|m - f\|^2,$$

by the same orthogonality, establishing optimality and the squared-loss excess risks.

For log loss, adding and subtracting the true log probabilities gives

$$\mathbb{E}[-\log \widehat{p}_D(Y) \mid X, D] = -\sum_y p_D(y) \log \widehat{p}_D(y) = H(p_D) + \mathrm{KL}(p_D \| \widehat{p}_D).$$

The nonnegativity of KL divergence shows that p_D is optimal. Average this equality over (X, D) and repeat it conditioned on X to obtain both optimal risks and excess-risk terms. Finally, iterated expectation gives

$$\mathbb{E}\,\mathrm{KL}(p_D \| p_0) = \mathbb{E} \log \frac{p_D(Y)}{p_0(Y)} = H(Y \mid X) - H(Y \mid X, D) = I(Y; D \mid X).$$

Subtracting the two reader-risk decompositions for each loss proves Eq. (24). □

What this separates. A factor can be perfectly accessible in D while irrelevant to Y: if F_1, F_2 are independent, $D = F_1$, and $Y = F_2$ with no recipient inputs, the code's task value is zero. Redundancy with X can also remove its incremental value. For squared loss, zero value means that the conditional *mean* is unchanged; for categorical log loss it means that the conditional *distribution* is unchanged. The criterion thus depends on the chosen loss. Equation (24) further separates attainable value from how well the particular readers realize it. The PokeWorld frozen-source study estimates realized risk for its fitted readers. Together with the accessibility probes, it separates readable information from the benefit those readers obtain.

An explicit cold-start example. Let $\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, $X = \theta + \epsilon_x$, and $D = \theta + \epsilon_d$, with centered Gaussian noises independent of each other and of θ, and positive prior and noise variances. For $Y = a_h\theta + \epsilon_y$, assume centered target noise of finite variance independent of $\theta, \epsilon_x, \epsilon_d$. Completing the square in the Gaussian conditional density gives recipient posterior variance $v_x = (\sigma_\theta^{-2} + \sigma_x^{-2})^{-1}$ and, after observing D, variance $(v_x^{-1} + \sigma_d^{-2})^{-1}$. Target noise cancels from the risk difference, giving

$$\Delta^\star_{\mathrm{sq}} = a_h^2 \left[v_x - (v_x^{-1} + \sigma_d^{-2})^{-1} \right] = a_h^2 \frac{v_x^2}{v_x + \sigma_d^2}. \tag{25}$$

Within this model, larger remaining recipient uncertainty v_x makes the same history more valuable. Averaging R independent history measurements replaces σ_d^2 by σ_d^2/R. Target sensitivity enters through a_h; a longer horizon helps only insofar as it changes the target's dependence on persistent information or the available evidence. The model explains the role of recipient uncertainty in SpringWorld cold start. A controlled change in recipient evidence holds population, target and fitted route fixed; horizon effects additionally depend on a_h.

Connection to training and generalization beyond JEPA. For fixed encoders, targets, and pairing law, the isolated Cross objective asks the reader for the recipient target's conditional mean under squared loss, or its conditional distribution under categorical log loss. Sharing an object, partner, or task constructs potentially informative pairs; the donor still needs to add target-relevant information beyond X. A numerical target can be a future state or a latent embedding held fixed for this reference; a categorical target can be an observed action. The complete SPRII objective fits these components jointly through a shared predictor.

A training law and an evaluation law define separate risks; each identity holds within its specified law and fixed representation. Action log loss also differs from rollout return, which depends on the policy-induced state distribution and environment response. Thus the generalization is a common relation-conditioned learning interface, with native targets and losses specifying what the reader learns to use. Native-task evaluation tests its realized benefit.

Connection to the downstream intervention. The M_1/M_2 study changes the consuming interface while keeping the source and prescribed query/base path fixed. The gain $\Delta_{\mathrm{realize}} = \mathrm{Err}(M_1) - \mathrm{Err}(M_2)$ measures a difference in realized risk, with positive values favoring M_2 (Section 4.3). Equation (24) attributes this difference to conditional value and reader error jointly. A positive adapter gain demonstrates additional realization under the compared routes; the unrestricted attainable value remains a separate quantity.

J Closest Methodological Connections

The closest precedents already use related experience, separate context from state, or test dependence on supplied context. Table 68 locates the additional questions studied here.

Table 68: **Closest methodological connections.**

Work	Relevant construction	What we study here
FCRL	Same-function observation sets; contrastive encoding; new decoders on a fixed encoder.	Relation training within predictive/action learners; selected shared factors; separate measurement of code access and predictor reliance.
NOD	A separate same-system trajectory supplies a low-dimensional code for state evolution.	Embedding, outcome, and action targets; Align/Cross components; controlled relation semantics and information conditions.
RIA	Predictive interventions estimate relations used to organize dynamics context.	Observed relations provide a controlled variable whose reliability and semantics can be changed independently.
GG-ODE	Initial-state/environment separation; same-environment contrastive regularization.	An interface instantiated across base objectives, with partial-factor sharing and task/use diagnostics.
Policy embeddings	Cross-episode imitation and identity objectives; downstream policies and wrong-embedding tests.	Relationship to physical and JEPA persistence; history-conditioned action applications and controlled information-use studies.

Relations and predictive context. FCRL learns from same-function observation sets and transfers fixed encoders (Gondal et al., 2021). NOD conditions an operator on a separate same-system trajectory, directly preceding the Cross-type construction (Chen et al., 2026). RIA uses an interventional prediction module to estimate whether two inferred contexts come from the same environment, then regularizes their similarity with a relational head. GG-ODE separates initial state from environmental context (Guo et al., 2022; Huang et al., 2023). Agent embeddings also use cross-episode imitation and incorrect-context tests (Grover et al., 2018); our Overcooked instance adds alignment to a history-conditioned action learner (Laskin et al., 2023; Jing et al., 2026). Here, controlled reliability and partial-factor relations make the selected shared content an experimental variable, while Align/Cross controls separate operations within each native predictive objective.

Context inference and test-time adaptation. CaDM learns dynamics context through forward/backward prediction (Lee et al., 2020); DALI learns latent dynamics context for adaptation (Röder et al., 2025). Our D-Clean experiment adapts the context-learning component to a common frozen-code reader and reports source-training budgets explicitly; it does not rank the original control agents. CoDA and GEPS adapt physical models with context codes (Kirchmeyer et al., 2022; Kassaï Koupaï et al., 2024). The Burgers reference runs reported here use 50 support-code optimization steps per query (Appendix C.4), whereas NOD and the SPRII endpoint use amortized code inference. The frozen-decoder CoDA extension in Appendix C.5 asks the separate question of whether relation supervision improves an amortized history encoder on this host. Its four control arms distinguish encoder learning, relation assignments and additional reconstruction computation.

Prediction, accessibility and reliance. Latent world models and joint-embedding predictors provide the predictive setting (Hafner et al., 2019; 2020; Assran et al., 2023; Bardes et al., 2024). Physical-information studies distinguish input evidence from learned access (Tan et al., 2026), while paired-view learning constrains shared information (Bardes et al., 2022; Tian et al., 2020). Our grouping supervision supplies relation assignments between histories. A frozen-code probe tests accessibility; a fitted reader tests realized task value; substitution through fixed weights tests model reliance under the stated donor law (Fisher et al., 2019; Molnar et al., 2023; Watson & Wright, 2021). These distinctions motivate the separate Formation, Use and Value analyses.